

CHAPTER 7

METHODS OF INTEGRATION

*God does not care about our mathematical difficulties—
he integrates empirically.*

Albert Einstein

PREVIEW

In this chapter we increase the number of techniques and procedures for integrating a function. One of the most important techniques, substitution, is reviewed in the first section and is then expanded in several different contexts. Other important integration procedures include using tables, integration by parts, and partial fractions. In addition, improper integrals, hyperbolic functions, and inverse hyperbolic functions are discussed, along with first-order differential equations.

PERSPECTIVE

It is possible to differentiate most functions that arise in practice by applying a fairly short list of rules and formulas, but integration is a more complicated process. The purpose of this chapter is to increase your ability to integrate a variety of different functions. Learning to integrate is like learning to play a musical instrument: at first, it may seem impossibly complicated, but if you persevere, after a while music starts to happen. It should be noted that as more powerful technology becomes available, *techniques* become less important and *ideas* become more important.

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7.1 REVIEW OF SUBSTITUTION AND INTEGRATION BY TABLE

IN THIS SECTION: Review of substitution, using tables of integrals

In this section, we review our integration formulas as well as the method of integration by substitution. Today's technology has greatly enhanced the ability of professional mathematicians and scientists in techniques of integration, thereby minimizing the need for spending hours learning rarely used integration techniques. Along with this new technology, the use of integral tables has increased in importance. Most mathematical handbooks contain extensive integral tables, and we have included such an integration table in Appendix D.

For easy reference, the most important integration rules and formulas are listed inside the back cover. Notice that the first four integration formulas are the procedural rules, which allow us to break up integrals into simpler forms, whereas those on the remainder of the page form the building blocks of integration.

Review of Substitution

Remember that when doing substitution you must choose u , calculate du , and then substitute to make the form you are integrating look *exactly like* one of the known integration formulas. We will review substitution by looking at different situations and special substitutions that prove useful.

Example 1 Integration by substitution

Find $\int \frac{x^2 dx}{(x^3 - 2)^5}$.

Solution Let u be the value in parentheses; that is, let $u = x^3 - 2$. Then $du = 3x^2 dx$, so by substitution,

$$\begin{aligned} \int \frac{x^2 dx}{(x^3 - 2)^5} &= \int \frac{\frac{du}{3}}{u^5} && \text{Using substitution} \\ &= \frac{1}{3} \int u^{-5} du \\ &= \frac{1}{3} \frac{u^{-4}}{-4} + C \\ &= -\frac{1}{12} (x^3 - 2)^{-4} + C \end{aligned}$$

As with all indefinite integrals, you can check your work by finding the derivative of your answer to see if you obtain the integrand. For this example,

$$\begin{aligned} \frac{d}{dx} \left[-\frac{1}{12} (x^3 - 2)^{-4} + C \right] &= -\frac{1}{12} [-4(x^3 - 2)^{-5}(3x^2) + 0] \\ &= \frac{x^2}{(x^3 - 2)^5} \end{aligned}$$

Example 2 Fitting the form of a known integration formula by substitution

Find $\int \frac{t dt}{\sqrt{1 - t^4}}$.

Solution We notice the similarity between this and the formula for inverse sine if we let $a = 1$ and $u = t^2$. Then $du = 2t dt$ and

$$\int \frac{t \, dt}{\sqrt{1-t^4}} = \int \frac{\frac{du}{2}}{\sqrt{1-u^2}}$$

$$\begin{array}{l} \text{If } u = t^2, \text{ then } du = 2t \, dt \text{ so } dt = \frac{du}{2}. \\ \text{Also, } \sqrt{1-t^4} = \sqrt{1-(t^2)^2} = \sqrt{1-u^2}. \end{array}$$

$$= \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}}$$

$$= \frac{1}{2} \sin^{-1} u + C \quad \text{Formula 22, Appendix D}$$

$$= \frac{1}{2} \sin^{-1} t^2 + C \quad \text{Rewrite in terms of original variable } t.$$

The method of substitution (Section 5.5) is very important, because many of the techniques developed in this chapter will be used in conjunction with substitution. Examples 3 and 4 illustrate additional ways substitution can be used in integration problems.

Example 3 Multiplication by 1 to derive an integration formula

Find $\int \sec x \, dx$.

Solution Multiply the integrand $\sec x$ by $\sec x + \tan x$ and divide by the same quantity:

$$\begin{aligned} \int \sec x \, dx &= \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx && \text{The advantage of this rearrangement} \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx && \text{is that the numerator is now the} \\ &= \int \frac{du}{u} && \text{derivative of the denominator.} \end{aligned}$$

$$\begin{array}{l} \text{If } u = \tan x + \sec x, \text{ then} \\ du = (\sec^2 x + \sec x \tan x) dx \end{array}$$

$$= \ln |u| + C$$

$$= \ln |\sec x + \tan x| + C$$

You may wonder why anyone would think to multiply and divide the integrand $\sec x$ in Example 3 by $\sec x + \tan x$. To say that we do it “because it works” is probably not a very satisfying answer. However, techniques like this are passed on from generation to generation, and it should be noted that multiplication by 1 is a common method in mathematics for changing the form of an expression to one that can be manipulated more easily.

Example 4 Substitution after an algebraic manipulation

Find $\int \frac{dx}{1+e^x}$.

Solution The straightforward substitution $u = 1 + e^x$ does not work:

$$\int \frac{dx}{1+e^x} = \int \frac{\frac{du}{e^x}}{u} = \int \frac{du}{ue^x} \quad \begin{array}{l} \text{This is not an appropriate form} \\ \text{because } x \text{ has not been eliminated.} \end{array}$$

Instead, rewrite the integrand as follows:

$$\int \frac{dx}{1+e^x} = \int \frac{1}{1+e^x} \cdot \frac{e^{-x}}{e^{-x}} dx \quad \text{Multiply by 1.}$$

$$= \int \frac{e^{-x} dx}{e^{-x} + 1}$$

$$\text{If } u = e^{-x} + 1, \text{ then } du = -e^{-x} dx$$

$$= \int \frac{-du}{u}$$

$$= -\ln |u| + C$$

$$= -\ln(e^{-x} + 1) + C \quad e^{-x} + 1 > 0 \text{ for all } x, \text{ so } \ln |e^{-x} + 1| = \ln(e^{-x} + 1)$$

When the integrand involves terms with fractional exponents, it is usually a good idea to choose the substitution $x = u^n$, where n is the smallest positive integer that is divisible by all the denominators of the exponents (that is, the least common multiple of the denominators). For example, if the integrand involves terms such as $x^{1/4}$, $x^{2/3}$, and $x^{1/6}$, then the substitution $x = u^{12}$ is suggested, because 12 is the smallest integer divisible by the exponential denominators 4, 3, and 6. The advantage of this policy is that it guarantees that each fractional power of x becomes an integral power of u . Thus,

$$x^{1/6} = (u^{12})^{1/6} = u^2, \quad x^{1/4} = (u^{12})^{1/4} = u^3, \quad x^{2/3} = (u^{12})^{2/3} = u^8$$

Example 5 Substitution with fractional exponents

Find $\int \frac{dx}{x^{1/3} + x^{1/2}}$.

Solution Because 6 is the smallest integer divisible by the denominators 2 and 3, we set $u = x^{1/6}$, so that $u^6 = x$ and $6u^5 du = dx$. We now use substitution

$$\begin{aligned} \int \frac{dx}{x^{1/3} + x^{1/2}} &= \int \frac{6u^5 du}{(u^6)^{1/3} + (u^6)^{1/2}} && \text{Let } x = u^6. \\ &= \int \frac{6u^5 du}{u^2 + u^3} \\ &= \int \frac{6u^5 du}{u^2(1 + u)} \\ &= \int \frac{6u^3 du}{1 + u} && \text{By long division, } \frac{6u^3}{1 + u} = 6u^2 - 6u + 6 + \frac{-6}{1 + u}. \\ &= \int \left(6u^2 - 6u + 6 + \frac{-6}{1 + u} \right) du \\ &= 2u^3 - 3u^2 + 6u - 6 \ln |1 + u| + C && \text{Substitute } u = x^{1/6}. \\ &= 2(x^{1/6})^3 - 3(x^{1/6})^2 \\ &\quad + 6(x^{1/6}) - 6 \ln |1 + x^{1/6}| + C \\ &= 2x^{1/2} - 3x^{1/3} + 6x^{1/6} \\ &\quad - 6 \ln(1 + x^{1/6}) + C && \text{Note: } 1 + x^{1/6} > 0. \end{aligned}$$

Substitution does not guarantee an obvious integrable form. When the degree of the numerator is greater than or equal to the degree of the denominator, division is often helpful.

Using Tables of Integrals

Example 5 (especially the part involving long division) seems particularly lengthy. When faced with the necessity of integrating a function such as

$$\int \frac{6u^3}{1 + u}$$

most people would turn to a computer, calculator, or a table of integrals. If you look, for example, at the short table of integrals (Appendix D), you will find (Formula 37)* that

$$\int \frac{u^3 du}{au + b} = \frac{(au + b)^3}{3a^4} - \frac{3b(au + b)^2}{2a^4} + \frac{3b^2(au + b)}{a^4} - \frac{b^3}{a^4} \ln |au + b| + C$$

*See the footnote at the beginning of the Integral Table that each entry in the table must add a constant of integration when using a formula from the table.

If we let $a = 1$ and $b = 1$, we find

$$\int \frac{6u^3 du}{1+u} = 6 \int \frac{u^3 du}{u+1} = 2(u+1)^3 - 9(u+1)^2 + 18(u+1) - 6 \ln |u+1| + C$$

When using integration tables, we note that the algebraic form does not always match the form we obtain by direct integration. You might wish to show algebraically that the form we have just found and the form

$$2u^3 - 3u^2 + 6u - 6 \ln |1+u| + C$$

from Example 5 differ by a constant and hence are equivalent.

To use an integral table, first classify the integral by form. To facilitate substitution of forms, we use u as the variable of integration, and let a, b, c, m , and n represent constants. The forms listed in Appendix D are as follows:

Elementary forms (Formulas 1-29; these were developed in the text)

Linear and quadratic forms (Formulas 30-76)

Forms involving $au + b$; $u^2 + a^2$; $u^2 - a^2$; $a^2 - u^2$; and $au^2 + bu + c$

Radical forms (Formulas 77-121)

Forms involving $\sqrt{au+b}$; $\sqrt{u^2+a^2}$; $\sqrt{u^2-a^2}$; $\sqrt{a^2-u^2}$

Trigonometric forms (Formulas 122-167)

Forms involving $\cos au$; $\sin au$; both $\sin au$ and $\cos au$; $\tan au$; $\cot au$; $\sec au$; $\csc au$

Inverse trigonometric forms (Formula 168-182)

Exponential and logarithmic forms (Formulas 183-200)

Forms involving e^{au} ; $\ln|u|$

There is a common misconception that integration will be easy if a table is provided, but even with a table available there can be a considerable amount of work. After deciding which form applies, match the individual type with the problem at hand by making appropriate choices for the arbitrary constants. More than one form may apply, but the results derived by using different formulas will be the same (except for the constants) even though they may look quite different. We will not include the constants in the table listing, but you should remember to include them with your answers when using the table for integration.

Take a few moments to look at the integration table Appendix D. Notice that the table has two basic types of integration formulas. The first gives a formula that is the antiderivative, whereas the second, called a *reduction formula*, simply rewrites the integral in another form.

Example 6 Integration using a table of integrals

Find $\int x^2(3-x)^5 dx$.

Solution We can find this integral using substitution:

$$\begin{aligned} \int x^2(3-x)^5 dx &= \int (3-u)^2 u^5 (-du) && \boxed{\text{If } u = 3-x, \text{ then } du = -dx} \\ &= \int (-u^7 + 6u^6 - 9u^5) du \\ &= -\frac{u^8}{8} + \frac{6u^7}{7} - \frac{9u^6}{6} + C \\ &= -\frac{1}{8}(3-x)^8 + \frac{6}{7}(3-x)^7 - \frac{3}{2}(3-x)^6 + C \end{aligned}$$

Even though this was not too difficult, it is a bit tedious, so we might think to find this integral by using a table of integrals. This is an integral involving an expression of the form $au + b$; we find that this is Formula 32 where $u = x$, $a = -1$, $b = 3$, and $n = 5$.

$$\begin{aligned}\int x^2(3-x)^5 dx &= \frac{(3-x)^{5+3}}{(5+3)(-1)^3} - \frac{2(3)(3-x)^{5+2}}{(5+2)(-1)^3} + \frac{3^2(3-x)^{5+1}}{(5+1)(-1)^3} + C \\ &= -\frac{1}{8}(3-x)^8 + \frac{6}{7}(3-x)^7 - \frac{3}{2}(3-x)^6 + C\end{aligned}$$

One of the difficult considerations when using tables or computer software to carry out integration is recognizing the variety of different forms for acceptable answers. Note that it is not easy to show that these forms are algebraically equivalent. If we use computer software for this evaluation, we find yet another algebraically equivalent form (remember you can check equivalence up to an arbitrary constant) by taking the derivative of both integrals and showing they are the same):

$$\int x^2(3-x)^5 dx = -\frac{x^8}{8} + \frac{15x^7}{7} - 15x^6 + 54x^5 - \frac{405x^4}{4} + 81x^3$$

Once again, remember that you must add the $+C$ to the computer software form.

Example 7 Integration using a reduction formula from a table of integrals

Find $\int (\ln x)^4 dx$.

Solution The integrand is in logarithmic form; from the table of integrals we see that Formula 198, Appendix D applies, where $u = x$ and $n = 4$. Take a close look at Formula 198, and note that it gives another integral as part of the result. This is called a **reduction formula** because it enables us to compute the given integral in terms of an integral of a similar type, only with a lower power in the integral.

$$\begin{aligned}\int (\ln x)^4 dx &= x(\ln x)^4 - 4 \int (\ln x)^{4-1} dx && \text{Formula 198} \\ &= x(\ln x)^4 - 4 \left[x(\ln x)^3 - 3 \int (\ln x)^{3-1} dx \right] && \text{Formula 198 again} \\ &= x(\ln x)^4 - 4x(\ln x)^3 + 12 \int (\ln x)^2 dx \\ &= x(\ln x)^4 - 4x(\ln x)^3 + 12 \left[x(\ln x)^2 \right. \\ &\quad \left. - 2x \ln x + 2x \right] + C && \text{This is formula 197.} \\ &= x(\ln x)^4 - 4x(\ln x)^3 + 12x(\ln x)^2 \\ &\quad - 24x \ln x + 24x + C\end{aligned}$$

Reduction formulas in a table of integrals, such as the one illustrated in Example 7, are usually obtained using substitution and integration by parts (which we will do in Section 7.2).

Note from the previous example that we follow the convention of adding the constant C only after eliminating the last integral sign (rather than being technically correct and writing $C_1, C_2, \dots$ for each integral). The reason we can do this is that $C_1 + C_2 + \dots = C$, for arbitrary constants.

It is often necessary to make substitutions before using one of the integration formulas, as shown in the following example.

Example 8 Using an integral table after substitution

Find $\int \frac{x \, dx}{\sqrt{8-5x^2}}$.

Solution This is an integral of the form $\int \frac{u \, du}{\sqrt{a^2-u^2}}$, but it does not exactly match any of the formulas. Note, however, that except for the coefficient of 5, it is like Formula 111. Let $u = \sqrt{5}x$ (so $u^2 = 5x^2$); then $du = \sqrt{5} \, dx$:

$$\begin{aligned} \int \frac{x \, dx}{\sqrt{8-5x^2}} &= \int \frac{\frac{u}{\sqrt{5}} \cdot \frac{du}{\sqrt{5}}}{\sqrt{8-u^2}} && \boxed{\begin{array}{l} \text{If } u = \sqrt{5}x, \text{ so that } x = \frac{u}{\sqrt{5}} \\ du = \sqrt{5} \, dx, \text{ so that } dx = \frac{du}{\sqrt{5}} \end{array}} \\ &= \frac{1}{5} \int \frac{u \, du}{\sqrt{8-u^2}} \\ &= \frac{1}{5}(-\sqrt{8-u^2}) + C \quad \text{Formula 111 where } a^2 = 8. \end{aligned}$$

As you can see from Example 8, using an integral table is not a trivial task. In fact, other methods of integration may be preferable. For Example 8, you can let $u = 8 - 5x^2$ and integrate by substitution:

$$\begin{aligned} \int \frac{x \, dx}{\sqrt{8-5x^2}} &= \int \frac{x \left(\frac{du}{-10x}\right)}{\sqrt{u}} && \boxed{\text{If } u = 8 - 5x^2, \text{ then } du = -10x \, dx} \\ &= -\frac{1}{10} \int u^{-1/2} \, du \\ &= -\frac{1}{10}(2u^{1/2}) + C \\ &= -\frac{1}{5}\sqrt{8-5x^2} + C \end{aligned}$$

Of course, this answer is the same as the one we obtained in Example 8. The point of this calculation is to emphasize that you should try simple methods of integration before turning to the table of integrals.

Example 9 Integration by table (multiple forms with substitution)

Find $\int 5x^2\sqrt{3x^2+1} \, dx$.

Solution This is similar to Formula 87, but you must take care of the 5 (constant multiple) and the 3 (by making a substitution).

$$\begin{aligned} \int 5x^2\sqrt{3x^2+1} \, dx &= 5 \int \left(\frac{u^2}{3}\right) \sqrt{u^2+1} \frac{du}{\sqrt{3}} && \boxed{\text{If } u = \sqrt{3}x, \text{ then } du = \sqrt{3} \, dx} \\ &= \frac{5}{3\sqrt{3}} \int u^2\sqrt{u^2+1} \, du \quad \text{Formula 87 with } a = 1 \\ &= \frac{5}{3\sqrt{3}} \left[\frac{u(u^2+1)^{3/2}}{4} - \frac{1^2 u \sqrt{u^2+1}}{8} \right. \\ &\quad \left. - \frac{1^4}{8} \ln(u + \sqrt{u^2+1}) \right] + C \\ &= \frac{5}{24\sqrt{3}} \left[2\sqrt{3}x(3x^2+1)^{3/2} - \sqrt{3}x\sqrt{3x^2+1} \right. \\ &\quad \left. - \ln(\sqrt{3}x + \sqrt{3x^2+1}) \right] + C \\ &= \frac{5}{24} \left[2x(3x^2+1)^{3/2} - x(3x^2+1)^{1/2} \right. \\ &\quad \left. - \frac{1}{\sqrt{3}} \ln(\sqrt{3}x + \sqrt{3x^2+1}) \right] + C \end{aligned}$$

If you use a calculator or computer, you will probably obtain an alternate, but equivalent, form:

$$\int 5x^2 \sqrt{3x^2 + 1} \, dx = \frac{5x \sqrt{3x^2 + 1} (6x^2 + 1)}{24} - \frac{5\sqrt{3} \ln(\sqrt{3x^2 + 1} + \sqrt{3}x)}{72}$$

You can verify that the two expressions are equivalent by differentiating both, as two functions whose derivatives are equal will differ by a constant (of integration, in this case).

PROBLEM SET 7.1

Level 1

Find each integral in Problems 1–12.

1. $\int \frac{2x+5}{\sqrt{x^2+5x}} \, dx$

2. $\int \frac{\ln x}{x} \, dx$

3. $\int \frac{dx}{x \ln x}$

4. $\int \cos x \, e^{\sin x} \, dx$

5. $\int \frac{x \, dx}{4+x^4}$

6. $\int \frac{t^2 \, dt}{9+t^6}$

7. $\int (1 + \cot x)^4 \csc^2 x \, dx$

8. $\int \frac{4x^3 - 4x}{x^4 - 2x^2 + 3} \, dx$

9. $\int \frac{x^3 - x}{(x^4 - 2x^2 + 3)^2} \, dx$

10. $\int \frac{2x+4}{x^2+4x+3} \, dx$

11. $\int \frac{2x+1}{x^2+x+1} \, dx$

12. $\int \frac{2x-1}{(4x^2-4x)^2} \, dx$

Integrate the expressions in Problems 13–24 using the short table of integrals given in Appendix D.

13. $\int \frac{dx}{x^2 \sqrt{x^2 - a^2}}$

14. $\int \frac{dx}{x^2 \sqrt{a^2 - x^2}}$

15. $\int x \ln x \, dx$

16. $\int \ln x \, dx$

17. $\int x e^{ax} \, dx$

18. $\int \frac{dx}{a + b e^{2x}}$

19. $\int \frac{x^2 \, dx}{\sqrt{x^2 + 1}}$

20. $\int \frac{dx}{x^2 \sqrt{x^2 + 16}}$

21. $\int \frac{x \, dx}{\sqrt{4x^2 + 1}}$

22. $\int \frac{dx}{x \sqrt{1 - 9x^2}}$

23. $\int e^{-4x} \sin 5x \, dx$

24. $\int x \sin^{-1} x \, dx$

Find the integrals in Problems 25–38. If you use the integral table, state the number of the formula used, and if you use substitution, show each step. If you use an alternative table of integrals, then cite the source as well as the formula number.

25. $\int (1 + bx)^{-1} \, dx$

26. $\int \frac{x \, dx}{\sqrt{a^2 - x^2}}$

27. $\int x(1+x)^3 \, dx$

28. $\int x \sqrt{1+x} \, dx$

29. $\int x e^{4x} \, dx$

30. $\int x \ln 2x \, dx$

31. $\int \frac{dx}{1 + e^{2x}}$

32. $\int \ln^3 x \, dx$

33. $\int \sec^3\left(\frac{x}{2}\right) \, dx$

34. $\int \frac{\sqrt{4x^2 + 1}}{x} \, dx$

35. $\int \sin^4 x \, dx$

36. $\int \frac{dx}{9x^2 + 6x + 1}$

37. $\int \sqrt{9 - x^2} \, dx$

38. $\int \frac{dx}{\sin^2 x \cos x}$

Level 2

39. Derive the **cosine squared** formula shown on the inside back cover

$$\int \cos^2 x \, dx = \frac{1}{2}x + \frac{1}{4} \sin 2x + C$$

Hint: Use the identity

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

40. Derive the **sine squared** formula shown on the inside back cover

$$\int \sin^2 x \, dx = \frac{1}{2}x - \frac{1}{4} \sin 2x + C$$

Hint: Use the identity

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

Problems 41–44 use substitution to integrate certain powers of sine and cosine.

41. $\int \sin^3 x \cos^4 x \, dx$ *Hint:* Let $u = \cos x$.

42. $\int \sin^4 x \cos x \, dx$ *Hint:* Let $u = \sin x$.

43. $\int \sin^2 x \cos^2 x \, dx$

Hint: Use the identities shown in Problems 41 and 42.

44. **Think Tank Problem** Using Problems 41–43 formulate a procedure for integrals of the form

$$\int \sin^m x \cos^n x \, dx$$

Find each integral in Problems 45–48.

45. $\int \frac{dx}{x^{1/2} + x^{1/4}}$

46. $\int \frac{e^x dx}{1 + e^{x/2}}$

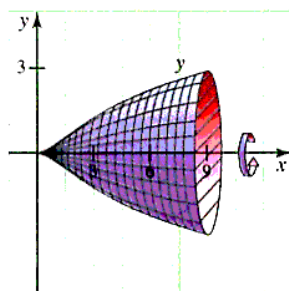
47. $\int \frac{18 \tan^2 t \sec^2 t}{(2 + \tan^3 t)^2} dt$

48. $\int \frac{4 dx}{x^{1/3} + 2x^{1/2}}$

49. Find the volume of the solid generated when the region under the curve

$$y = \frac{x^{3/2}}{\sqrt{x^2 + 9}}$$

between $x = 0$ and $x = 9$ is revolved about the x -axis.



Interactive

50. Find the volume of the solid generated when the curve $y = x(1 - x^2)^{1/4}$ from $x = 0$ to $x = 1$ is revolved about the x -axis.

51. Find the volume of the solid generated when the region between the curve

$$y = \frac{1}{\sqrt{x}}(1 + \sqrt{x})^{1/3}$$

and the x -axis between $x = 1$ and $x = 4$ is revolved about the y -axis.

52. Find the volume of the solid generated when the region between the curve

$$x = \sqrt[4]{4 - y^2}$$

and the y -axis between $y = 1$ and $y = 2$ is revolved about the y -axis.

53. Let $y = f(x)$ be a function that satisfies the differential equation

$$xy' = \sqrt{(\ln x)^2 - x^2}$$

Find the arc length of $y = f(x)$ between $x = \frac{1}{3}$ and $x = \frac{1}{2}$.

54. Find the arc length of the curve $y = \ln(\cos x)$ on the interval $[0, \frac{\pi}{4}]$.

55. Find the area of the surface generated when the curve $y = x^2$ on the interval $[0, 1]$ is revolved about the x -axis.

56. Find the area of the surface generated when the curve $y = x^2$ on the interval $[0, 1]$ is revolved about the y -axis.

57. Show that

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C$$

Hint: Multiply the integrand by $\frac{\csc x + \cot x}{\csc x + \cot x}$.

58. Find

$$\int 2 \sin x \cos x \, dx$$

by using the indicated substitution.

a. Let $u = \cos x$.

b. Let $u = \sin x$.

c. Write $2 \sin x \cos x = \sin 2x$ and carry out the integration.

d. Show that the answers you obtained for parts a–c are the same.

Level 3

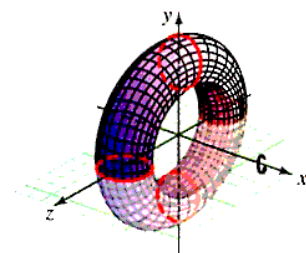
59. Derive the formula

$$\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$$

60. Find the surface area of the torus generated when the circle

$$x^2 + (y - b)^2 = 1$$

$b > 1$ is revolved about the x -axis.



Interactive

7.2 INTEGRATION BY PARTS

IN THIS SECTION: *Integration by parts formula, repeated use of integration by parts, definite integration by parts*

Integration by parts is a procedure based on reversing the product rule for differentiation. We present this section not only as a technique of integration but also as a procedure that is necessary in a variety of useful applications.

Integration by Parts Formula

Recall the formula for differentiation of a product. If u and v are differentiable functions, then

$$d(uv) = u dv + v du$$

Integrate both sides of this equation to find the formula for integration by parts:

$$\begin{aligned}\int d(uv) &= \int u dv + \int v du \\ uv &= \int u dv + \int v du\end{aligned}$$

If we rewrite this last equation, we obtain the formula summarized in the following box.

FORMULA FOR INTEGRATION BY PARTS

$$\int u dv = uv - \int v du$$

Integration by parts is one of the most important techniques of integration, with many applications in engineering and physics problems. Due to a lack of proper understanding of the formula and integration process, students often face obstacles in applying this procedure correctly, so it is important to take your time, ask questions, and work through each example carefully.

Example 1 Integration by parts

Find $\int xe^x dx$.

Solution To use integration by parts, we must choose u and dv so that the new integral is easier to integrate than the original.

$$\begin{aligned}\int \underbrace{x}_u \underbrace{e^x dx}_{dv} &= \underbrace{x}_u \underbrace{e^x}_v - \int \underbrace{e^x}_v \underbrace{dx}_{du} \\ \text{Let } u &= x, \quad dv = e^x dx \\ du &= dx \quad v = \int e^x dx = e^x \\ &= xe^x - e^x + C\end{aligned}$$

You may wonder why an arbitrary constant (call it K) was not included when performing the integration associated with $\int dv$ in integration by parts. The reason is that when applying integration by parts, we need just one function v whose differential is dv , so we take the simplest one—the one with $K = 0$. You may find it instructive to see that taking $v = e^x + K$ gives the same result.

You can check your work with integration by parts by differentiating, using software, or using the integration table found in Appendix D (Formula 184, with $a = 1$).

Integration by parts is often difficult the first time you try to do it because there is no absolute choice for u and dv . In Example 1, you might have chosen

$$\begin{aligned}\text{Let } u &= e^x, \quad dv = x dx \\ du &= e^x dx \quad v = \int x dx = \frac{x^2}{2}\end{aligned}$$

Then

$$\int x e^x dx = \underbrace{e^x}_u \underbrace{\frac{x^2}{2}}_v - \int \underbrace{\frac{x^2}{2}}_v \underbrace{e^x}_{du} dx$$

Note, however, that this choice of u and dv leads to a more complicated form than the original. In general, when you are integrating by parts, if you make a choice for u and dv that leads to a more complicated form than when you started, consider going back and making another choice for u and dv .

Generally, you want to choose dv to be as difficult as possible (and still be something you can integrate), with the remainder being left for the u -factor.

Example 2 When the differentiable part is the entire integrand

Find $\int \ln x dx$ for $x > 0$.

Solution

$$\begin{array}{ll} \text{Let } u = \ln x, & dv = dx \\ du = \frac{1}{x} dx & v = x \end{array}$$

$$\begin{aligned} \int \ln x dx &= (\ln x)x - \int x \left(\frac{1}{x} dx \right) \\ &= x \ln x - x + C \end{aligned}$$

Check with Formula 196 (Appendix D), which we worked as Problem 16 of Problem Set 7.1. ■

Repeated Use of Integration by Parts

Sometimes integration by parts must be applied several times to find a given integral.

Example 3 Repeated integration by parts

Find $\int x^2 e^{-x} dx$.

Solution

$$\begin{array}{ll} \text{Let } u = x^2, & dv = e^{-x} dx \\ du = 2x dx & v = -e^{-x} \end{array}$$

$$\int x^2 e^{-x} dx = x^2(-e^{-x}) - \int (-e^{-x})(2x dx)$$

$$= -x^2 e^{-x} + 2 \int x e^{-x} dx$$

$$\begin{array}{ll} \text{Let } u = x, & dv = e^{-x} dx \\ du = dx & v = -e^{-x} \end{array}$$

$$= -x^2 e^{-x} + 2 \left[x(-e^{-x}) - \int (-e^{-x}) dx \right] \quad \text{Do not forget the grouping symbols.}$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + C$$

$$= -e^{-x}(x^2 + 2x + 2) + C$$

Check with Formula 185, where $a = -1$. ■

In the following example, it is necessary to apply integration by parts more than once, but as you will see, when we do so a second time we return to the original integral. Note carefully how this situation can be handled algebraically.

Example 4 Repeated integration by parts with algebraic manipulation

Find $\int e^{2x} \sin x \, dx$.

Solution For this problem you will see that it will be useful to call the original integral I . That is, we let

$$I = \int e^{2x} \sin x \, dx$$

$$\begin{array}{ll} \text{Let } u = e^{2x}, & dv = \sin x \, dx \\ du = 2e^{2x} \, dx & v = -\cos x \end{array}$$

$$= e^{2x}(-\cos x) - \int (-\cos x)(2e^{2x} \, dx)$$

$$= -e^{2x} \cos x + 2 \int e^{2x} \cos x \, dx$$

$$\begin{array}{ll} \text{Let } u = e^{2x}, & dv = \cos x \, dx \\ du = 2e^{2x} \, dx & v = \sin x \end{array}$$

$$= -e^{2x} \cos x + 2 \left[e^{2x}(\sin x) - \int \sin x (2e^{2x} \, dx) \right]$$

$$= -e^{2x} \cos x + 2e^{2x} \sin x - 4 \int e^{2x} \sin x \, dx \quad \text{Notice } \int e^{2x} \sin x \, dx = I.$$

$$I = -e^{2x} \cos x + 2e^{2x} \sin x - 4I$$

Solve for I .

$$5I = -e^{2x} \cos x + 2e^{2x} \sin x + C$$

$$I = \frac{1}{5} e^{2x} (2 \sin x - \cos x) + C$$

Thus, $\int e^{2x} \sin x \, dx = \frac{1}{5} e^{2x} (2 \sin x - \cos x) + C$. Check with the integration table (Formula 192, where $a = 2$ and $b = 1$), or by taking the derivative. 

Definite Integration by Parts

The integration by parts formula can be used for definite integrals, as summarized in the following box.

INTEGRATION BY PARTS FOR DEFINITE INTEGRALS

$$\int_a^b u \, dv = uv \Big|_a^b - \int_a^b v \, du$$

You should recognize that this formula for definite integrals is the same as the formula for indefinite integrals, where the first term after the equal sign has been evaluated at the appropriate limits of integration. This is illustrated by the following example.

Example 5 Integration by parts with a definite integral

Evaluate $\int_0^1 x e^{2x} \, dx$

Solution

$$\begin{array}{ll} \text{Let } u = x, & dv = e^{2x} \, dx \\ du = dx & v = \frac{1}{2} e^{2x} \end{array}$$

$$\int_0^1 x e^{2x} dx = \left. \frac{1}{2} x e^{2x} \right|_0^1 - \frac{1}{2} \int_0^1 e^{2x} dx$$

↓ *It is sometimes easier to simplify the algebra and then do one evaluation here at the end of the integration part of the problem.*

$$\begin{aligned} &= \left[\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} \right]_0^1 \\ &= \frac{1}{4} e^2 + \frac{1}{4} \end{aligned}$$

Check in Appendix D (Formula 184, with $a = 2$). ■

Example 6 Integration by parts with a definite integral followed by substitution

Evaluate $\int_0^1 \tan^{-1} x \, dx$.

Solution

Let $u = \tan^{-1} x$,	$dv = dx$
$du = \frac{dx}{1+x^2}$	$v = x$

$$\begin{aligned} \int_0^1 \underbrace{\tan^{-1} x}_u \underbrace{dx}_{dv} &= \underbrace{(\tan^{-1} x)}_u \underbrace{x}_v \bigg|_0^1 - \int_0^1 \underbrace{\frac{x}{1+x^2}}_{v \, du} dx \\ &= \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 \quad \begin{array}{l} \text{Use substitution where } w = 1+x^2 \text{ and } dw = 2x \, dx. \\ \int \frac{x \, dx}{1+x^2} = \int \frac{dw}{2w} = \frac{1}{2} \ln w = \frac{1}{2} \ln(1+x^2) \end{array} \\ &= \left[1(\tan^{-1} 1) - \frac{1}{2} \ln(1+1) \right] - \left[0 - \frac{1}{2} \ln 1 \right] \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

Check in Appendix D (Formula 180, with $a = 1$). ■

PROBLEM SET 7.2

Level 1

Find each integral in Problems 1–16 using integration by parts.

1. $\int x e^{-2x} dx$

2. $\int x \sin x \, dx$

3. $\int x \ln x \, dx$

4. $\int x \tan^{-1} x \, dx$

5. $\int \sin^{-1} x \, dx$

6. $\int x^2 \sin x \, dx$

7. $\int e^{-3x} \cos 4x \, dx$

8. $\int e^{2x} \sin 3x \, dx$

9. $\int x^2 \ln x \, dx$

10. $\int (x + \sin x)^2 dx$

11. $\int \sin(\ln x) dx$

12. $\int x \sin x \cos x \, dx$

13. $\int \ln(x^2 + 1) dx$

14. $\int \sin \sqrt{x} \, dx$

15. $\int \frac{x e^{-x}}{(x-1)^2} dx$

16. $\int \frac{\ln(\sin x) dx}{\tan x}$

Find the exact value of the definite integrals in Problems 17–22 using integration by parts.

17. $\int_1^4 \sqrt{x} \ln x \, dx$

18. $\int_1^e x^3 \ln x \, dx$

$$19. \int_1^e (\ln x)^2 dx \quad 20. \int_{1/3}^e 3(\ln 3x)^2 dx$$

$$21. \int_0^\pi e^{2x} \cos 2x dx \quad 22. \int_0^\pi x(\sin x + \cos x) dx$$

Level 2

23. ■ **What does this say?** Describe the process known as *integration by parts*.
24. ■ **What does this say?** Contrast using integration by parts for definite and for indefinite integrals.

In Problems 25–28, first use an appropriate substitution and then integrate by parts to find the integral. Remember to give your answers in terms of x .

$$25. \int [\sin 2x \ln(\cos x)] dx$$

$$26. \int \frac{\ln x \sin(\ln x)}{x} dx$$

$$27. \int [\sin x \ln(2 + \cos x)] dx$$

$$28. \int e^{2x} \sin e^x dx$$

29. a. Find

$$\int \frac{x^3 dx}{x^2 - 1}$$

using integration by parts.

- b. Find the integral in part a by first dividing the integrand.

30. a. Find

$$\int \frac{x^5 dx}{x^3 - 1}$$

using integration by parts.

- b. Find the integral in part a by first dividing the integrand.

31. Find $\int \cos^2 x dx$.

32. Find $\int \frac{x dx}{\sqrt{x^2 + 1}}$

33. Use Problem 31 to find

$$\int x \cos^2 x dx$$

using integration by parts.

34. Use Problem 32 to find

$$\int \frac{x^3 dx}{\sqrt{x^2 + 1}}$$

using integration by parts.

35. Find $\int x^n \ln x dx$, where n is any positive real number.
36. Show that

$$\int_0^1 x^m (1-x)^n dx = \int_0^1 x^n (1-x)^m dx$$

for positive integers m, n .

37. After t hours on the job, a factory worker can produce $100te^{-0.5t}$ units per hour. How many units does the worker produce during the first 3 hours?
38. After t seconds, an object is moving along a line with velocity of $te^{-t/2}$ meters per second. Express the position of the moving object as a function of time.
39. After t weeks, contributions in response to a local fund-raising campaign were coming in at the rate of $2,000te^{-0.2t}$ dollars per week. How much money was raised during the first 5 weeks?
40. Find the volume of the solid generated when the region under the curve $y = \sin x + \cos x$ on the interval $[0, \frac{\pi}{4}]$ is revolved about the y -axis.
41. Find the volume of the solid generated when the region under the curve $y = e^{-x}$ on the interval $[0, 2]$ is revolved about the y -axis.
42. Find the volume of the solid generated when the region under the curve $y = \ln x$ on the interval $[1, e]$ is revolved about the indicated axis:
- a. x -axis b. y -axis
43. Find the centroid (with coordinates rounded to the nearest hundredth) of the region bounded by the curves $y = e^x$, $y = e^{-x}$, and the line $x = 1$.
44. Find the centroid (with coordinates rounded to the nearest hundredth) of the region in the first quadrant bounded by the curves $y = \sin x$ and $y = \cos x$ and the y -axis.

In Problems 45–46, solve the given separable differential equations.

45. $\frac{dy}{dx} = \sqrt{xy} \ln x$ 46. $\frac{dy}{dx} = xe^{y-x}$

47. Find a function $y = f(x)$ whose graph passes through $(0, 1)$ and has the property that the normal line at each point (x, y) on the graph has slope $\frac{\sec x}{xy}$.
48. Find a function $y = f(x)$ whose graph passes through $(1, 1)$ and has the property that at each point (x, y) on the graph, the slope of the tangent line is $y \tan^{-1} x$.
49. Suppose it is known that $f(0) = 3$ and

$$\int_0^\pi [f(x) + f''(x)] \sin x dx = 0$$

What is $f(\pi)$?

50. In physics, it is known that loudness L of a sound is related to its intensity I by the equation

$$L = 10 \log \frac{I}{I_0}$$

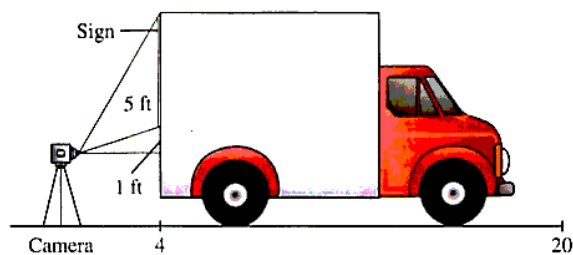
decibels where $I_0 = 10^{-12}$ watt/m² is the threshold of audibility (the lowest intensity that can be heard). What is the average value of L as the intensity of a TV show ranges between I_0 and $I_1 = 3 \cdot 10^{-5}$ watt/m²?

51. Because a rocket burns fuel in flight, its mass decreases with time, and this in turn affects its velocity. It can be shown that the velocity $v(t)$ of the rocket at time t in its flight is given by

$$v(t) = -r \ln \frac{w - kt}{w} - gt$$

where w is the initial weight of the rocket (including its fuel) and r and k are, respectively, the expulsion speed and the rate of consumption of the fuel, which are assumed to be constant. As usual, $g = 32$ ft/s² is the acceleration due to gravity (assumed to be constant). Suppose $w = 30,000$ lb, $r = 8,000$ ft/s, and $k = 200$ lb/sec. What is the height of the rocket after 2 minutes (120 seconds)?

52. A photographer is taking a picture of a clever sign on the back of a truck. The sign is 5 ft high and its lower edge is 1 ft above the lens of the camera. At first the truck is 4 ft away from the photographer, but then it begins to move away. What is the average value of the angle θ (correct to two decimal places) subtended by the camera lens as the truck moves from 4 ft to 20 ft away from the photographer?



53. The displacement from equilibrium of a mass oscillating at the end of a spring hanging from the ceiling is given by

$$y = 2.3e^{-0.25t} \cos 5t$$

feet. What is the average displacement (rounded to the nearest hundredth) of the mass between times $t = 0$ and $t = \pi/5$ seconds?



54. If n moles of an ideal gas expand at constant temperature T , then its pressure p and volume V satisfy the equation $pV = nRT$, for constant R . It can be shown that the work done by the gas in expanding from volume V_1 to V is

$$W = nRT \ln \frac{V}{V_1}$$

What is the average work done as V increases from V_1 to $V_2 = 10V_1$?

55. Find $\int \frac{\sin x \, dx}{\sqrt{\cos 2x}}$.

56. Find $\int \frac{(2x-1)}{x^2} e^{2x} \, dx$.

57. Derive the reduction formula

$$\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$$

(This is Formula 186, with $a = 1$.)

58. Derive the reduction formula

$$\int (\ln x)^n \, dx = x (\ln x)^n - n \int (\ln x)^{n-1} \, dx$$

(This is Formula 198.)

Level 3

59. **Wallis's formula** If n is an even positive integer, use reduction formulas to show

$$\begin{aligned} \int_0^{\pi/2} \sin^n x \, dx &= \int_0^{\pi/2} \cos^n x \, dx \\ &= \left[\frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdot 6 \cdots n} \right] \frac{\pi}{2} \end{aligned}$$

60. **EXPLORATION PROBLEM** State and prove a result similar to the one in Problem 59 for the case where n is an odd positive integer.

7.3 TRIGONOMETRIC METHODS

IN THIS SECTION: Powers of sine and cosine, powers of secant and tangent, trigonometric substitutions, quadratic-form integrals

If it has been awhile since you studied trigonometry, you might wish to review Appendix E.

Powers of Sine and Cosine

Problems 41–44 of Problem Set 7.1 anticipated the results of this section. We begin by considering a product of powers of sine and cosine, which we represent in the form

$$\int \sin^m x \cos^n x \, dx$$

There are essentially two cases that must be considered, depending on whether the powers m and n are both even or not. We will state the general strategy for handling each case and then illustrate the procedure with an example.

Case I: Either m or n is odd (or both)

General strategy: Suppose, for simplicity, that m is odd. Separate a factor of $\sin x$ from the rest of the integrand, so the remaining power of $\sin x$ is even; use the identity $\sin^2 x = 1 - \cos^2 x$ to express everything but the term $(\sin x \, dx)$ in terms of $\cos x$. Substitute $u = \cos x$, $du = -\sin x \, dx$ to convert the integral into a polynomial in u and integrate using the power rule. The case where n is odd is handled in analogous fashion, as shown in the following example by reversing the roles of $\sin x$ and $\cos x$.

Example 1 Power of cosine is odd

Find $\int \sin^4 x \cos^3 x \, dx$.

Solution Since $n = 3$ is odd, peel off a factor of $\cos x$ and use $\cos^2 x = 1 - \sin^2 x$ to express the integral as a polynomial in $\sin x$.

$$\begin{aligned} \int \sin^4 x \cos^3 x \, dx &= \int \sin^4 x \cos^2 x (\cos x \, dx) \\ &= \int \sin^4 x (1 - \sin^2 x) (\cos x \, dx) && \text{Let } u = \sin x, \text{ then } du = \cos x \, dx \\ &= \int u^4 (1 - u^2) du \\ &= \frac{1}{5} u^5 - \frac{1}{7} u^7 + C \\ &= \frac{1}{5} \sin^5 x - \frac{1}{7} \sin^7 x + C \end{aligned}$$

Case II: Both m and n are even

General strategy: Convert this into Case I by using the identities

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \text{and} \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

These are sometimes called *half-angle identities*.

Example 2 All powers are even

Find $\int \sin^2 x \cos^4 x \, dx$.

Solution

$$\begin{aligned}
\int \sin^2 x \cos^4 x \, dx &= \int \frac{1}{2}(1 - \cos 2x) \left(\frac{1}{4}\right) (1 + \cos 2x)^2 \, dx \\
&= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \, dx \quad \text{Half-angle identities} \\
&= \frac{1}{8} \int \left[1 + \cos 2x - \frac{1}{2}(1 + \cos 4x) - (1 - \sin^2 2x)(\cos 2x) \right] \, dx \quad \text{Half-angle identities again} \\
&= \frac{1}{8} \int \left[1 + \cos 2x - \frac{1}{2} - \frac{1}{2} \cos 4x - \cos 2x + (\sin^2 2x) \cos 2x \right] \, dx \\
&= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{2} \cos 4x \right) \, dx + \frac{1}{8} \int \sin^2 2x (\cos 2x) \, dx \\
&= \frac{1}{16} x - \frac{1}{64} \sin 4x + \frac{1}{48} \sin^3 2x + C
\end{aligned}$$

Let $u = \sin 2x$,
 then $du = 2 \cos 2x \, dx$

TECHNOLOGY NOTE: One computer algebra system applied to this example yields

$$\int \sin^2 x \cos^4 x \, dx = -\frac{1}{6} \sin x \cos^5 x + \frac{1}{24} \sin x \cos^3 x + \frac{1}{16} \sin x \cos x + \frac{1}{16} x$$

(Recall, technology does not show the “+C” term.) Both answers are correct. They are related by trigonometric identities, and it can be shown that the two answers differ by a constant.

Powers of Secant and Tangent

The simplest integrals of this form are

$$\int \tan x \, dx = \ln |\sec x| + C \quad \text{and} \quad \int \sec x \, dx = \ln |\sec x + \tan x| + C$$

For the more general situation, which we write as

$$\int \tan^m x \sec^n x \, dx$$

there are three essentially different cases to consider.

Case 1: n is even

General strategy: Peel off a factor of $\sec^2 x$ from the integrand and use the identity $\sec^2 x = \tan^2 x + 1$ to express the integrand in powers of $\tan x$ except for $(\sec^2 x \, dx)$; substitute $u = \tan x$, $du = \sec^2 x \, dx$, and integrate by using the power rule.

Example 3 Power of the secant is even

Find $\int \tan^2 x \sec^4 x \, dx$.

Solution

$$\begin{aligned}
 \int \tan^2 x \sec^4 x \, dx &= \int \tan^2 x \sec^2 x (\sec^2 x \, dx) \\
 &= \int \tan^2 x (\tan^2 x + 1) \sec^2 x \, dx && \text{Let } u = \tan x, \text{ then } du = \sec^2 x \, dx \\
 &= \int u^2(u^2 + 1) du \\
 &= \frac{1}{5}u^5 + \frac{1}{3}u^3 + C \\
 &= \frac{1}{5}\tan^5 x + \frac{1}{3}\tan^3 x + C
 \end{aligned}$$

Case II: m is odd

General strategy: Peel off a factor of $\sec x \tan x$ from the integrand and use the identity $\tan^2 x = \sec^2 x - 1$ to express the integrand in powers of $\sec x$, except for $(\sec x \tan x \, dx)$; substitute $u = \sec x$, $du = \sec x \tan x \, dx$, and integrate using the power rule.

Example 4 Power of the tangent is odd

Find $\int \tan x \sec^6 x \, dx$.

Solution

$$\begin{aligned}
 \int \tan x \sec^6 x \, dx &= \int \sec^5 x (\sec x \tan x \, dx) && \text{Let } u = \sec x, \text{ then } du = \sec x \tan x \, dx \\
 &= \int u^5 \, du \\
 &= \frac{1}{6}u^6 + C \\
 &= \frac{1}{6}\sec^6 x + C
 \end{aligned}$$

Case III: m is even and n is odd

General strategy: Use the identity $\tan^2 x = \sec^2 x - 1$ to express the integrand in terms of powers of $\sec x$; then use the reduction Formula 161:

$$\int \sec^n u \, du = \frac{\sec^{n-2} u \tan u}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} u \, du$$

Example 5 Power of the tangent is even and power of the secant is odd

Find $\int \tan^2 x \sec^3 x \, dx$.

Solution

$$\begin{aligned}
 \int \tan^2 x \sec^3 x \, dx &= \int (\sec^2 x - 1) \sec^3 x \, dx \\
 &= \int \sec^5 x \, dx - \int \sec^3 x \, dx \\
 &= \left[\frac{\sec^3 x \tan x}{4} + \frac{3}{4} \int \sec^3 x \, dx \right] - \int \sec^3 x \, dx \\
 &= \frac{\sec^3 x \tan x}{4} - \frac{1}{4} \int \sec^3 x \, dx
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{\sec^3 x \tan x}{4} - \frac{1}{4} \left[\frac{\sec x \tan x}{2} + \frac{1}{2} \int \sec x \, dx \right] \\
 &= \frac{\sec^3 x \tan x}{4} - \frac{\sec x \tan x}{8} - \frac{1}{8} \ln |\sec x + \tan x| + C
 \end{aligned}$$

Trigonometric Substitutions

Trigonometric substitutions can also be useful. For instance, suppose an integrand contains the term $\sqrt{a^2 - u^2}$, where $a > 0$. Then by setting $u = a \sin \theta$ for an acute angle θ , and using the identity $\cos^2 \theta = 1 - \sin^2 \theta$, we obtain

$$\sqrt{a^2 - u^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = a \sqrt{1 - \sin^2 \theta} = a \cos \theta$$

Thus, the substitution $u = a \sin \theta$, $du = a \cos \theta \, d\theta$ eliminates the square root and may convert the given integral into one involving only sine and cosine. This substitution can best be remembered by setting up a reference triangle. This process is illustrated in the following example.

Example 6 Trigonometric substitution with form $\sqrt{a^2 - u^2}$

Find $\int \sqrt{4 - x^2} \, dx$.

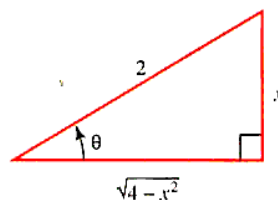
Solution First, using a table of integration, we find (Formula 117; $a = 2$)

$$\int \sqrt{4 - x^2} \, dx = \frac{x\sqrt{4 - x^2}}{2} + 2 \sin^{-1} \left(\frac{x}{2} \right) + C$$

Our goal with this example is to show how we might obtain this formula with a trigonometric substitution. Refer to the triangle shown in Figure 7.1.

Let $x = 2 \sin \theta$, so $dx = 2 \cos \theta \, d\theta$. Then

$$\begin{aligned}
 \int \sqrt{4 - x^2} \, dx &= \int \sqrt{4 - 4 \sin^2 \theta} (2 \cos \theta \, d\theta) \\
 &= 4 \int \cos^2 \theta \, d\theta && \text{Since } \sqrt{1 - \sin^2 \theta} = \cos \theta \\
 &= 4 \int \frac{1 + \cos 2\theta}{2} \, d\theta && \text{Half-angle identity} \\
 &= 2\theta + \sin 2\theta + C \\
 &= 2\theta + 2 \sin \theta \cos \theta + C
 \end{aligned}$$



$$\cos \theta = \frac{\sqrt{4 - x^2}}{2}; \quad \sin \theta = \frac{x}{2}$$

Figure 7.1 Reference triangle with form $\sqrt{a^2 - u^2}$

The final step is to convert this answer back to terms involving x . Using the reference triangle in Figure 7.1, we find that

$$\sin \theta = \frac{x}{2} \quad \text{and} \quad \cos \theta = \frac{1}{2} \sqrt{4 - x^2}$$

Thus, we have $\theta = \sin^{-1} \frac{x}{2}$

$$\begin{aligned}
 \int \sqrt{4 - x^2} \, dx &= 2 \sin^{-1} \left(\frac{x}{2} \right) + 2 \left(\frac{x}{2} \right) \left(\frac{\sqrt{4 - x^2}}{2} \right) + C \\
 &= 2 \sin^{-1} \left(\frac{x}{2} \right) + \frac{x}{2} \sqrt{4 - x^2} + C
 \end{aligned}$$

Similar methods can be used to convert integrals that involve terms of the form $\sqrt{a^2 + u^2}$ or $\sqrt{u^2 - a^2}$ into trigonometric integrals, as shown in Table 7.1. For this table, we require $0 \leq \theta < \frac{\pi}{2}$.

Table 7.1 Trigonometric substitution for integrands involving a radical form

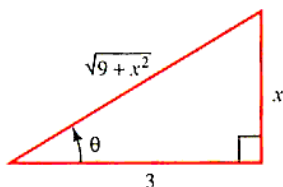
If the integrand involves...	substitute	to obtain...
$\sqrt{a^2 - u^2}$	$u = a \sin \theta$	$\sqrt{a^2 - u^2} = a \cos \theta$
$\sqrt{a^2 + u^2}$	$u = a \tan \theta$	$\sqrt{a^2 + u^2} = a \sec \theta$
$\sqrt{u^2 - a^2}$	$u = a \sec \theta$	$\sqrt{u^2 - a^2} = a \tan \theta$

Example 7 Trigonometric substitution with form $\sqrt{a^2 + u^2}$

Find $\int x^2 \sqrt{9 + x^2} dx$.

Solution Let $x = 3 \tan \theta$, $dx = 3 \sec^2 \theta d\theta$; then

$$\begin{aligned} \int x^2 \sqrt{9 + x^2} dx &= \int (3 \tan \theta)^2 \sqrt{9 + 9 \tan^2 \theta} (3 \sec^2 \theta d\theta) \\ &= \int (9 \tan^2 \theta) (3 \sec \theta) (3 \sec^2 \theta d\theta) \quad 1 + \tan^2 \theta = \sec^2 \theta \\ &= 81 \int \tan^2 \theta \sec^3 \theta d\theta \end{aligned}$$



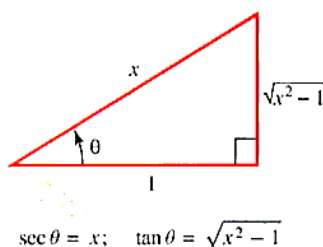
$$\tan \theta = \frac{x}{3}; \quad \sec \theta = \frac{\sqrt{9 + x^2}}{3}$$

Figure 7.2 Reference triangle with form $\sqrt{a^2 + u^2}$

This problem can now be completed by looking at Example 5. In order to express the antiderivative in terms of the original variable x , you can use the reference triangle in Figure 7.2.

Since $\tan \theta = \frac{x}{3}$, we have $\sec \theta = \frac{\sqrt{9 + x^2}}{3}$, and by substituting into the solution shown in Example 5, we find

$$\begin{aligned} \int x^2 \sqrt{9 + x^2} dx &= 81 \left[\frac{\sec^3 \theta \tan \theta}{4} - \frac{\sec \theta \tan \theta}{8} - \frac{1}{8} \ln |\sec \theta + \tan \theta| \right] + C \\ &= \frac{81}{4} \left(\frac{\sqrt{9 + x^2}}{3} \right)^3 \left(\frac{x}{3} \right) - \frac{81}{8} \left(\frac{\sqrt{9 + x^2}}{3} \right) \left(\frac{x}{3} \right) \\ &\quad - \frac{81}{8} \ln \left| \frac{\sqrt{9 + x^2}}{3} + \frac{x}{3} \right| + C \\ &= \frac{x}{4} (9 + x^2)^{3/2} - \frac{9x}{8} (9 + x^2)^{1/2} - \frac{81}{8} \ln \left| \frac{(9 + x^2)^{1/2}}{3} + \frac{x}{3} \right| + C \end{aligned}$$



$$\sec \theta = x; \quad \tan \theta = \sqrt{x^2 - 1}$$

Figure 7.3 Reference triangle with form $\sqrt{u^2 - a^2}$

Example 8 Trigonometric substitution with form $\sqrt{u^2 - a^2}$

Find $\int x^3 \sqrt{x^2 - 1} dx$.

Solution Let $x = \sec \theta$, $dx = \sec \theta \tan \theta d\theta$; we use the reference triangle shown in Figure 7.3.

$$\begin{aligned}
\int x^3 \sqrt{x^2 - 1} \, dx &= \int \sec^3 \theta \sqrt{\sec^2 \theta - 1} (\sec \theta \tan \theta \, d\theta) \\
&= \int \sec^4 \theta \tan^2 \theta \, d\theta && \tan^2 \theta = \sec^2 \theta - 1 \\
&= \int \sec^2 \theta \tan^2 \theta (\sec^2 \theta \, d\theta) && \text{Rearranging in view of substitution} \\
&= \int (\tan^2 \theta + 1) \tan^2 \theta (\sec^2 \theta \, d\theta) \\
&= \int (\tan^4 \theta + \tan^2 \theta) (\sec^2 \theta \, d\theta) \\
&= \int (u^4 + u^2) du && \text{Let } u = \tan \theta. \\
&= \frac{1}{5} u^5 + \frac{1}{3} u^3 + C \\
&= \frac{1}{5} \tan^5 \theta + \frac{1}{3} \tan^3 \theta + C \\
&= \frac{1}{5} (x^2 - 1)^{5/2} + \frac{1}{3} (x^2 - 1)^{3/2} + C
\end{aligned}$$

Quadratic-Form Integrals

An integral involving an expression of the form $Ax^2 + Bx + C$, with $A \neq 0$, $B \neq 0$, can often be evaluated by completing the square and making an appropriate substitution to convert it to one of the forms we have previously analyzed.

Example 9 Integration by completing the square

Find $\int \sqrt{16x - 2x^2 - 23} \, dx$.

Solution Complete the square within the radical:

$$\begin{aligned}
16x - 2x^2 - 23 &= -2(x^2 - 8x) - 23 \\
&= -2(x^2 - 8x + 4^2) + 2 \cdot 4^2 - 23 \\
&= -2(x - 4)^2 + 9
\end{aligned}$$

Thus,

$$\begin{aligned}
\int \sqrt{16x - 2x^2 - 23} \, dx &= \int \sqrt{9 - 2(x - 4)^2} \, dx \\
&= \int \sqrt{9 - u^2} \left(\frac{du}{\sqrt{2}} \right) && \text{Where } u = \sqrt{2}(x - 4) \\
&= \int \sqrt{9 - (3 \sin \theta)^2} \left(\frac{3}{\sqrt{2}} \cos \theta \, d\theta \right) && \text{Where } u = 3 \sin \theta \\
&= \frac{3}{\sqrt{2}} \int 3 \sqrt{1 - \sin^2 \theta} \cos \theta \, d\theta \\
&= \frac{9}{\sqrt{2}} \int \cos^2 \theta \, d\theta
\end{aligned}$$

$$\begin{aligned}
 &= \frac{9}{2\sqrt{2}} \left[\theta + \frac{\sin 2\theta}{2} \right] + C && \text{Half-angle identity} \\
 &= \frac{9}{2\sqrt{2}} \sin^{-1} \left[\frac{\sqrt{2}}{3}(x-4) \right] + \frac{x-4}{2} \sqrt{16x-2x^2-23} + C
 \end{aligned}$$

This last step requires back-substituting from θ to u and then from u to x . Details are left as an exercise. 

PROBLEM SET 7.3

Level 1

1. **What does this say?** Explain how to integrate $\int \sin^m x \cos^n x \, dx$ when m and n are both even.
2. **What does this say?** Explain how to integrate $\int \tan^m x \sec^n x \, dx$ when n is even.
3. **What does this say?** Explain the process of using a trigonometric substitution on integrals of the form $\sqrt{a^2 + u^2}$.
4. **What does this say?** Explain the process of using a trigonometric substitution on integrals of the form $\sqrt{a^2 - u^2}$. How is this different from handling an integral involving $\sqrt{u^2 - a^2}$?

Evaluate the integrals in Problems 5-50.

- | | |
|--|--|
| 5. $\int \cos^3 x \, dx$ | 6. $\int \sin^5 x \, dx$ |
| 7. $\int \sin^2 x \cos^3 x \, dx$ | 8. $\int \sin^3 x \cos^3 x \, dx$ |
| 9. $\int \sqrt{\cos t} \sin t \, dt$ | 10. $\int \frac{\cos x \, dx}{1 + 3 \sin x}$ |
| 11. $\int e^{\cos x} \sin x \, dx$ | 12. $\int \cos^2(2t) \, dt$ |
| 13. $\int \sin^2 x \cos^2 x \, dx$ | 14. $\int \frac{\sin x \, dx}{\cos^5 x}$ |
| 15. $\int \tan 2\theta \, d\theta$ | 16. $\int \sec \frac{x}{2} \, dx$ |
| 17. $\int \tan^3 x \sec^4 x \, dx$ | 18. $\int \sec^5 x \tan x \, dx$ |
| 19. $\int (\tan^2 x + \sec^2 x) \, dx$ | 20. $\int (\sin x + \cos x)^2 \, dx$ |
| 21. $\int \tan^2 u \sec u \, du$ | 22. $\int \sec^4 x \, dx$ |
| 23. $\int \sqrt[3]{\tan x} \sec^2 x \, dx$ | 24. $\int e^x \sec e^x \, dx$ |
| 25. $\int x \sin x^2 \cos x^2 \, dx$ | 26. $\int x \sec^2 x \, dx$ |
| 27. $\int \tan^4 t \sec t \, dt$ | 28. $\int \csc 2\theta \, d\theta$ |
| 29. $\int \csc^3 x \cot x \, dx$ | 30. $\int \csc^2 x \cot^2 x \, dx$ |

- | | |
|--|---|
| 31. $\int \csc^2 x \cos x \, dx$ | 32. $\int \tan x \csc^3 x \, dx$ |
| 33. $\int \sqrt{9 - 4t^2} \, dt$ | 34. $\int \frac{dx}{\sqrt{9 - x^2}}$ |
| 35. $\int \frac{x+1}{\sqrt{4+x^2}} \, dx$ | 36. $\int \sqrt{9+x^2} \, dx$ |
| 37. $\int \frac{dx}{\sqrt{x^2-7}}$ | 38. $\int \frac{dx}{5+2x^2}$ |
| 39. $\int \frac{dx}{\sqrt{5-x^2}}$ | 40. $\int \frac{dx}{x\sqrt{7x^2-4}}$ |
| 41. $\int \frac{dx}{x^2\sqrt{4-x^2}}$ | 42. $\int \frac{dx}{x\sqrt{x^2+9}}$ |
| 43. $\int \frac{\sqrt{x^2-4}}{x} \, dx$ | 44. $\int \frac{dx}{(x-1)^2+4}$ |
| 45. $\int \frac{dx}{9-(x+1)^2}$ | 46. $\int \sqrt{2x-x^2} \, dx$ |
| 47. $\int \frac{dx}{\sqrt{x^2-2x+6}}$ | 48. $\int \frac{dx}{\sqrt{x^2+8x+3}}$ |
| 49. $\int \frac{\sin^3 u \, du}{\cos^5 u}$ | 50. $\int \frac{\sec^2 x \, dx}{\tan^2 x + \sec^2 x}$ |

Level 2

51. Find the average value of $f(x) = \sin^2 x$ over the interval $[0, \pi]$.
52. Find the centroid of the region (correct to two decimal places) bounded by the curve $y = \cos^2 x$, the x -axis, and the vertical lines $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$.
53. Find the volume (correct to four decimal places) of the solid generated when the region bounded by the curve $y = \sin^2 x$ and the x -axis is revolved about the y -axis, $0 \leq x \leq \pi$.
54. A particle moves along the x -axis in such a way that the acceleration at time t is $a(t) = \sin^2 t$. What is the total distance traveled by the particle over the time interval $[0, \pi]$ if its initial velocity is $v(0) = 2$ units per second?

In Problems 55–58, use the following identities:

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

55. $\int \sin 3x \sin 5x \, dx$

56. $\int \cos \frac{x}{2} \sin 2x \, dx$

57. $\int \sin^2 3x \cos 4x \, dx$

58. $\int \cos 7x \cos(-3x) \sin 4x \, dx$

Level 3

59. Let f be a twice differentiable function that satisfies the initial value problem

$$f''(x) = -\frac{1}{2} \tan x f'(x) \quad f'(0) = f(0) = 1$$

on the interval $[0, \frac{\pi}{2}]$. Find the arc length of the curve $y = f(x)$ over this interval.

60. Find $\int \frac{x \, dx}{9 - x^2 - \sqrt{9 - x^2}}$.

7.4 METHOD OF PARTIAL FRACTIONS

IN THIS SECTION: *Partial fraction decomposition, integrating rational functions, rational functions of sine and cosine*

Partial fraction decomposition has great value as a tool of integration. This process may be thought of as the “reverse” of adding fractional algebraic expressions, and it allows us to break up rational expressions into simpler terms.

Partial Fraction Decomposition

You are familiar with the algebraic procedure of adding a string of rational expressions to form a combined rational function with a common denominator. For example,

$$\frac{2}{x+1} + \frac{-3}{x+2} = \frac{2(x+2) + (-3)(x+1)}{(x+1)(x+2)} = \frac{-x+1}{x^2+3x+2}$$

In **partial fraction decomposition**, we do just the opposite: We start with the reduced fraction

$$\frac{-x+1}{x^2+3x+2}$$

and write it as the sum of fractions

$$\frac{2}{x+1} + \frac{-3}{x+2}$$

This procedure has great value for integration because the terms $\frac{2}{x+1}$ and $\frac{-3}{x+2}$ are easy to integrate. In particular,

$$\begin{aligned} \int \frac{-x+1}{x^2+3x+2} dx &= \int \frac{2}{x+1} dx + \int \frac{-3}{x+2} dx \\ &= 2 \ln|x+1| - 3 \ln|x+2| + C \end{aligned}$$

In the following discussion, we shall consider rational functions

$$f(x) = \frac{P(x)}{D(x)}$$

that are reduced in the sense that $P(x)$ and $D(x)$ are polynomials in x which have no common factors and the degree of P is less than the degree of D . In algebra, it is shown that if $P(x)/D(x)$ is such an expression, then

$$\frac{P(x)}{D(x)} = F_1(x) + F_2(x) + \cdots + F_N(x)$$

where the $F_k(x)$ are expressions of the form

$$\frac{A}{(x-r)^n} \quad \text{or} \quad \frac{Ax+B}{(x^2+sx+t)^m}$$

If $\frac{P(x)}{D(x)}$ is not reduced, then we simply divide until a reduced form is obtained. For example ($x \neq 1$),

$$\begin{aligned} \frac{(2x^3 + 7x^2 + 6x + 3)(x-1)}{(x-1)(x^2 + 3x + 2)} &= 2x + 1 + \frac{-x + 1}{x^2 + 3x + 2} \\ &= 2x + 1 + \frac{2}{x+1} + \frac{-3}{x+2} \end{aligned}$$

TECHNOLOGY NOTE: Software programs will decompose rational expressions quite handily using the direction “expand.”

We will examine a nonreduced rational expression in Example 5.

We begin by focusing on the case where $D(x)$ can be expressed as a product of linear powers.

PARTIAL FRACTION DECOMPOSITION A Single Linear Power

Let $f(x) = \frac{P(x)}{(x-r)^n}$, where $P(x)$ is a polynomial of degree less than n and $P(r) \neq 0$. Then $f(x)$ can be **decomposed** into partial fractions in the following “cascading form”

$$\frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_n}{(x-r)^n}$$

Example 1 Partial fraction decomposition with a single linear power

Decompose $\frac{x^2 - 6x + 3}{(x-2)^3}$ into a sum of partial fractions.

Solution

$$\frac{x^2 - 6x + 3}{(x-2)^3} = \frac{A_1}{x-2} + \frac{A_2}{(x-2)^2} + \frac{A_3}{(x-2)^3} \quad \text{Partial fraction decomposition.}$$

$$x^2 - 6x + 3 = A_1(x-2)^2 + A_2(x-2) + A_3 \quad \text{Multiply both sides by } (x-2)^3.$$

$$\begin{aligned} \text{Let } x = 2: \quad (2)^2 - 6(2) + 3 &= A_1(2-2)^2 + A_2(2-2) + A_3 && \text{This causes most terms to drop out.} \\ -5 &= A_3 \end{aligned}$$

We now substitute this value, $A_3 = -5$ and expand the right side to obtain

$$\begin{aligned} x^2 - 6x + 3 &= A_1(x-2)^2 + A_2(x-2) + (-5) \\ &= A_1x^2 + (-4A_1 + A_2)x + (4A_1 - 2A_2 - 5) \end{aligned}$$

This implies (by equating the coefficients of the similar terms) that

$$\begin{array}{rcl} 1 & = & A_1 \\ -6 & = & -4A_1 + A_2 \\ 3 & = & 4A_1 - 2A_2 - 5 \end{array}$$

x^2 terms
 x terms
 constants

Because $A_1 = 1$, we find $A_2 = -2$, so the decomposition is

$$\frac{x^2 - 6x + 3}{(x - 2)^3} = \frac{1}{x - 2} + \frac{-2}{(x - 2)^2} + \frac{-5}{(x - 2)^3}$$

If there are two or more linear factors in the factorization of $D(x)$, there must be a separate cascade for each power. In particular, if $D(x)$ can be expressed as the product of n distinct linear factors, then

$$\frac{P(x)}{(x - r_1)(x - r_2) \cdots (x - r_n)}$$

is decomposed into separate terms

$$\frac{A_1}{x - r_1} + \frac{A_2}{x - r_2} + \cdots + \frac{A_n}{x - r_n}$$

This process is illustrated with the following example.

Example 2 Partial fraction decomposition with distinct linear factors

Decompose $\frac{8x - 1}{x^2 - x - 2}$.

Solution

$$\frac{8x - 1}{x^2 - x - 2} = \frac{8x - 1}{(x - 2)(x + 1)}$$

First, factor the denominator.

$$= \frac{A_1}{x - 2} + \frac{A_2}{x + 1}$$

Break up the fraction into parts, each with a linear denominator. The task is to find A_1 and A_2 .

$$= \frac{A_1(x + 1) + A_2(x - 2)}{(x - 2)(x + 1)}$$

Obtain the least common denominator on the right.

$$8x - 1 = A_1(x + 1) + A_2(x - 2)$$

Multiply both sides by the least common denominator.

Note that the degree of the denominator is the same as the number of arbitrary constants, A_1 and A_2 . This provides a quick intermediate check on the correct procedure.

Substitute, one at a time, the values that cause each of the factors in the least common denominator to be zero.

$$8x - 1 = A_1(x + 1) + A_2(x - 2)$$

$$\text{Let } x = -1 \quad 8(-1) - 1 = A_1(-1 + 1) + A_2(-1 - 2)$$

$$-9 = -3A_2$$

$$3 = A_2$$

$$\text{Let } x = 2 \quad 8(2) - 1 = A_1(2 + 1) + A_2(2 - 2)$$

$$15 = 3A_1$$

$$5 = A_1$$

$$\frac{8x - 1}{x^2 - x - 2} = \frac{5}{x - 2} + \frac{3}{x + 1}$$

If there is a mixture of distinct and repeated linear factors, we combine the procedures illustrated in the preceding examples. For example,

$$\frac{5x^2 + 21x + 4}{(x+1)^3(x-3)} \text{ is decomposed as } \frac{A_1}{x+1} + \frac{A_2}{(x+1)^2} + \frac{A_3}{(x+1)^3} + \frac{A_4}{x-3}$$

Note that the degree of the denominator is 4, so we use four constants.

If the denominator $D(x)$ in the rational expression $P(x)/D(x)$ contains an irreducible quadratic power, the partial fraction decomposition contains a different kind of cascading sum.

PARTIAL FRACTION DECOMPOSITION A Single Quadratic Factor

Let $f(x) = \frac{P(x)}{(x^2 + sx + t)^m}$, where $P(x)$ is a polynomial of degree less than $2m$. Then $f(x)$ can be decomposed into partial fractions in the following “cascading form”

$$\frac{A_1x + B_1}{x^2 + sx + t} + \frac{A_2x + B_2}{(x^2 + sx + t)^2} + \cdots + \frac{A_mx + B_m}{(x^2 + sx + t)^m}$$

Because the degree of the denominator is $2m$, we have $2m$ arbitrary constants, namely $A_1, A_2, \dots, A_m, B_1, B_2, \dots, B_m$.

Note that the numerator in each term of a linear cascade is a constant A_k , while each numerator in a quadratic cascade is a linear term of the form $A_kx + B_k$.

Example 3 Partial fraction decomposition with a single quadratic power

Decompose $\frac{-3x^3 - x}{(x^2 + 1)^2}$.

Solution The decomposition gives

$$\frac{-3x^3 - x}{(x^2 + 1)^2} = \frac{A_1x + B_1}{x^2 + 1} + \frac{A_2x + B_2}{(x^2 + 1)^2}$$

Multiply both sides of this equation by $(x^2 + 1)^2$, and simplify algebraically.

$$\begin{aligned} -3x^3 - x &= (A_1x + B_1)(x^2 + 1) + (A_2x + B_2) \\ &= A_1x^3 + B_1x^2 + (A_1 + A_2)x + (B_1 + B_2) \end{aligned}$$

Next, equate the corresponding coefficients on each side of this equation and solve the resulting system of equations to find $A_1 = -3$, $A_2 = 2$, $B_1 = 0$, and $B_2 = 0$. This means that

$$\frac{-3x^3 - x}{(x^2 + 1)^2} = \frac{-3x}{x^2 + 1} + \frac{2x}{(x^2 + 1)^2}$$

Many of the examples we encounter will offer a mixture of linear and quadratic factors. For example,

$$\frac{x^2 + 4x - 23}{(x^2 + 4)(x + 3)^2} \text{ is decomposed as } \frac{A_1x + B_1}{x^2 + 4} + \frac{A_2}{x + 3} + \frac{A_3}{(x + 3)^2}$$

The degree of the denominator is four, and there are four constants.

In algebra, the theory of equations tells us that any polynomial P with real coefficients can be expressed as a product of linear and irreducible quadratic powers, some of which may be repeated. This fact can be used to justify the following general procedure for obtaining the partial fraction decomposition of a rational function.

PARTIAL FRACTION DECOMPOSITION Let $f(x) = P(x)/D(x)$, where $P(x)$ and $D(x)$ are polynomials with no common factors and $D(x) \neq 0$.

Step 1 If the degree of P is greater than or equal to the degree of D , use long (or synthetic) division to express $\frac{P(x)}{D(x)}$ as the sum of a polynomial and a fraction $\frac{R(x)}{D(x)}$ in which the degree of the remainder polynomial $R(x)$ is less than the degree of the denominator polynomial $D(x)$.

Step 2 Factor the denominator $D(x)$ into the product of linear and irreducible quadratic powers.

Step 3 Express $\frac{P(x)}{D(x)}$ as a cascading sum of partial fractions of the form

$$\frac{A_i}{(x-r)^n} \quad \text{and} \quad \frac{A_k x + B_k}{(x^2 + sx + t)^m}$$

Integrating Rational Functions

We will now apply the procedure of partial fraction decomposition to integration.

Example 4 Integrating a rational function with a repeated linear factor

Find $\int \frac{x^2 - 6x + 3}{(x-2)^3} dx$.

Solution From Example 1, we have

$$\begin{aligned} \int \frac{x^2 - 6x + 3}{(x-2)^3} dx &= \int \left[\frac{1}{x-2} + \frac{-2}{(x-2)^2} + \frac{-5}{(x-2)^3} \right] dx \\ &= \int (x-2)^{-1} dx - 2 \int (x-2)^{-2} dx - 5 \int (x-2)^{-3} dx \\ &= \ln|x-2| + \frac{2}{x-2} + \frac{5}{2(x-2)^2} + C \end{aligned}$$

Example 5 Integrating a rational expression with distinct linear factors

Find $\int \frac{x^4 + 2x^3 - 4x^2 + x - 3}{x^2 - x - 2} dx$.

Solution We have $P(x) = x^4 + 2x^3 - 4x^2 + x - 3$ and $D(x) = x^2 - x - 2$; because the degree of P is greater than the degree of D , we carry out the long division and write

$$\int \frac{x^4 + 2x^3 - 4x^2 + x - 3}{x^2 - x - 2} dx = \int \left(x^2 + 3x + 1 + \frac{8x - 1}{x^2 - x - 2} \right) dx$$

The polynomial part is easy to integrate. The rational expression was decomposed into partial fractions in Example 2.

$$\begin{aligned} \int \frac{x^4 + 2x^3 - 4x^2 + x - 3}{x^2 - x - 2} dx &= \int \left(x^2 + 3x + 1 + \frac{8x - 1}{x^2 - x - 2} \right) dx \\ &= \int \left(x^2 + 3x + 1 + \frac{5}{x-2} + \frac{3}{x+1} \right) dx \end{aligned}$$

$$\begin{aligned}
&= \int x^2 dx + 3 \int x dx + \int dx + 5 \int (x-2)^{-1} dx + 3 \int (x+1)^{-1} dx \\
&= \frac{x^3}{3} + \frac{3x^2}{2} + x + 5 \ln|x-2| + 3 \ln|x+1| + C
\end{aligned}$$

Example 6 Integrating a rational function with repeated quadratic factors

Find $\int \frac{-3x^3 - x}{(x^2 + 1)^2} dx$.

Solution From Example 3,

$$\begin{aligned}
\int \frac{-3x^3 - x}{(x^2 + 1)^2} dx &= \int \frac{2x}{(x^2 + 1)^2} dx + \int \frac{-3x}{x^2 + 1} dx && \text{Let } u = x^2 + 1, \text{ then } du = 2x dx \\
&= \int u^{-2} du - \frac{3}{2} \int u^{-1} du \\
&= -u^{-1} - \frac{3}{2} \ln|u| + C \\
&= \frac{-1}{x^2 + 1} - \frac{3}{2} \ln(x^2 + 1) + C
\end{aligned}$$

Example 7 Repeated linear factors

Find $\int \frac{5x^2 + 21x + 4}{(x+1)^2(x-3)} dx$.

Solution The partial fraction decomposition of the integrand is

$$\frac{5x^2 + 21x + 4}{(x+1)^2(x-3)} = \frac{A_1}{(x+1)^2} + \frac{A_2}{x+1} + \frac{A_3}{x-3}$$

Multiply both sides by $(x+1)^2(x-3)$:

$$5x^2 + 21x + 4 = A_1(x-3) + A_2(x+1)(x-3) + A_3(x+1)^2$$

As in Example 2, we substitute $x = -1$ and $x = 3$ on both sides of the equation to obtain $A_1 = 3$ and $A_3 = 7$, but A_2 cannot be obtained in this fashion. To find A_2 , multiply out the polynomial on the right:

$$5x^2 + 21x + 4 = (A_2 + A_3)x^2 + (A_1 - 2A_2 + 2A_3)x + (-3A_1 - 3A_2 + A_3)$$

Equate the coefficients of x^2 , x , and 1 (the constant term) on each side of the equation:

$$\begin{array}{ll}
5 = A_2 + A_3 & x^2 \text{ terms} \\
21 = A_1 - 2A_2 + 2A_3 & x \text{ terms} \\
4 = -3A_1 - 3A_2 + A_3 & \text{constants}
\end{array}$$

Because we already know that $A_1 = 3$ and $A_3 = 7$, we use the equation $5 = A_2 + A_3$ to obtain $A_2 = -2$. We now turn to the integration:

$$\begin{aligned}
\int \frac{5x^2 + 21x + 4}{(x+1)^2(x-3)} dx &= \int \frac{3}{(x+1)^2} dx + \int \frac{-2}{x+1} dx + \int \frac{7}{x-3} dx \\
&= -3(x+1)^{-1} - 2 \ln|x+1| + 7 \ln|x-3| + C
\end{aligned}$$

Example 8 Distinct linear and quadratic factors

Find $\int \frac{x^2 + 4x - 23}{(x^2 + 4)(x + 3)} dx$.

Solution The partial fraction decomposition of the integrand has the form

$$\frac{x^2 + 4x - 23}{(x^2 + 4)(x + 3)} = \frac{A_1x + B_1}{x^2 + 4} + \frac{A_2}{x + 3}$$

Multiply both sides by $(x^2 + 4)(x + 3)$ and then combine the terms on the right:

$$\begin{aligned} x^2 + 4x - 23 &= (A_1x + B_1)(x + 3) + A_2(x^2 + 4) \\ &= (A_1 + A_2)x^2 + (3A_1 + B_1)x + (3B_1 + 4A_2) \end{aligned}$$

Equate the coefficients to set up the following system of equations:

$$\begin{cases} A_1 + A_2 = 1 & x^2 \text{ terms} \\ 3A_1 + B_1 = 4 & x \text{ terms} \\ 3B_1 + 4A_2 = -23 & \text{constants} \end{cases}$$

Solve this system (the details are not shown) to find $A_1 = 3$, $B_1 = -5$, and $A_2 = -2$. We now turn to the integration:

$$\begin{aligned} \int \frac{x^2 + 4x - 23}{(x^2 + 4)(x + 3)} dx &= \int \frac{3x - 5}{x^2 + 4} dx + \int \frac{-2}{x + 3} dx \\ &= 3 \int \frac{x dx}{x^2 + 4} - 5 \int \frac{dx}{x^2 + 4} - 2 \int \frac{dx}{x + 3} \end{aligned}$$

Let $u = x^2 + 4$, then $du = 2x dx$

$$= 3 \left[\frac{1}{2} \ln(x^2 + 4) \right] - 5 \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right] - 2 \ln|x + 3| + C$$

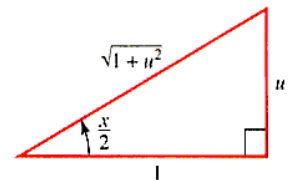
Rational Functions of Sine and Cosine

The German mathematician Karl Weierstraß (1815-1897) noticed that the substitution

$$u = \tan \frac{x}{2} \quad -\pi < x < \pi$$

will convert any rational function of $\sin x$ and $\cos x$ into a rational function of x . To see why this is true, we use the double-angle identities for sine and cosine (see Figure 7.4):

$$\begin{aligned} \sin x &= 2 \sin \frac{x}{2} \cos \frac{x}{2} & \text{and} & & \cos x &= \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \\ &= 2 \left(\frac{u}{\sqrt{1+u^2}} \right) \left(\frac{1}{\sqrt{1+u^2}} \right) & & & &= \left(\frac{1}{\sqrt{1+u^2}} \right)^2 - \left(\frac{u}{\sqrt{1+u^2}} \right)^2 \\ &= \frac{2u}{1+u^2} & & & &= \frac{1-u^2}{1+u^2} \end{aligned}$$



$$\begin{aligned} \tan \frac{x}{2} &= u \\ \cos \frac{x}{2} &= \frac{1}{\sqrt{1+u^2}} \\ \sin \frac{x}{2} &= \frac{u}{\sqrt{1+u^2}} \end{aligned}$$

Figure 7.4 Weierstrass substitution

Finally, because $u = \tan \frac{x}{2}$, we have $x = 2 \tan^{-1} u$, so $dx = \frac{2}{1+u^2} du$.

WEIERSTRASS SUBSTITUTION For $-\pi < x < \pi$, let $u = \tan \frac{x}{2}$ so that

$$\sin x = \frac{2u}{1+u^2}, \cos x = \frac{1-u^2}{1+u^2}, \text{ and } dx = \frac{2}{1+u^2} du$$

This is called the **Weierstrass substitution**.

The following example illustrates how this substitution can be used along with partial fractions to integrate a rational trigonometric function.

Example 9 Integrating a rational trigonometric function

Find $\int \frac{dx}{3 \cos x - 4 \sin x}$.

Solution Use the Weierstrass substitution — that is, let $u = \tan \frac{x}{2}$. Remember,

$$dx = \frac{2 du}{1+u^2}; \cos x = \frac{1-u^2}{1+u^2}; \text{ and } \sin x = \frac{2u}{1+u^2}$$

$$\begin{aligned} \int \frac{dx}{3 \cos x - 4 \sin x} &= \int \frac{\frac{2 du}{1+u^2}}{3 \left(\frac{1-u^2}{1+u^2} \right) - 4 \left(\frac{2u}{1+u^2} \right)} && \text{Weierstrass substitution} \\ &= \int \frac{-2 du}{3u^2 + 8u - 3} && \text{Simplify} \\ &= \int \frac{-2 du}{(3u-1)(u+3)} \end{aligned}$$

This integral can be handled by the method of partial fractions.

$$\frac{-2}{(3u-1)(u+3)} = \frac{A_1}{3u-1} + \frac{A_2}{u+3}$$

Solve this to find $A_1 = -\frac{3}{5}$ and $A_2 = \frac{1}{5}$. We continue with the integration.

$$\begin{aligned} \int \frac{dx}{3 \cos x - 4 \sin x} &= \int \frac{-2 du}{(3u-1)(u+3)} \\ &= \int \frac{-\frac{3}{5} du}{3u-1} + \int \frac{\frac{1}{5} du}{u+3} \\ &= -\frac{3}{5} \cdot \frac{1}{3} \ln |3u-1| + \frac{1}{5} \cdot \ln |u+3| + C \\ &= -\frac{1}{5} \ln \left| 3 \tan \frac{x}{2} - 1 \right| + \frac{1}{5} \ln \left| \tan \frac{x}{2} + 3 \right| + C \end{aligned}$$

Problem 59 asks you to derive the formula

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

from scratch. You might recall that we derived this formula using an unusual algebraic step in Example 3 of Section 7.1. You can now derive it by using a Weierstrass substitution.

PROBLEM SET 7.4

Level 1

Write each rational function given in Problems 1-14 as a sum of partial fractions.

1. $\frac{1}{x(x-3)}$
2. $\frac{3x-1}{x^2-1}$
3. $\frac{3x^2+2x-1}{x(x+1)}$
4. $\frac{2x^2+5x-1}{x(x^2-1)}$
5. $\frac{4}{2x^2+x}$
6. $\frac{x^2-x+3}{x^2(x-1)}$
7. $\frac{4x^3+4x^2+x-1}{x^2(x+1)^2}$
8. $\frac{x^2-5x-4}{(x^2+1)(x-3)}$
9. $\frac{x^3+3x^2+3x-4}{x^2(x+3)^2}$
10. $\frac{1}{x^3-1}$
11. $\frac{1}{1-x^4}$
12. $\frac{x^4-x^2+2}{x^2(x-1)}$
13. $\frac{x^2+x-1}{x(x+1)(2x-1)}$
14. $\frac{x^3-2x^2+x-5}{x(x^2-1)(3x+5)}$

Find the indicated integrals in Problems 15-30.

15. $\int \frac{2x^3+9x-1}{x^2(x^2-1)} dx$
16. $\int \frac{x^4-x^2+2}{x^2(x-1)} dx$
17. $\int \frac{x^2+1}{x^2+x-2} dx$
18. $\int \frac{dx}{x^3-8}$
19. $\int \frac{x^4+1}{x^4-1} dx$
20. $\int \frac{x^3+1}{x^3-1} dx$
21. $\int \frac{x dx}{(x+1)^2}$
22. $\int \frac{2x dx}{(x-2)^2}$
23. $\int \frac{dx}{x(x+1)(x-2)}$
24. $\int \frac{x+2}{x(x-1)^2} dx$
25. $\int \frac{x dx}{(x+1)(x+2)^2}$
26. $\int \frac{x+1}{x(x^2+2)} dx$
27. $\int \frac{5x+7}{x^2+2x-3} dx$
28. $\int \frac{5x dx}{x^2-6x+9}$
29. $\int \frac{3x^2-2x+4}{x^3-x^2+4x-4} dx$
30. $\int \frac{3x^2+4x+1}{x^3+2x^2+x-2} dx$

Level 2

31. ■ **What does this say?** Describe the process of partial fraction decomposition.

32. ■ **What does this say?** What is the Weierstrass substitution and when would you use it?

Find the indicated integrals in Problems 33-50.

33. $\int \frac{e^x dx}{2e^{2x}-5e^x-3}$
34. $\int \frac{\cos x dx}{\sin^2 x - \sin x - 2}$
35. $\int \frac{\sin x dx}{(1+\cos x)^2}$
36. $\int \frac{e^x dx}{e^{2x}-1}$
37. $\int \frac{\sec^2 x dx}{\tan x + 4}$
38. $\int \frac{\tan x dx}{\sec^2 x + 4}$
39. $\int \frac{dx}{x^{2/3} - x^{1/2}}$
40. $\int \frac{dx}{x^{1/4} - x}$
41. $\int \frac{dx}{3 \cos x + 4 \sin x}$
42. $\int \frac{dx}{\sin x - \cos x}$
43. $\int \frac{\sin x - \cos x}{\sin x + \cos x} dx$
44. $\int \frac{dx}{5 \sin x + 4}$
45. $\int \frac{dx}{\sec x - \tan x}$
46. $\int \frac{dx}{4 \cos x + 5}$
47. $\int \frac{dx}{4 \sin x - 3 \cos x - 5}$
48. $\int \frac{dx}{3 \sin x + 4 \cos x + 5}$
49. $\int \frac{dx}{x(3-\ln x)(1-\ln x)}$
50. $\int \frac{dx}{2 \csc x - \cot x + 2}$

51. Find the area of the region bounded by the curve

$$y = \frac{1}{6-5x+x^2}$$

and the lines $x = \frac{4}{3}$, $x = \frac{7}{4}$, and $y = 0$.

52. Find the area under the curve

$$y = \frac{1}{x^2+5x+4}$$

between $x = 0$ and $x = 3$.

53. Find the volume (to four decimal places) of the solid generated when the region under the curve

$$y = \frac{1}{\sqrt{x^2+4x+3}}$$

on the interval $[0, 3]$ is revolved about

- a. the x -axis b. the y -axis

54. Find the volume (to four decimal places) of the solid generated when the curve

$$y = \frac{1}{x^2+5x+4}, \quad 0 \leq x \leq 1$$

is revolved about

- a. the y -axis
b. the x -axis
c. the line $x = -1$

55. *Historical Quest*

George Pólya was born in Hungary and attended universities in Budapest, Vienna, Göttingen, and Paris. He was a professor of mathematics at Stanford University. Pólya's research and winning personality earned him a place of honor not only among

mathematicians, but among students and teachers as well. His discoveries spanned an impressive range of mathematics, real and complex analysis, probability, combinatorics, number theory, and geometry. Pólya's book, *How to Solve It*, has been translated into 20 languages. His books have a clarity and elegance seldom seen in mathematics, making them a joy to read. For example, he explained why he became a mathematician by saying that essentially, he thought he was not good enough for physics but on the other hand, too good for philosophy, and mathematics is "in between". A story told by Pólya provides our next *Quest*. He said that he had once intentionally put the problem

$$\int \frac{x \, dx}{x^2 - 9}$$

first on a test of techniques of integration, thinking that it would make his students confident at the start of the exam. If you substitute $u = x^2 - 9$, you can solve the problem very quickly and he assumed his students would do just that. Half of them did use it, but a quarter of them used the procedure of partial fractions, which is the correct method but it used up much of the time allowed, so they were not able to finish the exam. Another group of students used the trig substitution $x = 3 \sin \theta$, which is also correct, but again, used up so much time that they weren't able to finish and did poorly on the exam. What Pólya found to be interesting was that the students who used the more time-consuming techniques demonstrated that they



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George Pólya
(1887–1985)

knew more difficult mathematics than those who opted for the easy technique. "But they showed that 'it's not just what you know; it's how and when you use it.' It's nice when what you do is right, but it's much better when it's also *appropriate*."* Carry out all three methods of solution of the given integral Pólya described in this quotation.

56. **EXPLORATION PROBLEM** Consider the integral

$$\int \sqrt{1 - \cos x} \, dx, \quad 0 < x < \pi$$

- If you have access to a CAS, report on your attempts to find this integral.
- Use a Weierstrass substitution to find this integral.
- Can you think of an even more clever way to perform the integration?

Level 3

57. Use partial fractions to derive the integration formula

$$\int \frac{dx}{x(ax+b)} = \frac{1}{b} \ln \left| \frac{x}{ax+b} \right| + C$$

58. Use partial fractions to derive the integration formula

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$$

59. Derive the formula

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

using a Weierstrass substitution.

60. Derive the formula

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C$$

using a Weierstrass substitution.

7.5 SUMMARY OF INTEGRATION TECHNIQUES

IN THIS SECTION: *Integration strategy*

We conclude our study of integration techniques with a procedure summarizing integration techniques.

*"Pólya, Problem Solving, and Education," by Alan H. Schoenfeld, *Mathematics Magazine*, Vol. 60, No. 5, December 1987, p. 290.

INTEGRATION STRATEGY In order to integrate a function, consider the following steps.

Step 1 Simplify.

Simplify the integrand, if possible, and use one of the procedural rules (see the inside back cover or the table of integrals).

Step 2 Use basic formulas.

Use the basic integration formulas (1-29 in the integration table, Appendix D). These are the fundamental building blocks for integration. Almost every integration will involve some basic formulas somewhere in the process, which means that you should memorize these formulas.

Step 3 Substitute.

Make any substitution that will transform the integral into one of the basic forms.

Step 4 Classify.

Classify the integrand according to form in order to use a table of integrals. You may need to use substitution to transform the integrand into a form contained in the integration table. Some special types of substitution are contained in the following list:

I. Integration by parts

A. Forms $\int x^n e^{ax} dx$, $\int x^n \sin ax dx$, $\int x^n \cos ax dx$

Let $u = x^n$.

B. Forms $\int x^n \ln x dx$, $\int x^n \sin^{-1} ax dx$, $\int x^n \tan^{-1} ax dx$

Let $dv = x^n dx$.

C. Forms $\int e^{ax} \sin bx dx$, $\int e^{ax} \cos bx dx$

Let $dv = e^{ax} dx$.

II. Trigonometric forms

A. $\int \sin^m x \cos^n x dx$

m odd:

Peel off a factor of $(\sin x dx)$ and let $u = \cos x$.

n odd:

Peel off a factor of $(\cos x dx)$ and let $u = \sin x$.

m and n both even:

Use **half-angle identities**:

$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

B. $\int \tan^m x \sec^n x dx$

n even:

Peel off a factor of $(\sec^2 x dx)$ and let $u = \tan x$.

m odd:

Peel off a factor of $(\sec x \tan x)$ and let $u = \sec x$.

m even, n odd:

Write using powers of secant and use a **reduction formula** (Formula 161 in the integration tables).

C. For a rational trigonometric integral, try the Weierstrass substitution. Let $u = \tan \frac{x}{2}$, so that

$\sin x = \frac{2u}{1+u^2}$, $\cos x = \frac{1-u^2}{1+u^2}$, and $dx = \frac{2}{1+u^2} du$

III. Radical forms; try a trigonometric substitution.

A. Form $\sqrt{a^2 - u^2}$: Let $u = a \sin \theta$.

B. Form $\sqrt{a^2 + u^2}$: Let $u = a \tan \theta$.

C. Form $\sqrt{u^2 - a^2}$: Let $u = a \sec \theta$.

IV. Rational forms; try partial fraction decomposition.

Step 5 Try again. Still stuck?

1. Manipulate the integrand:
Multiply by 1 (clever choice of numerator or denominator).
Rationalize the numerator.
Rationalize the denominator.
2. Relate the problem to a previously worked problem.
3. Look at another table of integrals or consult computer software that does integration.
4. Some integrals do not have simple antiderivatives, so all these methods may fail. We will look at some of these forms later in the text.
5. If dealing with a definite integral, an approximation may suffice. It may be appropriate to use a calculator, computer, or approximate integration.

Example 1 Deciding on an integration procedure

Indicate a procedure to set up each integral. It is not necessary to carry out the integration.

$$\begin{array}{llll} \text{a. } \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx & \text{b. } \int (1 + \tan^2 \theta) d\theta & \text{c. } \int \sin^3 x \cos^2 x dx & \text{d. } \int 4x^2 \cos 3x dx \\ \text{e. } \int \frac{x dx}{\sqrt{9-x^2}} & \text{f. } \int \frac{x^2 dx}{\sqrt{9-x^2}} & \text{g. } \int e^{3x} \sin 2x dx & \text{h. } \int \frac{\cos^4 x dx}{1 - \sin^2 x} \end{array}$$

Solution Keep in mind that there may be several correct approaches, and the way you proceed from problem to solution is often a matter of personal preference. However, to give you some practice, we will show several hints and suggestions in the context of this example.

$$\text{a. } \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$$

The integrand is simplified and is not a fundamental type. Substitution will work well for this problem if you let $u = \sqrt{x}$. After you make this substitution you can integrate using a basic formula.

$$\text{b. } \int (1 + \tan^2 \theta) d\theta$$

The integrand can be simplified using a trigonometric identity ($1 + \tan^2 \theta = \sec^2 \theta$), so that it can now be integrated using a basic formula.

$$\text{c. } \int \sin^3 x \cos^2 x dx$$

The integrand is simplified, it is not a basic formula, and there does not seem to be an easy substitution. Next, try to classify the integrand. We see that it involves powers of trigonometric functions of the type $\int \sin^m x \cos^n x dx$, where $m = 3$ (odd) and $n = 2$ (even), so we make the substitution $u = \cos x$.

$$\text{d. } \int 4x^2 \cos 3x dx$$

The integrand is simplified (you can bring the 4 out in front of the integral, if you wish), it is not a basic formula, and there is no easy substitution. You might try integration by parts (several steps) or you can check the table of integrals and find it to be Formula 124.

$$\text{e. } \int \frac{x dx}{\sqrt{9-x^2}}$$

The integrand is simplified and is not a basic formula. However, we might try the substitution $u = 9 - x^2$. It looks like this will work because the degree of the numerator is one less than the degree of the denominator.

f. $\int \frac{x^2 dx}{\sqrt{9-x^2}}$

The integrand is simplified, it is not a basic formula, and it looks as though an algebraic substitution will not work because of the degree of the numerator. Since the integrand involves $\sqrt{9-x^2}$, we use the trigonometric substitution $x = 3 \sin \theta$, $dx = 3 \cos \theta d\theta$. If the integral had been $\int \frac{x^2 dx}{\sqrt{x^2-9}}$, you would have used the substitution $x = 3 \sec \theta$, $dx = 3 \tan \theta \sec \theta d\theta$.

g. $\int e^{3x} \sin 2x dx$

The integrand is simplified and it is not a basic formula. If we try to classify the integrand, we see that integration by parts will work with $u = \sin 2x$ and $dv = e^{3x} dx$. We also find that this form is Formula 192 in the integration table.

h. $\int \frac{\cos^4 x dx}{1 - \sin^2 x}$

The integration can be simplified by writing $1 - \sin^2 x = \cos^2 x$. After doing this substitution, you will obtain

$$\int \frac{\cos^4 x dx}{1 - \sin^2 x} = \int \frac{\cos^4 x dx}{\cos^2 x} = \int \cos^2 x dx$$

This form can be integrated by using a half-angle identity or Formula 127 in the table of integrals. It can also be integrated by parts. ■

PROBLEM SET 7.5

Level 1

Find each integral in Problems 1–34.

1. $\int \frac{2x-1}{(x-x^2)^3} dx$

2. $\int \frac{2x+3}{\sqrt{x^2+3x}} dx$

3. $\int (x \sec 2x^2) dx$

4. $\int (x^2 \csc^2 2x^3) dx$

5. $\int (e^x \cot e^x) dx$

6. $\int \frac{\tan \sqrt{x} dx}{\sqrt{x}}$

7. $\int \frac{\tan(\ln x) dx}{x}$

8. $\int \sqrt{\cot x} \csc^2 x dx$

9. $\int \frac{e^{2t} dt}{1+e^{4t}}$

10. $\int \frac{\sin 2x dx}{1+\sin^4 x}$

11. $\int \frac{x^2+x+1}{x^2+9} dx$

12. $\int \frac{3x+2}{\sqrt{4-x^2}} dx$

13. $\int \frac{1+e^x}{1-e^x} dx$

14. $\int \frac{t^3 dt}{2^8+t^8}$

15. $\int \frac{dx}{1+e^{2x}}$

16. $\int \frac{dx}{4-e^{-x}}$

17. $\int \frac{dx}{x^2+2x+2}$

18. $\int \frac{dx}{x^2+x+4}$

19. $\int e^{-x} \cos x dx$

20. $\int e^{2x} \sin 3x dx$

21. $\int \sin^3 x dx$

22. $\int \cos^5 x dx$

23. $\int \sin^3 x \cos^2 x dx$

24. $\int \sin^3 x \cos^3 x dx$

25. $\int \sin^2 x \cos^4 x dx$

26. $\int \sin^2 x \cos^5 x dx$

27. $\int \tan^5 x \sec^4 x dx$

28. $\int \tan^4 x \sec^4 x dx$

29. $\int \frac{\sqrt{1-x^2}}{x} dx$

30. $\int \frac{dx}{\sqrt{x^2-16}}$

31. $\int \frac{\cos x dx}{\sqrt{1+\sin^2 x}}$

32. $\int \frac{\sec^2 x dx}{\sqrt{\sec^2 x - 2}}$

33. $\int \sin^5 x dx$

34. $\int \cos^6 x dx$

Find the exact value of the definite integrals in Problems 35-42.

$$35. \int_0^2 \sqrt{4-x^2} dx$$

$$36. \int_0^1 \frac{dx}{\sqrt{9-x^2}}$$

$$37. \int_0^{\ln 2} e^t \sqrt{1+e^{2t}} dt$$

$$38. \int_0^1 \frac{dt}{4t^2+4t+5}$$

$$39. \int_1^2 \frac{dx}{x^4 \sqrt{x^2+3}}$$

$$40. \int_0^2 \frac{x^3}{(3+x^2)^{3/2}} dx$$

$$41. \int_{-2}^{2\sqrt{3}} x^3 \sqrt{x^2+4} dx$$

$$42. \int_0^{\sqrt{5}} x^2 \sqrt{5-x^2} dx$$

Level 2

Find each integral in Problems 43-50.

$$43. \int \frac{e^x dx}{\sqrt{1+e^{2x}}}$$

$$44. \int \frac{(2x+1)dx}{\sqrt{4x^2+4x+2}}$$

$$45. \int \frac{x^2+4x+3}{x^3+x^2+x} dx$$

$$46. \int \frac{5x^2+3x-2}{x^3+2x^2} dx$$

$$47. \int \frac{3x+5}{x^2+2x+1} dx$$

$$48. \int \frac{3x^2+2x+1}{x^3+x^2+x} dx$$

$$49. \int \frac{5x^2+18x+34}{(x-7)(x+2)^2} dx$$

$$50. \int \frac{-3x^2+9x+21}{(x+2)^2(2x+1)} dx$$

51. ■ **What does this say?** Outline a method of approaching integration problems. Comment on the strategy presented in this section.

52. ■ **What does this say?** Integrals of the general form

$$\int \cot^m x \csc^n x dx$$

are handled in much the same way as those of the form

$$\int \tan^m x \sec^n x dx$$

- What substitution would you use in the case where m is odd?
- What substitution would you use if n is even?
- What would you do if n is odd and m is even?

53. Find the arc length (correct to four decimal places) of the curve $y = x^2$ from $x = -1$ to $x = 1$.

54. Find the arc length (correct to four decimal places) of the curve $y = \ln x$ from $x = 2$ to $x = 3$.

55. Find the volume of the solid generated when the region under the curve $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{2}$ is revolved about the x -axis.

56. What is the volume of the solid obtained when the region bounded by $y = \sqrt{x} e^{-x^2}$, $y = 0$, $x = 0$, and $x = 2$ is revolved about the x -axis?

Level 3

57. Generalize the result of Example 4, Section 7.2 by showing that

$$\int e^{ax} \sin bx dx = \frac{(a \sin bx - b \cos bx)e^{ax}}{a^2 + b^2} + C$$

for constants a and b . (This is Formula 192.)

58. Let f'' be continuous on the closed interval $[a, b]$. Use integration by parts to show that

$$\int_a^b x f''(x) dx = b f'(b) - f(b) + f(a) - a f'(a)$$

59. Derive the reduction formula

$$\begin{aligned} \int x^m (\ln x)^n dx \\ = \frac{x^{m+1} (\ln x)^n}{m+1} - \frac{n}{m+1} \int x^m (\ln x)^{n-1} dx \end{aligned}$$

where m and n are positive integers. Use the formula to find

$$\int x^2 (\ln x)^3 dx$$

60. Derive the reduction formula

$$\begin{aligned} \int \sin^n Ax dx \\ = \frac{-\sin^{n-1} Ax \cos Ax}{An} + \frac{n-1}{n} \int \sin^{n-2} Ax dx \end{aligned}$$

Use this formula to evaluate

$$\int \sin^4 4x dx$$

7.6 FIRST-ORDER DIFFERENTIAL EQUATIONS

IN THIS SECTION: *First-order linear differential equations, applications of first-order equations*

In this section, we see how differential equations can be used to model several real-world applications.

First-Order Linear Differential Equations

We introduced separable differential equations in Section 5.6, but not all first-order differential equations have the separable form

$$\frac{dy}{dx} = \frac{f(x)}{g(y)}$$

The goal of this section is to examine a second class of differential equations, called **first-order linear**, that have the general form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

For example, the differential equation

$$\frac{dy}{dx} - \frac{y}{x} = e^{-x}$$

is first-order linear with $P(x) = -\frac{1}{x}$ and $Q(x) = e^{-x}$, and

$$x^2 \frac{dy}{dx} - (x^2 + 2)y = x^5$$

can be put into this form by dividing by x^2 , the coefficient of dy/dx , to obtain

$$\frac{dy}{dx} - \left(\frac{x^2 + 2}{x^2}\right)y = \frac{x^5}{x^2}$$

[Note that $P(x) = -\left(\frac{x^2 + 2}{x^2}\right) = -1 - 2x^{-2}$ and $Q(x) = \frac{x^5}{x^2} = x^3$.]

First-order linear equations can be used to model a variety of important situations in the physical, social, and life sciences. We shall examine a few such models as well as additional models involving separable equations later in this section, but first we need to know how first-order linear differential equations may be solved.

Theorem 7.1 General solution of a first-order linear differential equation

The general solution of the first-order linear differential equation

$$\frac{dy}{dx} + P(x)y = Q(x)$$

is given by

$$y = \frac{1}{I(x)} \left[\int I(x)Q(x)dx + C \right]$$

where $I(x) = e^{\int P(x)dx}$, the exponent is any antiderivative of $P(x)$, and C is an arbitrary constant.

Proof: First, note that

$$\frac{dI(x)}{dx} = e^{\int P(x)dx} \left[\frac{d}{dx} \int P(x)dx \right] = e^{\int P(x)dx} P(x) = I(x)P(x)$$

*Coefficient of $\frac{dy}{dx}$ must be 1.
The $\int P(x)dx$ that appears
in the exponent of $I(x)$ is any
antiderivative of $P(x)$.*

because $\frac{d}{dx}e^u = e^u \frac{du}{dx}$. Thus,

$$\frac{d}{dx}[I(x)y] = I(x)\frac{dy}{dx} + y\frac{d}{dx}I(x) = I(x)\frac{dy}{dx} + yI(x)P(x) \quad \text{Product rule}$$

We use this computation as the third step in the following.

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \text{Given}$$

$$I(x)\left[\frac{dy}{dx} + P(x)y\right] = I(x)Q(x) \quad \text{Multiply both sides by } I(x).$$

$$I(x)\frac{dy}{dx} + I(x)P(x)y = I(x)Q(x)$$

$$\frac{d}{dx}[I(x)y] = I(x)Q(x) \quad \text{Substitute}$$

$$\int \frac{d}{dx}[I(x)y] dx = \int I(x)Q(x) dx \quad \text{Integrate both sides.}$$

$$I(x)y = \int I(x)Q(x) dx + C \quad \text{As usual, the constant of integration } C \text{ is added after the last integration.}$$

$$y = \frac{1}{I(x)} \left[\int I(x)Q(x) dx + C \right]$$

Because multiplying both sides of the differential equation

$$\frac{dy}{dx} + P(x)y = Q(x)$$

by $I(x)$ makes the left side an exact derivative, the function $I(x)$ is called an **integrating factor** of the differential equation.

A first-order **initial value problem** involves a first-order equation and the value of y at a particular value $x = x_0$. Here is an example of such a problem.

Example 1 First-order linear initial value problem

Solve $\frac{dy}{dx} = e^{-x} - 2y$, $x \geq 0$, subject to the initial condition $y = 2$ when $x = 0$.

Solution A graphical solution (using technology) is found by looking at the direction field as shown in Figure 7.5.

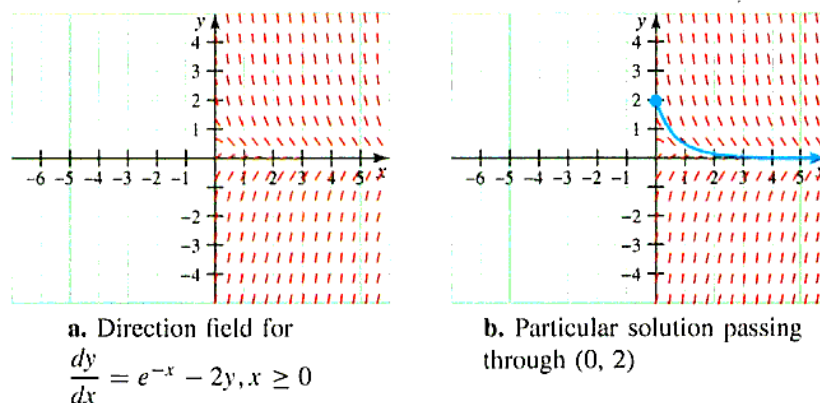


Figure 7.5 Interactive Graphical solution using a direction field

To find an analytic solution, we use Theorem 7.1. The differential equation can be expressed in first-order linear form by adding $2y$ to both sides:

$$\frac{dy}{dx} + 2y = e^{-x} \quad \text{for } x \geq 0$$

We have $P(x) = 2$ and $Q(x) = e^{-x}$. Both $P(x)$ and $Q(x)$ are continuous in the domain $x \geq 0$. An integrating factor is given by

$$I(x) = e^{\int P(x) dx} = e^{\int 2 dx} = e^{2x} \quad \text{for } x \geq 0$$

We now use the first-order linear differential equation theorem, where $I(x) = e^{2x}$ and $Q(x) = e^{-x}$, to find y .

$$\begin{aligned} y &= \frac{1}{e^{2x}} \left[\int e^{2x} e^{-x} dx + C \right] \\ &= \frac{1}{e^{2x}} [e^x + C] \\ &= e^{-x} + Ce^{-2x} \quad \text{for } x \geq 0 \end{aligned}$$

To find C , note that because $y = 2$ when $x = 0$ it follows that $1 = C$. Thus,

$$y = e^{-x} + e^{-2x}, \quad x \geq 0$$

You might wish to compare the analytic and graphical solutions by noting that the graph of the equation matches the solution shown in Figure 7.5b. ■

Integration by parts is often used in solving first-order linear differential equations. Here is an example.

Example 2 First-order linear differential equation

Find the general solution of the differential equation

$$x \frac{dy}{dx} + y = xe^{2x}, \quad x > 0$$

Solution As a first step, divide each term in the given equation by x so that we can use an integrating factor, as in Theorem 7.1.

$$\frac{dy}{dx} + \frac{y}{x} = e^{2x}$$

For this equation, $P(x) = \frac{1}{x}$ and $Q(x) = e^{2x}$. The integrating factor is

$$I(x) = e^{\int (1/x) dx} = e^{\ln x} = x$$

Thus, the general solution is

$$\begin{aligned} y &= \frac{1}{x} \left[\int xe^{2x} dx + C \right] && \text{Integrate } \int xe^{2x} dx \text{ by parts.} \\ &= \frac{1}{x} \left[\frac{1}{2} xe^{2x} - \frac{1}{4} e^{2x} + C \right] && \text{See Example 5, Section 7.2.} \\ &= \frac{1}{2} e^{2x} - \frac{e^{2x}}{4x} + \frac{C}{x} \end{aligned}$$

Applications of First-Order Equations

Modeling Logistic Growth

When the population $Q(t)$ of a colony of living organisms (humans, bacteria, etc.) is small, it is reasonable to expect the relative rate of change of the population to be constant. In other words,

$$\frac{\frac{dQ}{dt}}{Q} = k \quad \text{or} \quad \frac{dQ}{dt} = kQ$$

where k is a constant (the **unrestricted growth rate**). This is called **exponential growth**. As long as the colony has plenty of food and living space, its population will obey this unrestricted growth rate formula and $Q(t)$ will grow exponentially.

However, in practice, there often comes a time when environmental factors begin to restrict the further expansion of the colony, and at this point, the growth ceases to be purely exponential in nature. To construct a population model that takes into account the effect of diminishing resources and crowding, we assume that the population of the species has a “cap” B , called the **carrying capacity** of the species. We further assume that the rate of change in population is jointly proportional to the current population $Q(t)$ and the difference $B - Q$, which can be thought of as the potential unborn population of the species. That is,

$$\frac{\frac{dQ}{dt}}{Q} = k(B - Q) \quad \text{or} \quad \frac{dQ}{dt} = kQ(B - Q)$$

This is called a **logistic equation**, and it arises not only in connection with population models, but also in a variety of other situations. The following example illustrates one way such an equation can arise.

Example 3 Logistic equation for the spread of an epidemic

The rate at which an epidemic spreads through a community is proportional to the product of the number of residents who have been infected and the number of susceptible residents. Express the number of residents who have been infected as a function of time.

Solution Let $Q(t)$ denote the number of residents who have been infected by time t and B the total number of residents. Then the number of susceptible residents who have not been infected is $B - Q$, and the differential equation describing the spread of the epidemic is

$$\frac{dQ}{dt} = kQ(B - Q) \quad k \text{ is the constant of proportionality.}$$

$$\frac{dQ}{Q(B - Q)} = k \, dt \quad \text{Use separation of variables.}$$

$$\int \frac{dQ}{Q(B - Q)} = \int k \, dt$$

Partial fraction decomposition:

$$\int \frac{1}{B} \left[\frac{1}{Q} + \frac{1}{B - Q} \right] dQ = \int k \, dt \quad \frac{1}{B} \left(\frac{1}{Q} + \frac{1}{B - Q} \right) = \frac{1}{B} \left[\frac{B - Q + Q}{Q(B - Q)} \right]$$

$$= \frac{1}{Q(B - Q)}$$

$$\frac{1}{B} \int \frac{dQ}{Q} + \frac{1}{B} \int \frac{dQ}{B - Q} = \int k \, dt$$

$$\frac{1}{B} \ln |Q| - \frac{1}{B} \ln |B - Q| = kt + C$$

Integrate each term.

$$\frac{1}{B} \ln \left| \frac{Q}{B - Q} \right| = kt + C$$

Division property of logarithms

$$\ln \left(\frac{Q}{B - Q} \right) = Bkt + BC$$

Multiply both sides by B
(note that $B > Q > 0$).

$$\frac{Q}{B - Q} = e^{Bkt + BC}$$

Definition of logarithm

$$Q = (B - Q)e^{Bkt} e^{BC}$$

Because e^{BC} is a constant, let $A_1 = e^{BC}$.

$$Q = A_1 B e^{Bkt} - A_1 Q e^{Bkt}$$

$$Q + A_1 Q e^{Bkt} = A_1 B e^{Bkt}$$

$$Q = \frac{A_1 B e^{Bkt}}{1 + A_1 e^{Bkt}}$$

To simplify the equation, we make another substitution; let $A = \frac{1}{A_1}$, so that (after several simplification steps) we have

$$Q = \frac{\frac{B e^{Bkt}}{A}}{1 + \frac{e^{Bkt}}{A}} = \frac{B}{1 + A e^{-Bkt}}$$

The graph of $Q(t)$ is shown in Figure 7.6.

Note that the curve has an inflection point where

$$Q(t) = \frac{B}{2}$$

since

$$\begin{aligned} \frac{d^2 Q}{dt^2} &= kQ \cdot (-1) + k(B - Q) \\ &= k(B - 2Q) \\ &= 0 \text{ when } Q = \frac{B}{2} \text{ and changes sign there.} \end{aligned}$$

This corresponds to the fact that the epidemic is spreading most rapidly when half the susceptible residents have been infected (dQ/dt has a maximum there).

Note also in Example 3 that $y = Q(t)$ approaches the line $y = B$ asymptotically as $t \rightarrow \infty$. Thus, the number of infected people approaches the number of those susceptible in the long run. ■

A summary of growth models is given in Table 7.2.

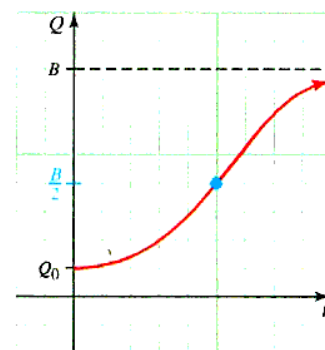
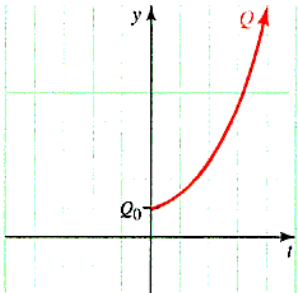
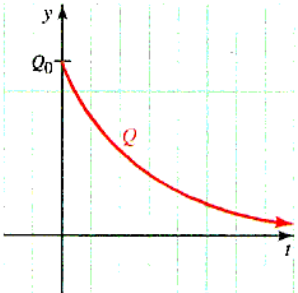
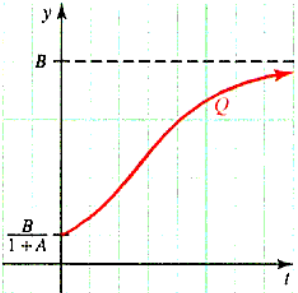
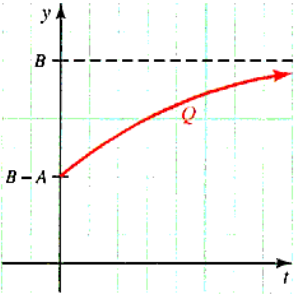


Figure 7.6 Interactive A logistic curve:

$$Q(t) = \frac{B}{1 + A e^{-Bkt}}$$

Table 7.2 Summary of Growth Models

Model	Graph	Equation and solution	Sample applications
$Q(t)$ is the population at time t , Q_0 is the initial population, and k is a constant of proportionality. If $Q(t)$ has a limiting value, let it be denoted by B .			
Uninhibited growth $(k > 0)$ Rate is proportional to the amount present.	J-curve 	$\frac{dQ}{dt} = kQ$ Solution $Q = Q_0 e^{kt}$	Exponential growth Short-term population growth Interest compounded continuously Inflation Price-supply curves
Uninhibited decay $(k < 0)$ Rate is proportional to the amount present.	L-curve 	$\frac{dQ}{dt} = kQ$ Solution $Q = Q_0 e^{kt}$	Radioactive decay Depletion of natural resources Price-demand curves
Inhibited growth or Logistic growth Rate is proportional to the amount present and to the difference between the amount present and a fixed amount.	S-curve 	$\frac{dQ}{dt} = kQ(B - Q)$ Solution $Q = \frac{B}{1 + Ae^{-Bkt}}$ A is an arbitrary constant.	Long-term population growth Sales fads Growth of a business
Limited growth $(k > 0)$ Rate is proportional to the difference between the amount present and a fixed limit.	C-curve 	$\frac{dQ}{dt} = k(B - Q)$ Solution $Q = B - Ae^{-kt}$	Learning curve Diffusion of mass media Intravenous infusion of medication Newton's law of cooling Sales of new products

Compartmental Analysis: Modeling Dilution

Example 4 A dilution problem

A tank contains 20 lb of salt dissolved in 50 gal of water. Suppose 3 gal of brine containing 2 lb of dissolved salt per gallon runs into the tank every minute and that the mixture (kept uniform by stirring) runs out of the tank at the rate of 2 gal/min. Find the amount of salt in the tank at any time t . How much salt is in the tank at the end of one hour?

Solution Let $S(t)$ denote the amount of salt in the tank at the end of t minutes. Because 3 gal of brine flows into the tank each minute and each gallon contains 2 lb of salt, it follows that $(3)(2) = 6$ lb of salt flows into the tank each minute (see Figure 7.7). This is the inflow rate.

For the outflow, note that at time t , there are $S(t)$ lb of salt and $50 + (3 - 2)t$ gallons of solution (because solution flows in at 3 gal/min and out at 2 gal/min). Thus, the concentration of salt in the solution at time t is

$$\frac{S(t)}{50 + t} \text{ lb/gal}$$

and the outflow rate of salt is

$$\left[\frac{S(t)}{50 + t} \text{ lb/gal} \right] [2 \text{ gal/min}] = \frac{2S(t)}{50 + t} \text{ lb/min}$$

Combining these observations, we see that the net rate of change of salt dS/dt is given by

$$\frac{dS}{dt} = \underbrace{6}_{\substack{\uparrow \\ \text{Inflow}}} - \underbrace{\frac{2S}{50 + t}}_{\text{Outflow}}$$

$$\frac{dS}{dt} + \frac{2S}{50 + t} = 6$$

which we recognize as a first-order linear differential equation with

$$P(t) = \frac{2}{50 + t}$$

and $Q(t) = 6$. An integrating factor is

$$I(t) = e^{\int P(t) dt} = e^{\int (2 dt)/(50+t)} = e^{2 \ln|50+t|} = (50 + t)^2$$

and the general solution is

$$\begin{aligned} S(t) &= \frac{1}{(50 + t)^2} \left[\int 6(50 + t)^2 dt + C \right] \\ &= \frac{1}{(50 + t)^2} [2(50 + t)^3 + C] \\ &= 2(50 + t) + \frac{C}{(50 + t)^2} \end{aligned}$$

To evaluate C , we recall that there are 20 lb of salt initially in the solution. This means $S(0) = 20$, so that

$$S(0) = 2(50 + 0) + \frac{C}{(50 + 0)^2}$$

$$20 = 100 + \frac{C}{50^2}$$

$$-80(50^2) = C$$

Inflow rate of salt: 6 lb/min



Outflow rate of salt:

$$\frac{S(t)}{50 + t} \text{ lb/min}$$

Figure 7.7 Rate of flow equals inflow rate minus outflow rate

Thus, the solution to the given differential equation, subject to the initial condition $S(0) = 20$, is

$$S(t) = 2(50 + t) - \frac{80(50^2)}{(50 + t)^2}$$

Specifically, at the end of 1 hour (60 min), the tank contains

$$S(60) = 2(50 + 60) - \frac{80(50)^2}{(50 + 60)^2} \approx 203.4710744$$

The tank contains about 200 lb of salt.

Modeling RL Circuits

Another application of first-order linear differential equations involves the current in an RL electric circuit. An RL circuit is one with a constant resistance, R , and a constant inductance, L . Figure 7.8 shows an electric circuit with an electromotive force (EMF), a resistor, and an inductor connected in series.

The EMF source, which is usually a battery or generator, supplies voltage that causes a current to flow in the circuit. According to Kirchhoff's second law, if the circuit is closed at time $t = 0$, then the applied electromotive force is equal to the sum of the voltage drops in the rest of the circuit. It can be shown that this implies that the current $I(t)$ that flows in the circuit at time t must satisfy the first-order linear differential equation

$$L \frac{dI}{dt} + RI = E$$

where L (the inductance), and R (the resistance) are positive constants.* Write

$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E}{L}$$

to see that $P(t) = \frac{R}{L}$ (a constant) and $Q(t) = \frac{E}{L}$ (a constant). Then

$$\begin{aligned} I(t) &= e^{-\int (R/L) dt} \left[\int \frac{E}{L} e^{\int (R/L) dt} dt + C \right] \\ &= e^{-(R/L)t} \left[\frac{E}{L} \int e^{(R/L)t} dt + C \right] \\ &= e^{-(R/L)t} \left[\frac{E}{L} \cdot \frac{L}{R} e^{(R/L)t} + C \right] \\ &= \frac{E}{R} + C e^{-(R/L)t} \end{aligned}$$

It is reasonable to assume that no current flows when $t = 0$. That is, $I = 0$ when $t = 0$, so we have $0 = E/R + C$ or $C = -E/R$. The solution of the initial value problem is

$$I = \frac{E}{R} (1 - e^{-Rt/L})$$

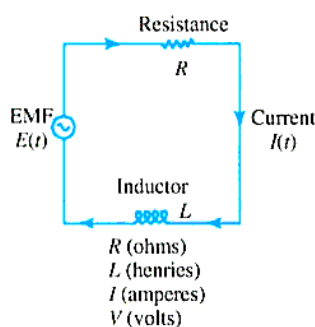


Figure 7.8 An RL circuit diagram

*It is common practice in physics and applied mathematics to use I as the symbol for current. Of course, this has nothing to do with the concept of integrating factor introduced earlier in this section.

Notice that because $e^{-(Rt/L)} \rightarrow 0$ as $t \rightarrow \infty$, we have

$$\begin{aligned}\lim_{t \rightarrow \infty} I(t) &= \lim_{t \rightarrow \infty} \frac{E}{R} (1 - e^{-Rt/L}) \\ &= \frac{E}{R} (1 - 0) \\ &= \frac{E}{R}\end{aligned}$$

This means that, in the long run, the current I must approach $\frac{E}{R}$. The solution of the differential equation consists of two parts, which are given special names:

$$\frac{E}{R} \text{ is the \textbf{steady-state current} } \quad -\frac{E}{R}e^{-Rt/L} \text{ is the \textbf{transient current}}$$

Figure 7.9 shows how the current $I(t)$ varies with time t .

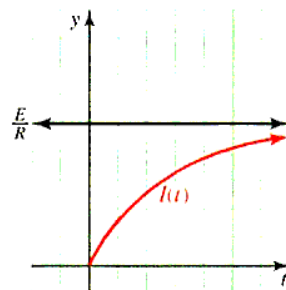


Figure 7.9 The current in an RL circuit with constant EMF

PROBLEM SET 7.6

Level 1

Solve the differential equations in Problems 1–16.

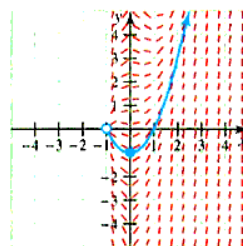
1. $\frac{dy}{dx} = x^2$
2. $\frac{dy}{dx} = 5$
3. $12 \frac{dy}{dx} = \sqrt{5x+1}$
4. $\frac{3}{2} \frac{dy}{dx} = e^{4x-3}$
5. $\frac{dP}{dt} = 0.02P$
6. $\frac{dP}{dt} = 1.25P$
7. $\frac{dy}{dx} + \frac{3y}{x} = x$
8. $\frac{dy}{dx} - \frac{2y}{x} = \sqrt{x} + 1$
9. $x^4 \frac{dy}{dx} + 2x^3y = 5$
10. $x^2 \frac{dy}{dx} + xy = 2$
11. $x \frac{dy}{dx} + 2y = xe^{x^3}$
12. $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$
13. $\frac{dy}{dx} + \frac{y}{x} = \tan^{-1}x$
14. $\frac{dy}{dx} + \frac{2y}{x} = \frac{\ln x}{x}$
15. $\frac{dy}{dx} + (\tan x)y = \sin x$
16. $\frac{dy}{dx} + (\sec x)y = \sin 2x$

Solve the differential equations in Problems 17–20 for a particular solution.

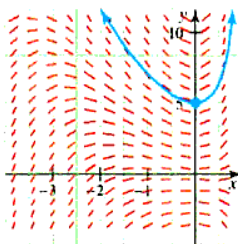
17. $\frac{dy}{dx} = x^2y^{-2}$ for $y = 5$ when $x = 0$
18. $x^2 \frac{dy}{dx} = y^{-2}$ for $y = 2$ when $x = 1$
19. $x \frac{dy}{dx} - y\sqrt{x} = 0$ for $y = 1$ when $x = 1$
20. $\frac{dy}{dx} = \frac{xy}{1+x^2}$ for $y = 2$ when $x = -1$

In Problems 21–26, solve the initial value problems. A graphical solution is shown as a check for your work.

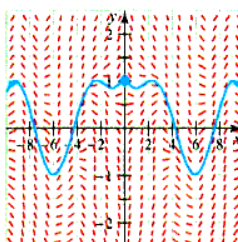
21. $\frac{dy}{dx} + \frac{xy}{1+x} = x(1+x)$, $x > -1$ with $y = -1$ when $x = 0$



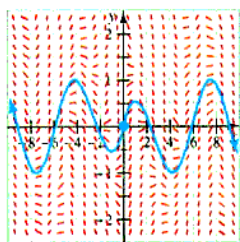
22. $\frac{dy}{dx} - \frac{xy}{1-x} = x(1+x)$, $x < 1$ with $y = 5$ when $x = 0$



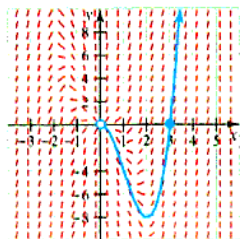
23. $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \sin x$ with $y = 1$ when $x = 0$



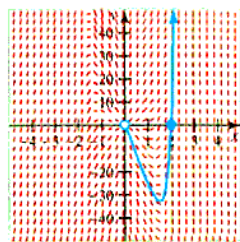
24. $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \cos x$ with $y = 0$ when $x = 0$



25. $x \frac{dy}{dx} - 2y = 2x^3$, $x > 0$ with $y = 0$ when $x = 3$



26. $\frac{dy}{dx} - \frac{y}{x} = x^2 e^{x^2}$, $x > 0$ with $y = 0$ when $x = 2$



Level 2

In Problems 27–30, find the orthogonal trajectories of the given family of curves. Recall from Section 5.6 that a curve is an orthogonal trajectory of a given family if it intersects each member of that family at right angles.

27. the family of parabolas $y^2 = 4kx$
28. the family of hyperbolas $xy = c$
29. the family of circles $x^2 + y^2 = r^2$
30. the family of exponential curves $y = Ce^{-x}$

Modeling Problems: In Problems 31–42, set up an appropriate model to answer the given question. Be sure to state your assumptions.

31. In 2010, the gross domestic product (GDP) of the United States was \$14.26 trillion. Suppose the growth rate from 2009 to 2010 was 1.80%. Predict the GDP in 2020.
32. In 2000, the gross domestic product (GDP) of the United States was \$9.38 trillion. Suppose the growth rate from 2000 to 2010 was 3.10%. Predict the GDP in 2010. Do some research to find the GDP in 2010. Comment on your prediction. Cite your sources.

33. According to the U.S. Census Bureau, the number of U.S. marriages in 2005 was 2,230,000. If the marriage rate is 0.71%, how many marriages (to the nearest ten thousand) would you expect to be reported in 2010 if the marriage rate is constant?
34. According to the U.S. Census Bureau, the number of U.S. divorces was 957,200 in 2000. If the divorce rate is 0.35%, how many divorces (to the nearest hundred) would you expect to be reported in 2010 if the divorce rate is constant?
35. A tank contains 10 lb of salt dissolved in 30 gal of water. Suppose 2 gal of brine containing 1 lb of dissolved salt per gallon runs into the tank every minute and that the mixture (kept uniform by stirring) runs out at the same rate.
 - a. Find the amount of salt in the tank at time t .
 - b. How long does it take (to the nearest second) for the tank to contain 15 lb of salt?
36. In Problem 35, suppose the tank has a capacity of 100 gal and that the mixture flows out at the rate of 1 gal/min (instead of 2 gal/min).
 - a. How long will it take for the tank to fill?
 - b. How much salt will be in the tank when it is full?
37. The rate at which a drug is absorbed into the blood system is given by

$$\frac{db}{dt} = \alpha - \beta b$$

where $b(t)$ is the concentration of the drug in the bloodstream at time t . What does $b(t)$ approach in the long run (that is, as $t \rightarrow \infty$)? At what time is $b(t)$ equal to half this limiting value? Assume that $b(0) = 0$.

38. An RL circuit has a resistance of R ohms, inductance of L henries, and EMF of E volts, where R , L , and E are constant. Suppose no current flows in the circuit at time $t = 0$. If L is doubled and E and R are held constant, what effect does this have on the “long run” current in the circuit (that is, the current as $t \rightarrow \infty$)?
39. In 2000 there were 31.1 million Hispanics in the United States. If the growth is proportional to the population, how many Hispanics should have been recorded in 2010 if the 1990 Hispanic population was 15.5 million?
40. A population of animals on Catalina Island is limited by the amount of food available. Studies show there were 1,800 animals present in 2005 and 2,000 in 2011 and suggest that 5,000 animals can be supported by the conditions present on the island. Use a logistic model to predict the animal population in the year 2030.
41. In 1986 the Chernobyl nuclear disaster in the Soviet Union contaminated the atmosphere. The buildup of radioactive material in the atmosphere satisfies the differential equation

$$\frac{dM}{dt} = r \left(\frac{k}{r} - M \right)$$

$M = 0$ when $t = 0$ where M = mass of radioactive material in the atmosphere after time t (in years); k is

the rate at which the radioactive material is introduced into the atmosphere; r is the annual decay rate of the radioactive material. Find the solution, $M(t)$, of this differential equation in terms of k and r .

42. The Motion of a Body Falling through a Resisting Medium

A body of mass m is dropped from a great height and falls in a straight line. Assume that the only forces acting on the body are the earth's gravitational attraction mg and air resistance kv . (Recall $g = -32 \text{ ft/s}^2$.)

- a. According to Newton's second law,

$$m \frac{dv}{dt} = mg - kv$$

Solve this equation, assuming the object has velocity $v_0 = 0$ at time $t = 0$.

- b. Find the distance $s(t)$ the body has fallen at time t . Assume $s = 0$ at time $t = 0$.
c. If the body weighs $W = 100 \text{ lb}$ and $k = 0.35$, how long does it take for the body to reach the ground from a height of 10,000 ft?

Recall from Section 5.6 that a tank filled with water drains at the rate

$$\frac{dV}{dt} = -4.8A_0\sqrt{h}$$

where h is the height of water at time t (in seconds) and A_0 is the area (in ft^2) of the drain hole. This formula, called Torricelli's law, is used in Problems 43 and 44.

- 43.** A full tank of water has a drain with area 0.07 ft^2 .

If the tank has a constant cross-sectional area of $A = 5 \text{ ft}^2$ and height of $h_0 = 4 \text{ ft}$, how long does it take to empty?

- 44.** A full tank of water of height 5 ft and constant cross-sectional area $A = 3 \text{ ft}^2$ has two drains, both of area 0.02 ft^2 . One drain is at the bottom and the other at height 2 ft. How long does it take for the tank to drain?

- 45. The Euler beam model.** For a rigid beam with uniform loading, the deflection $y(x)$ is modeled by the differential equation $y^{(4)} = -k$, where $k > 0$ is a constant and x measures the distance along the beam from one of its ends. Assume that $y(0) = y(L) = 0$, where L is the length of the beam, and that $y''(0) = y''(L) = 0$.

- a. Solve the beam equation to find $y(x)$.
b. Where does the maximum deflection occur? What is the maximum deflection?
c. Suppose the beam is cantilevered, so that $y(0) = y(L) = 0$ and $y''(0) = y'(L) = 0$. Now where does the maximum deflection occur? Is the maximum deflection greater or less than the case considered in part b? Would you expect the graph of

the deflection $y = y(x)$ to be concave up or concave down on $[0, L]$? Prove your conjecture.

- 46. Modeling Problem** A tank initially contains 5 lb of salt in 50 ft^3 of solution. At time $t = 0$, brine begins to enter the tank at the rate of $2 \text{ ft}^3/\text{h}$, and the mixed solution drains at the same rate. The brine coming into the tank has concentration

$$C(t) = 1 - e^{-0.2t} \text{ lb/ft}^3$$

t hours after the dilution begins.

- a. Set up and solve a differential equation for the amount of salt $S(t)$ in the solution at time t .
b. How much salt is eventually in the tank (as $t \rightarrow \infty$)?
c. At what time is $S(t)$ maximized? What is the maximum amount of salt in the tank?
47. Modeling Problem Two 100-gallon tanks initially contain pure water. Brine containing 2 lb of salt per gallon enters the first tank at the rate of 1 gal/min, and the mixed solution drains into the second tank at the same rate. There, it is again thoroughly mixed and drains at the same rate, 1 gal/min.
a. Set up and solve a differential equation of the amount of salt $S_1(t)$ in the first tank at time t (minutes).
b. Set up and solve a second differential equation for the amount of salt $S_2(t)$ in the second tank at time t .
c. Let $S(t) = S_1 - S_2$. Intuitively, $S(t) \geq 0$ for all t . At what time is the excess $S(t)$ maximized? What is the maximum excess?

- 48. Modeling Problem** A chemical in a solution diffuses from a compartment with known concentration $C_1(t)$ across a membrane to a second compartment whose concentration $C_2(t)$ changes at a rate proportional to the difference $C_1 - C_2$. Set up and solve the differential equation for $C_2(t)$ in the following cases:

- a. $C_1(t) = 5e^{-2t}$; $k = 1.7$; $C_2(0) = 0$
b. $k = 4$; $C_2(0) = 3$, and

$$C_1(t) = \begin{cases} 4 & \text{if } 0 \leq t \leq 3 \\ 5 & \text{if } t > 3 \end{cases}$$

- 49.** The Gompertz equation for a population $P(t)$ is

$$\frac{dP}{dt} = kP(B - \ln P)$$

where k and B are positive constants. If the initial population is $P(0) = P_0$, and the ultimate population is

$$\lim_{t \rightarrow \infty} P(t) = P_\infty$$

find $P(t)$.

- 50. Modeling Problem** Uranium-234 (half-life 2.48×10^5 yr) decays to thorium-230 (half-life 80,000 yr).

- a. If $U(t)$ and $T(t)$ are the amounts of uranium and thorium at time t , then

$$\frac{dU}{dt} = -k_1 U, \quad \frac{dT}{dt} = -k_2 T + k_1 U$$

Solve this system of differential equations to obtain $U(t)$ and $T(t)$.

- b. If we start with 100 g of pure U-234, how much Th-230 will there be after $t = 5,000$ yr?

51. Solve the differential equation

$$\frac{dy}{dx} = \frac{1+y}{xy + e^y(1+y)}$$

by regarding y as the independent variable (i.e., reverse the roles of x and y).

52. Certain biological processes occur periodically over the 24 hours of a day. A patient's metabolic excretion rate is modified by

$$R(t) = \frac{1}{4} - \frac{1}{8} \cos \frac{\pi}{12}(t - 6)$$

and the rate of intake is $I(t)$ for $0 \leq t \leq 24$. The patient's body contains 200 g of the substance when $t = 0$. The amount of substance $Q(t)$ in a patient's body at time t satisfies $Q(t)$ is equal to the amount of intake minus the amount of input.

- a. If $I(t) = 0$ (the patient intakes only water), the amount of substance $Q(t)$ in the patient's body at time t satisfies

$$\frac{dQ}{dt} = 0 - R(t)$$

Solve this equation for $Q(t)$.

- b. Suppose the patient intakes (in gal/hr) the substance at a constant rate for part of the day. Specifically,

$$I(t) = \begin{cases} 0.4 & \text{for } 10 \leq t \leq 20 \\ 0 & \text{otherwise} \end{cases}$$

Set up and solve a differential equation for the amount of substance $Q(t)$ in the patient's body at time t . When is $Q(t)$ maximized?

53. An RL circuit has a resistance of $R = 10$ ohms and an inductance of $L = 5$ henries. Find the current $I(t)$ in the circuit at time t if $I(0) = 0$ and the electromotive force (EMF) is

- a. $E = 15$ volts b. $E = 5e^{-2t} \sin t$

54. An RL circuit has an inductance of $L = 3$ henries and a resistance $R = 6$ ohms in series with an EMF of $E = 50 \sin 30t$. Assume $I(0) = 0$.

- a. What is the current $I(t)$ at time t ?
b. What is the transient current? The steady-state current?

55. **Modeling Problem** A lake has a volume of 6 billion ft^3 , and its initial pollutant content is 0.22%. A river whose waters contain only 0.06% pollutants flows into the lake at the rate of 350 million ft^3/day , and another river flows out of the lake also carrying 350 million ft^3/day . Assume that the water in the two rivers and the lake is always well mixed. How long does it take for the pollutant content to be reduced to 0.15%?

Level 3

56. A **Bernoulli equation** is a differential equation of the form

$$y' + P(x)y = Q(x)y^n$$

where n is a real number, $n \neq 0$, $n \neq 1$.

- a. Show that the change of variable $u = y^{1-n}$ transforms such an equation into one of the form

$$u'(x) + (1-n)P(x)u(x) = (1-n)Q(x)$$

This transformed equation is a first-order linear differential equation in u and can be solved by the methods of this section, yielding a solution to the given Bernoulli equation.

- b. Use the change of variable suggested to solve the Bernoulli equation

$$y' + \frac{y}{x} = 2y^2$$

57. Consider a curve with the property that when horizontal and vertical lines are drawn from each point $P(x, y)$ on the curve to the coordinate axes, the area A_1 under the curve is *twice* the area A_2 above the curve, as shown in Figure 7.10

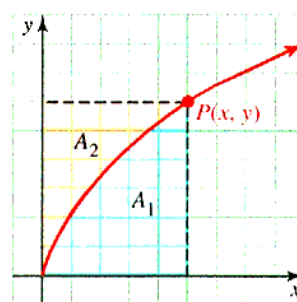


Figure 7.10 Problem 57

Show that x and y satisfy the differential equation

$$\frac{dy}{dx} = \frac{y}{2x}$$

Solve the equation and characterize the family of all curves that satisfy the given geometric condition.

58. *Historical Quest*

Daniel Bernoulli was a member of the famous Bernoulli family (see *Historical Quest* Problem 50, Section 4.5). Between 1725 and 1749 he won ten prizes for his work in astronomy, gravity, tides, magnetism, ocean currents, and the behavior of ships at sea. While modeling the effects of a smallpox epidemic, he obtained the differential equation



Daniel Bernoulli
(1700–1782)

$$\frac{dS}{dt} = -pS + \left(\frac{S}{N}\right) \frac{dN}{dt} + \frac{pS^2}{mN}$$

In this equation $S(t)$ is the number of people at age t that are susceptible to smallpox; $N(t)$ is the number of people at age t who survive; p is the probability of a susceptible person getting the disease, and $1/m$ is the proportion of those who die from the disease. Let $y = S/N$, then solve the resulting equation for $y(t)$ and then write N as a function of S . (Hint: You will need the result of Problem 56.)

59. An object of mass m is projected upward from ground level with initial velocity v_0 against air resistance proportional to its velocity $v(t)$. Thus,

$$m \frac{dv}{dt} = -mg - kv$$

- Solve this equation for $v(t)$, and then find $s(t)$ the height of the object above the ground at time t .
 - When does the object reach its maximum height? What is the maximum height?
 - Suppose $k = \frac{3}{4}$. For an object weighing 20 lb with initial velocity $v_0 = 150$ ft/s, what is the maximum height to the nearest foot? When to the nearest second does the object hit the ground?
 - Suppose the same object as in part c is launched with the same initial velocity but in a vacuum, where there is no air resistance ($k = 0$). Would you expect the object to hit the ground sooner or later than the object in part c? Prove your conjecture.
60. A body of mass m falls from a height of s_0 ft against air resistance modeled to be proportional to the square of its velocity v , so that

$$m \frac{dv}{dt} = mg - kv^2$$

- Solve this differential equation to find $v(t)$. Then find the height $s(t)$ of the object at time t (in seconds).
- If the object weighs 1,000 lb, $s_0 = 100$ ft, and $k = 0.01$, how long to the nearest hundredth of a second does it take for the object to hit the ground? Interpret this result.

7.7 IMPROPER INTEGRALS

IN THIS SECTION: Improper integrals with infinite limits of integration, improper integrals with unbounded integrands

We have defined the definite integral $\int_a^b f(x) dx$ on a closed bounded interval $[a, b]$, where the integrand $f(x)$ is bounded. In this section, we extend the concept of integral to the case where the interval of integration is infinite and also to the case where f is unbounded at a finite number of points on the interval of integration. Collectively, these are called **improper integrals**.

Improper Integrals with Infinite Limits of Integration

In physics, economics, probability and statistics, and other applied areas, it is useful to have a concept of an integral that is defined on the entire real line or on half-lines of the form $x \geq a$ or $x \leq a$. If $f(x) \geq 0$, the integral of f on the interval $x \geq a$ can be thought of as the area under the curve $y = f(x)$ on this unbounded interval, as shown in Figure 7.11.

A reasonable strategy for finding this area is first to use a definite integral to compute the area from $x = a$ to some finite number $x = N$ and then to let N approach infinity in the resulting expression. Here is a definition.

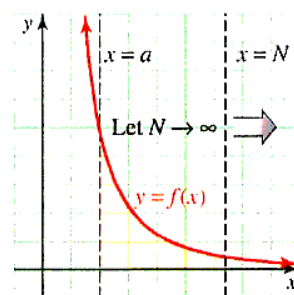


Figure 7.11 Area of an unbounded region

IMPROPER INTEGRAL TYPE I Let a be a fixed number and assume $\int_a^N f(x) dx$ exists for all $N \geq a$. Then if $\lim_{N \rightarrow \infty} \int_a^N f(x) dx$ exists, we define the **improper integral** $\int_a^\infty f(x) dx$ by

$$\int_a^\infty f(x) dx = \lim_{N \rightarrow \infty} \int_a^N f(x) dx$$

The improper integral is said to **converge** if this limit is a finite number and to **diverge** otherwise.

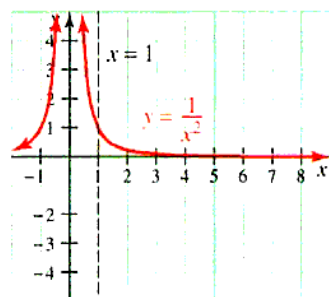


Figure 7.12 Graph of $y = \frac{1}{x^2}$

Example 1 Convergent improper integral

Evaluate $\int_1^\infty \frac{dx}{x^2}$.

Solution The region under the curve $y = 1/x^2$ for $x \geq 1$ is shown in Figure 7.12.

This region is unbounded, so it might seem reasonable to conclude that the area is also infinite. Begin by computing the integral from 1 to N :

$$\int_1^N \frac{dx}{x^2} = -\frac{1}{x} \Big|_1^N = -\frac{1}{N} + 1$$

Let us consider the situation more carefully, as shown in Figure 7.13.

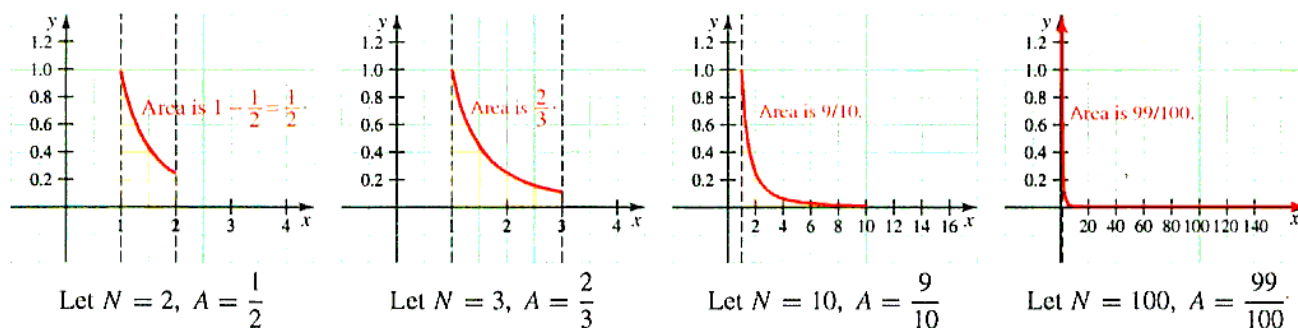


Figure 7.13 Area enclosed by $y = \frac{1}{x^2}$, the x -axis, $x = 1$, and $x = N$

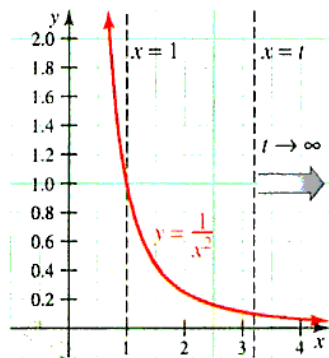


Figure 7.14 The region to the right of $x = 1$ bounded by the curve $y = \frac{1}{x^2}$ and the x -axis, has area 1

For $N = 2, 3, 10$, or 100 , this integral has values $\frac{1}{2}$, $\frac{2}{3}$, $\frac{9}{10}$, and $\frac{99}{100}$, respectively, suggesting that the region under $y = \frac{1}{x^2}$ for $x \geq 1$ actually has finite area approaching 1 as N gets larger and larger.

To find this area analytically, we take the limit as $N \rightarrow \infty$, as shown in Figure 7.14. Thus, the improper integral converges and has the value 1.

Example 2 A divergent improper integral

Evaluate $\int_1^\infty \frac{dx}{x}$.

Solution The graph of $y = \frac{1}{x}$, shown in Figure 7.15, looks very much like the graph of $y = \frac{1}{x^2}$ in Figure 7.14.

We compute the integral from 1 to N and then let N go to infinity.

$$\begin{aligned}\int_1^\infty \frac{dx}{x} &= \lim_{N \rightarrow \infty} \int_1^N \frac{dx}{x} \\ &= \lim_{N \rightarrow \infty} \ln|x| \Big|_1^N \\ &= \lim_{N \rightarrow \infty} \ln N \\ &= \infty\end{aligned}$$

The limit is not a finite number, which means the improper integral diverges.

We have shown that the improper integral $\int_1^\infty \frac{dx}{x^2}$ converges and the improper integral $\int_1^\infty \frac{dx}{x}$ diverges. In geometric terms, this says that the area to the right of $x = 1$ under the curve $y = \frac{1}{x^2}$ is finite, whereas the corresponding area under the curve $y = \frac{1}{x}$ is infinite. The reason for the difference is that as x increases, $\frac{1}{x^2}$ approaches zero more quickly than does $\frac{1}{x}$, as shown in Figure 7.16.

Example 3 Determine convergence of an improper integral

Show that the improper integral $\int_1^\infty \frac{dx}{x^p}$ converges only for constant $p > 1$.

Solution For this example, remember that p is a constant and x is the variable. We have seen in Example 1 that the integral converges for $p = 2$. We also know from Example 2 that the integral diverges for $p = 1$, so we can assume $p \neq 1$ in the following computation:

$$\begin{aligned}\int_1^\infty \frac{dx}{x^p} &= \lim_{N \rightarrow \infty} \int_1^N \frac{dx}{x^p} \\ &= \lim_{N \rightarrow \infty} \left[\frac{x^{-p+1}}{-p+1} \right]_1^N \\ &= \lim_{N \rightarrow \infty} \frac{1}{1-p} [N^{1-p} - 1]\end{aligned}$$

If $p > 1$, then $1 - p < 0$, so $N^{1-p} \rightarrow 0$ as $N \rightarrow \infty$, and

$$\int_1^\infty \frac{dx}{x^p} = \lim_{N \rightarrow \infty} \frac{1}{1-p} [N^{1-p} - 1] = \frac{1}{1-p} [-1] = \frac{1}{p-1}$$

If $p < 1$, then $1 - p > 0$, so $N^{1-p} \rightarrow \infty$ as $N \rightarrow \infty$, and

$$\int_1^\infty \frac{dx}{x^p} = \lim_{N \rightarrow \infty} \frac{1}{1-p} [N^{1-p} - 1] = \infty$$

Thus, the improper integral diverges for $p \leq 1$ and converges for $p > 1$.

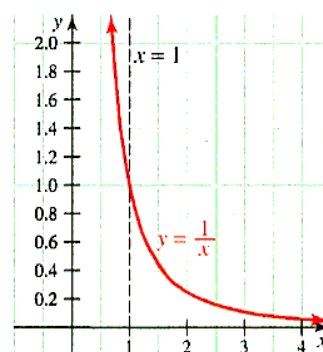


Figure 7.15 The area to the right of the line $x = 1$, bounded by the curve $y = \frac{1}{x}$ and the x -axis, is infinite

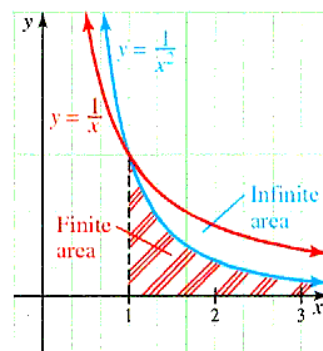


Figure 7.16 An unbounded region may have either a finite or an infinite area

■ **What this says** The improper integral

$$\int_1^\infty \frac{dx}{x^p} = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \\ \infty & \text{if } p \leq 1 \end{cases}$$

Example 4 Improper integral using l'Hôpital's rule and integration by parts

Evaluate $\int_0^{\infty} xe^{-2x} dx$.

Solution

$$\int_0^{\infty} xe^{-2x} dx = \lim_{N \rightarrow \infty} \int_0^N \underbrace{x}_u \underbrace{e^{-2x}}_{dv} dx$$

By parts,
Let $u = x$, $dv = e^{-2x} dx$
 $du = dx$ $v = \frac{1}{-2} e^{-2x}$

$$= \lim_{N \rightarrow \infty} \left[\left(\underbrace{x}_u \underbrace{\frac{e^{-2x}}{-2}}_v \right) \Big|_0^N - \int_0^N \underbrace{\frac{1}{-2}}_v \underbrace{e^{-2x}}_{du} dx \right]$$

$$= \lim_{N \rightarrow \infty} \left[\frac{-xe^{-2x}}{2} - \frac{e^{-2x}}{4} \right]_0^N$$

$$= \lim_{N \rightarrow \infty} \left[-\frac{1}{2} Ne^{-2N} - \frac{1}{4} e^{-2N} + 0 + \frac{1}{4} \right]$$

$$\lim_{N \rightarrow \infty} e^{-2N} = 0$$

$$= -\frac{1}{2} \lim_{N \rightarrow \infty} \left(\frac{N}{e^{2N}} \right) + \frac{1}{4}$$

l'Hôpital's rule

$$= -\frac{1}{2} \lim_{N \rightarrow \infty} \left(\frac{1}{2e^{2N}} \right) + \frac{1}{4}$$

$$= \frac{1}{4}$$

Example 5 Gabriel's horn: a solid with a finite volume but infinite surface area

Gabriel's horn is the name given to the solid formed by revolving about the x -axis the unbounded region under the curve $y = \frac{1}{x}$ for $x \geq 1$. Show that this solid has finite volume but infinite surface area.

You can fill Gabriel's horn with a finite amount of paint, but it takes an infinite amount of paint to color its inside surface!

Solution We will find the volume, V , by using disks, as shown in Figure 7.17.

$$V = \pi \int_1^{\infty} (x^{-1})^2 dx = \pi \lim_{N \rightarrow \infty} \int_1^N x^{-2} dx = \pi \lim_{N \rightarrow \infty} \left(-\frac{1}{N} + 1 \right) = \pi$$

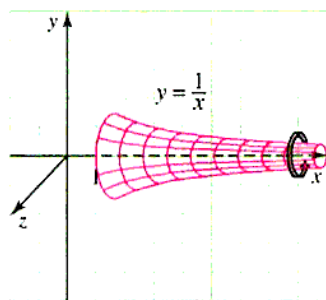


Figure 7.17 Interactive
Gabriel's horn

We also find the surface area, S :

$$S = 2\pi \int_1^{\infty} f(x) \sqrt{1 + [f'(x)]^2} dx$$

$$= 2\pi \int_1^{\infty} \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx$$

$$= 2\pi \int_1^{\infty} \frac{1}{x} \sqrt{\frac{x^4 + 1}{x^4}} dx$$

Let $u = x^2$, then $du = 2x dx$

$$= 2\pi \int_1^{\infty} \frac{\sqrt{u^2 + 1}}{2u^2} du$$

$$= \pi \lim_{N \rightarrow \infty} \int_1^N \frac{\sqrt{u^2 + 1}}{u^2} du$$

$$= \pi \lim_{N \rightarrow \infty} \left[\frac{-\sqrt{u^2 + 1}}{u} + \ln(u + \sqrt{u^2 + 1}) \right]_1^N \quad \text{Formula 97}$$

$$= \infty$$

We can also define an improper integral on an interval that is unbounded to the left or on the entire real number line.

IMPROPER INTEGRAL TYPE I (Extended) Let b be a fixed number and assume $\int_t^b f(x) dx$ exists for all $t < b$. Then if $\lim_{t \rightarrow -\infty} \int_t^b f(x) dx$ exists, we define the **improper integral**

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

The improper integral $\int_{-\infty}^b f(x) dx$ is said to **converge** if this limit is a finite number and to **diverge** otherwise. If both

$$\int_a^{\infty} f(x) dx \quad \text{and} \quad \int_{-\infty}^a f(x) dx$$

converge for some number a , the improper integral of $f(x)$ on the entire x -axis is defined by

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$$

We say the integral on the left converges if both the integrals on the right converge, and it diverges otherwise.

Example 6 Improper integrals

Evaluate the following integrals.

a. $\int_1^{\infty} 2x^{-3} dx$ b. $\int_{-\infty}^{-3} \frac{dx}{x}$ c. $\int_{-\infty}^{\infty} xe^{-x^2} dx$

Solution

a. This is an improper integral at the upper limit.

$$\begin{aligned} \int_1^{\infty} 2x^{-3} dx &= \lim_{b \rightarrow \infty} \int_1^b 2x^{-3} dx \\ &= \lim_{b \rightarrow \infty} (-x^{-2}) \Big|_1^b \\ &= \lim_{b \rightarrow \infty} (-b^{-2} + 1) \\ &= 1 \end{aligned}$$

b. This is an improper integral at the lower limit.

$$\begin{aligned} \int_{-\infty}^{-3} \frac{dx}{x} &= \lim_{a \rightarrow -\infty} \int_a^{-3} x^{-1} dx \\ &= \lim_{a \rightarrow -\infty} (\ln |x|) \Big|_a^{-3} \\ &= \lim_{a \rightarrow -\infty} (\ln 3 - \ln |a|) \end{aligned}$$

This limit does not exist, so we say that the integral diverges.

- c. This is an improper integral at both limits, so we choose some point, say 0, between $-\infty$ and ∞ .

Be sure that when an integral is improper for more than one reason (in this case because it starts at minus infinity and ends at (positive) infinity), you must break the integral into pieces. Each of these improper integrals must converge for the entire integral to converge.

$$\begin{aligned}
 \int_{-\infty}^{\infty} xe^{-x^2} dx &= \int_{-\infty}^0 xe^{-x^2} dx + \int_0^{\infty} xe^{-x^2} dx \\
 &= \lim_{a \rightarrow -\infty} \int_a^0 xe^{-x^2} dx + \lim_{b \rightarrow \infty} \int_0^b xe^{-x^2} dx \\
 &= \lim_{a \rightarrow -\infty} \int_{x=a}^{x=0} \frac{-1}{2} e^u du + \lim_{b \rightarrow \infty} \int_{x=0}^{x=b} \frac{-1}{2} e^u du && \boxed{\text{Let } u = -x^2 \\ du &= -2x \, dx} \\
 &= \lim_{a \rightarrow -\infty} \left(-\frac{1}{2} e^{-x^2} \right) \Big|_a^0 + \lim_{b \rightarrow \infty} \left(-\frac{1}{2} e^{-x^2} \right) \Big|_0^b \\
 &= \lim_{a \rightarrow -\infty} \left(-\frac{1}{2} + \frac{1}{2} e^{-a^2} \right) + \lim_{b \rightarrow \infty} \left(-\frac{1}{2} e^{-b^2} + \frac{1}{2} \right) \\
 &= -\frac{1}{2} + 0 - 0 + \frac{1}{2} \\
 &= 0
 \end{aligned}$$

Example 7 Improper integral to negative infinity

- a. Evaluate $\int_{-\infty}^0 \frac{dx}{1+x^2}$.
- b. How would you expect the value of $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ to be related to the improper integral in a? Verify your conjecture by evaluating this integral.

Solution

$$\begin{aligned}
 \text{a. } \int_{-\infty}^0 \frac{dx}{1+x^2} &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{dx}{1+x^2} \\
 &= \lim_{t \rightarrow -\infty} \tan^{-1} x \Big|_t^0 \\
 &= \lim_{t \rightarrow -\infty} [\tan^{-1} 0 - \tan^{-1} t] \\
 &= 0 - \left(-\frac{\pi}{2} \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

- b. Since $y(x) = y(-x)$, we see that $y = \frac{1}{1+x^2}$ is symmetric about the y-axis, and it is reasonable to conjecture that the integral in part b is twice the integral in part a. We have

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{dx}{1+x^2} &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{dx}{1+x^2} + \lim_{N \rightarrow \infty} \int_0^N \frac{dx}{1+x^2} \\
 &= \frac{\pi}{2} + \lim_{N \rightarrow \infty} (\tan^{-1} N - \tan^{-1} 0) \\
 &= \frac{\pi}{2} + \left(\frac{\pi}{2} - 0 \right) \\
 &= \pi
 \end{aligned}$$

We see this integral is indeed twice that in part a, as conjectured.

Example 8 An improper polar integral

Evaluate $\int_0^\infty \frac{1}{2} e^{-2\theta} d\theta$.

$$\begin{aligned}
 \text{Solution } \int_0^\infty \frac{1}{2} e^{-2\theta} d\theta &= \lim_{N \rightarrow \infty} \int_0^N \frac{1}{2} e^{-2\theta} d\theta \\
 &= \lim_{N \rightarrow \infty} \left(-\frac{1}{4} e^{-2\theta} \right) \Big|_0^N \\
 &= \lim_{N \rightarrow \infty} \left[-\frac{1}{4} e^{-2N} - \left(-\frac{1}{4} \right) \right] \\
 &= \frac{1}{4}
 \end{aligned}$$

Improper Integrals with Unbounded Integrands

A function f is **unbounded** at c if it has arbitrarily large values near c . Geometrically, this occurs when the line $x = c$ is a vertical asymptote to the graph of f at c , as shown in Figure 7.18.

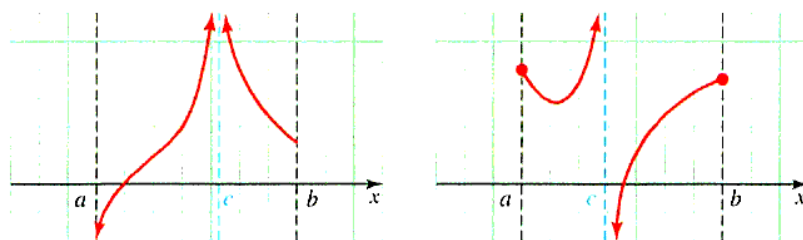


Figure 7.18 Two functions that are unbounded at c

If f is unbounded at c and $a \leq c \leq b$, the Riemann integral $\int_a^b f(x) dx$ is not even defined (only bounded functions are Riemann-integrable). However, it may still be possible to define $\int_a^b f(x) dx$ as an improper integral in certain cases.

Let us examine a specific problem. Suppose

$$f(x) = \frac{1}{\sqrt{x}} \quad \text{for } 0 < x \leq 1$$

Then f is unbounded at $x = 0$ and $\int_0^1 f(x) dx$ is not defined. However,

$$f(x) = \frac{1}{\sqrt{x}}$$

is continuous on every interval $[t, 1]$ for $t > 0$, as shown in Figure 7.19.

For any such interval $[t, 1]$, we have

$$\int_t^1 \frac{dx}{\sqrt{x}} = \int_t^1 x^{-1/2} dx = 2\sqrt{x} \Big|_t^1 = 2 - 2\sqrt{t}$$

If we let $t \rightarrow 0$ through positive values, we see that

$$\lim_{t \rightarrow 0^+} \int_t^1 \frac{dx}{\sqrt{x}} = \lim_{t \rightarrow 0^+} (2 - 2\sqrt{t}) = 2$$

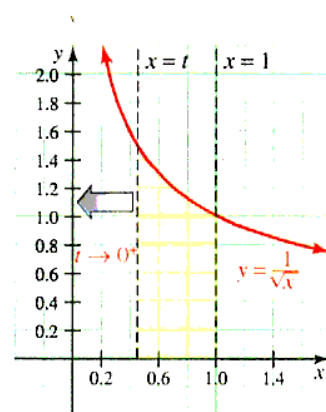


Figure 7.19 Interactive

Graph of $y = \frac{1}{\sqrt{x}}$

This is called a **convergent improper integral** with value 2, and it seems reasonable to say

$$\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{t \rightarrow 0^+} \int_t^1 \frac{dx}{\sqrt{x}} = 2$$

In this example, f is unbounded at the left endpoint of the interval of integration, but similar reasoning would apply if it were unbounded at the right endpoint or at an interior point. Here is a definition of this kind of improper integral.

IMPROPER INTEGRAL TYPE II If f is unbounded at a and $\int_t^b f(x) dx$ exists for all t such that $a < t \leq b$, then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

If the limit exists (as a finite number), we say that the improper integral **converges**; otherwise, the improper integral **diverges**. Similarly, if f is unbounded at b and $\int_a^t f(x) dx$ exists for all t such that $a \leq t < b$, then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

If f is unbounded at c , where $a < c < b$, and the improper integrals $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ both converge, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

We say that the integral on the left diverges if *either* (or both) of the integrals on the right diverges.

■ **What this says:** If a function is continuous on the open interval (a, b) but is unbounded at one of its endpoints of the interval, then replace that endpoint by t , evaluate the integral, and take the limit as t approaches that endpoint. On the other hand, if f is continuous on the interval $[a, b]$, except for some c in (a, b) where f has an infinite discontinuity, rewrite the integral as

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

You will then need limits to evaluate each of the integrals on the right.

Example 9 Improper integral at a right endpoint

Evaluate $\int_0^1 \frac{dx}{(x-1)^{2/3}}$.

Solution $f(x) = (x-1)^{-2/3}$ is unbounded at the right endpoint of the interval of integration and is continuous on $[0, t]$ for any t with $0 \leq t < 1$. We find that

$$\begin{aligned} \int_0^1 \frac{dx}{(x-1)^{2/3}} &= \lim_{t \rightarrow 1^-} \int_0^t \frac{dx}{(x-1)^{2/3}} \\ &= \lim_{t \rightarrow 1^-} \left[3(x-1)^{1/3} \right]_0^t \\ &= 3 \lim_{t \rightarrow 1^-} [(t-1)^{1/3} - (-1)] \\ &= 3 \end{aligned}$$

That is, the improper integral converges and has the value 3.

Example 10 Improper integral at a left endpoint

Evaluate $\int_{\pi/2}^{\pi} \sec x \, dx$.

Solution Because $\sec x$ is unbounded at the left endpoint $\frac{\pi}{2}$ of the interval of integration and is continuous on $[t, \pi]$ for any t with $\frac{\pi}{2} < t \leq \pi$, we find that

$$\begin{aligned} \int_{\pi/2}^{\pi} \sec x \, dx &= \lim_{t \rightarrow (\pi/2)^+} \int_t^{\pi} \sec x \, dx \\ &= \lim_{t \rightarrow (\pi/2)^+} \ln |\sec x + \tan x| \Big|_t^{\pi} \\ &= \lim_{t \rightarrow (\pi/2)^+} [\ln |-1 + 0| - \ln |\sec t + \tan t|] \\ &= -\infty \end{aligned}$$

Thus, the integral diverges. ■

Example 11 Improper integral at an interior point

Evaluate $\int_0^3 (x-2)^{-1} \, dx$.

Solution The integral is improper because the integrand is unbounded at $x = 2$. If the improper integral converges, we have

$$\begin{aligned} \int_0^3 (x-2)^{-1} \, dx &= \int_0^2 (x-2)^{-1} \, dx + \int_2^3 (x-2)^{-1} \, dx \\ &= \lim_{t \rightarrow 2^-} \int_0^t (x-2)^{-1} \, dx + \lim_{t \rightarrow 2^+} \int_t^3 (x-2)^{-1} \, dx \\ &= \lim_{t \rightarrow 2^-} \ln |x-2| \Big|_0^t + \lim_{t \rightarrow 2^+} \int_t^3 (x-2)^{-1} \, dx \\ &= \lim_{t \rightarrow 2^-} [\ln |t-2| - \ln 2] + \lim_{t \rightarrow 2^+} \int_t^3 (x-2)^{-1} \, dx \\ &= -\infty + \lim_{t \rightarrow 2^+} \int_t^3 (x-2)^{-1} \, dx \end{aligned}$$

If either of these limits fails to exist, then the original integral diverges.

The original integral diverges. ■

✖ A common mistake is to fail to notice the discontinuity at $x = 2$. It is **WRONG** to write

$$\begin{aligned} \int_0^3 (x-2)^{-1} \, dx &= \ln |x-2| \Big|_0^3 \\ &= \ln 1 - \ln 2 = -\ln 2 \end{aligned}$$

It is also **WRONG** to write

$$\begin{aligned} \int_0^3 (x-2)^{-1} \, dx &= \int_0^2 (x-2)^{-1} \, dx + \int_2^3 (x-2)^{-1} \, dx \\ &= \ln |x-2| \Big|_0^2 + \ln |x-2| \Big|_2^3 \\ &= \ln 0 - \ln 2 + \ln 1 - \ln 0 \\ &= -\ln 2 + \infty - \infty, \text{ since } \end{aligned}$$

$\infty - \infty$ is indeterminate. Notice that either of these two mistakes leads to the conclusion that the integral converges, which, as you can see, is incorrect.

You must also be cautious in using computer software with improper integrals, because it may not detect that the integral is improper. ✖

Comparison Test for Convergence or Divergence

Sometimes it is difficult or even impossible to find the exact value of an improper integral, yet it is important to know whether or not it converges. If it converges we can often use numerical techniques to estimate its value, but first we must find out whether it converges. In those cases, the following result, stated for Improper Integrals of the First Type, can be useful. A similar theorem for Improper Integrals of the Second Type can be stated.

Theorem 7.2 Comparison test for improper integrals

Suppose f and g are continuous functions such that $f(x) \geq g(x) \geq 0$ for $x \geq a$. If $\int_a^\infty f(x) dx$ converges, then $\int_a^\infty g(x) dx$ converges to a value less than or equal to the integral of $f(x)$. If $\int_a^\infty g(x) dx$ diverges then $\int_a^\infty f(x) dx$ diverges.

Proof: We omit the proof of this but it is similar to the proof of the dominance rule (Theorem 5.6 in Section 5.3). See the illustration in the next example for an indication why it works. ♦

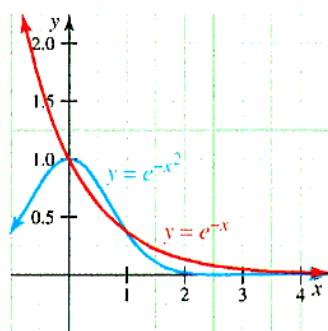


Figure 7.20 The graph of $y = e^{-x^2}$ is under the graph of $y = e^{-x}$ for $x > 1$

Example 12 Comparison test

Show that $\int_1^\infty e^{-x^2} dx$ converges.

Solution We can't evaluate this integral directly as the antiderivative of e^{-x^2} cannot be stated as an elementary function. However, $e^{-x^2} < e^{-x}$ for $x > 1$ (see Figure 7.20), so

$$\int_1^\infty e^{-x^2} dx < \int_1^\infty e^{-x} dx = \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx = \lim_{t \rightarrow \infty} (-e^{-t} + e^{-1}) = e^{-1}$$

Notice from Figure 7.20 that the graph of $y = e^{-x^2}$ is under the graph of $y = e^{-x}$ for $x > 1$, so that Theorem 7.2 gives the result. ■

PROBLEM SET 7.7**Level 1**

- What does this say? What is an improper integral?
- What does this say? Discuss the different types of improper integrals.

In Problems 3–42, either show that the improper integral converges and find its value, or show that it diverges.

- $\int_1^\infty \frac{dx}{x^3}$
- $\int_1^\infty \frac{dx}{\sqrt[3]{x}}$
- $\int_1^\infty \frac{dx}{x^{0.99}}$
- $\int_1^\infty \frac{dx}{\sqrt{x}}$
- $\int_1^\infty \frac{dx}{x^{1.01}}$
- $\int_1^\infty x^{-2/3} dx$
- $\int_3^\infty \frac{dx}{2x-1}$
- $\int_3^\infty \frac{dx}{\sqrt[3]{2x-1}}$
- $\int_3^\infty \frac{dx}{(2x-1)^2}$
- $\int_0^\infty e^{-x} dx$
- $\int_1^\infty \frac{x^2 dx}{(x^3+2)^2}$
- $\int_1^\infty \frac{x^2 dx}{\sqrt{x^3+2}}$
- $\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$
- $\int_0^\infty 5xe^{10-x} dx$
- $\int_1^\infty \frac{x^2 dx}{x^3+2}$
- $\int_0^\infty xe^{-x^2} dx$
- $\int_0^\infty xe^{-x} dx$
- $\int_1^\infty \frac{\ln x dx}{x}$
- $\int_2^\infty \frac{dx}{x \ln x}$
- $\int_{-\infty}^0 \frac{2x dx}{x^2+1}$
- $\int_{-\infty}^0 \frac{dx}{\sqrt{2-x}}$
- $\int_{-\infty}^\infty xe^{-|x|} dx$
- $\int_0^1 \frac{dx}{x^{1/5}}$
- $\int_0^1 \frac{dx}{(1-x)^{1/2}}$
- $\int_{-1}^1 \frac{e^x}{\sqrt[3]{1-e^x}} dx$
- $\int_0^1 \ln x dx$
- $\int_e^\infty \frac{dx}{x(\ln x)^2}$
- $\int_0^1 e^{-\frac{1}{2} \ln x} dx$
- $\int_0^{\pi/3} \frac{\sec^2 x dx}{1-\tan x}$
- $\int_2^\infty \frac{dx}{x\sqrt{\ln x}}$
- $\int_1^\infty \frac{x dx}{(1+x^2)^2}$
- $\int_{-\infty}^4 \frac{dx}{(5-x)^2}$
- $\int_{-\infty}^\infty \frac{dx}{x^2+1}$
- $\int_0^4 \frac{dx}{x\sqrt{x}}$
- $\int_0^2 \frac{dx}{(1-x)^2}$
- $\int_{-\infty}^\infty \frac{3x dx}{(3x^2+2)^3}$
- $\int_1^\infty \ln x dx$
- $\int_0^1 \frac{x dx}{1-x^2}$
- $\int_0^\infty \frac{dx}{e^x + e^{-x}}$
- $\int_0^{\pi/2} \frac{\sin x dx}{\sqrt[3]{1-2\cos x}}$

Level 2

43. Find the area of the unbounded region between the x -axis and the curve

$$y = \frac{2}{(x-4)^3}$$

for $x \geq 6$.

44. Find the area of the unbounded region between the x -axis and the curve

$$y = \frac{2}{(x-4)^3}$$

for $x \leq 2$.

45. The total amount of radioactive material present in the atmosphere at time T is modeled by

$$A = \int_0^T P e^{-rt} dt$$

where P is a constant and t is the number of years. Suppose a recent United Nations publication indicates that, at the present time, $r = 0.002$ and $P = 200$ millirads. Estimate the total future buildup of radioactive material in the atmosphere if these values remain constant.

46. Suppose that an oil well produces $P(t)$ thousand barrels of crude oil per month according to the formula

$$P(t) = 100e^{-0.02t} - 100e^{-0.1t}$$

where t is the number of months the well has been in production. What is the total amount of oil produced by the oil well?

47. Find all values of p for which

$$\int_2^{\infty} \frac{dx}{x(\ln x)^p}$$

converges, and find the value of the integral when it exists.

48. Find all values of p for which

$$\int_0^1 \frac{dx}{x^p}$$

converges, and find the value of the integral when it exists.

49. Find all values of p for which

$$\int_0^{1/2} \frac{dx}{x(\ln x)^p}$$

converges, and find the value of the integral when it exists.

50. **Think Tank Problem** Discuss the calculation

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x^2} &= \left. \frac{-1}{x} \right|_{-1}^1 \\ &= -[1 - (-1)] \\ &= -2 \end{aligned}$$

Is the calculation correct? Explain.

51. **Journal Problem** (*College Mathematics Journal**) Peter Lindstrom of North Lake College in Irving, Texas, had a student who handled an ∞/∞ form as follows:

$$\begin{aligned} \int_1^{\infty} (x-1)e^{-x} dx &= \int_1^{\infty} \frac{x-1}{e^x} dx \\ &= \int_1^{\infty} \frac{1}{e^x} dx \quad \text{L'Hôpital's rule} \\ &= \frac{1}{e} \end{aligned}$$

What is wrong, if anything, with this student's solution?

52. Find $\int_0^2 f(x) dx$, where

$$f(x) = \begin{cases} \frac{1}{\sqrt[4]{x^3}} & \text{for } 0 \leq x \leq 1 \\ \frac{1}{\sqrt[4]{(x-1)^3}} & \text{for } 1 < x < 2 \end{cases}$$

Level 3

The **Laplace transform** of the function f is defined by the improper integral

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

where s is a constant. This notation is used in Problems 53-60.

53. Show that $\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}$.

54. Show that for constant a (with $s - a > 0$):

a. $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

b. $\mathcal{L}\{a\} = \frac{a}{s}$

c. $\mathcal{L}\{t\} = \frac{1}{s^2}$

d. $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

e. $\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$

f. $\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$

*Vol. 24, No. 4, September, 1993, p. 343.

55. If $F(s) = \mathcal{L}\{f(t)\}$, show that

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

56. Let $\mathcal{L}^{-1}\{F(s)\}$ denote the *inverse* Laplace transform of $F(s)$; that is, the function $f(t)$ such that $\mathcal{L}\{f(t)\} = F(s)$. Find the following inverse Laplace transformations (Problems 53 and 54 may help).

a. $\mathcal{L}^{-1}\left\{\frac{5}{s}\right\}$ b. $\mathcal{L}^{-1}\left\{\frac{s+2}{s^2+4}\right\}$
 c. $\mathcal{L}^{-1}\left\{\frac{2s^2-3s+3}{s^2(s-1)}\right\}$

Hint: Use partial fraction decomposition.

d. $\mathcal{L}^{-1}\left\{\frac{3s^3+2s^2-3s-17}{(s^2+4)(s^2-1)}\right\}$

57. a. If $F(s) = \mathcal{L}\{f(t)\}$, show that

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

You may assume that

$$\frac{d}{ds} \int_0^\infty e^{-st}f(t)dt = \int_0^\infty \frac{d}{ds} [e^{-st}f(t)] dt$$

- b. Use the result to find $\mathcal{L}\{t \cos 2t\}$.

58. Use the result in Problem 55, along with those of Problem 54, to find the following Laplace transforms and inverse transforms.

a. $\mathcal{L}\{t^3 e^{-2t}\}$ b. $\mathcal{L}\{e^{-3t} \cos 2t\}$
 c. $\mathcal{L}^{-1}\left\{\frac{5}{(s-1)^2}\right\}$ d. $\mathcal{L}^{-1}\left\{\frac{4s}{s^2+4s+5}\right\}$

Hint: Complete the square.

59. What is $\mathcal{L}\{t^2 f(t)\}$? What about $\mathcal{L}\{t^n f(t)\}$?

60. Let $\mathcal{L}\{f\} = F(s)$. Suppose that f is continuous on $[0, \infty)$ and that f' is piecewise continuous on $[0, k]$ for every positive number k . Further suppose that

$$\lim_{k \rightarrow \infty} e^{-sk} f(k) = 0$$

if $s > 0$.

- a. Show that $\mathcal{L}\{f'\} = sF(s) - f(0)$.

Hint: Use integration by parts.

- b. Apply the result in part a to find $\mathcal{L}\{y\}$, where y satisfies the initial value problem

$$y' + 3y = \sin 2t$$

with $y(0) = 2$.

- c. Solve the initial value problem in part b by using inverse Laplace transforms to obtain y .

7.8 HYPERBOLIC AND INVERSE HYPERBOLIC FUNCTIONS

IN THIS SECTION: *Hyperbolic functions, derivatives and integrals involving hyperbolic functions, inverse hyperbolic functions*

Hyperbolic and inverse hyperbolic functions are introduced and studied; integration formulas to be used throughout the rest of the text are derived.

Hyperbolic Functions

In physics, it is shown that a heavy, flexible cable (for example, a power line) that is suspended between two points at the same height assumes the shape of a curve called a **catenary** (see Figure 7.21), with an equation of the form

$$y = \frac{a}{2}(e^{x/a} + e^{-x/a})$$

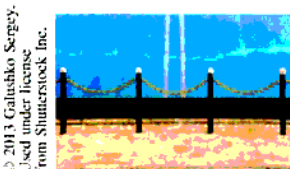


Figure 7.21 The hanging cable problem

This is one of several important applications that involve sums of exponential functions. The goal of this section is to study such sums and their inverses.

In certain ways, the functions we shall study are analogous to the trigonometric functions, and they have essentially the same relationship to the hyperbola that the trigonometric functions have to the circle. For this reason, these functions are called **hyperbolic functions**. Three basic hyperbolic functions are the **hyperbolic sine** (denoted $\sinh x$ and pronounced “cinch”), the **hyperbolic cosine** ($\cosh x$; pronounced “kosh”), and the **hyperbolic tangent** ($\tanh x$; pronounced “tansh”). They are defined as follows.

HYPERBOLIC FUNCTIONS

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{for all } x$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \text{for all } x$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \text{for all } x$$

Graphs of the three basic hyperbolic functions are shown in Figure 7.22.

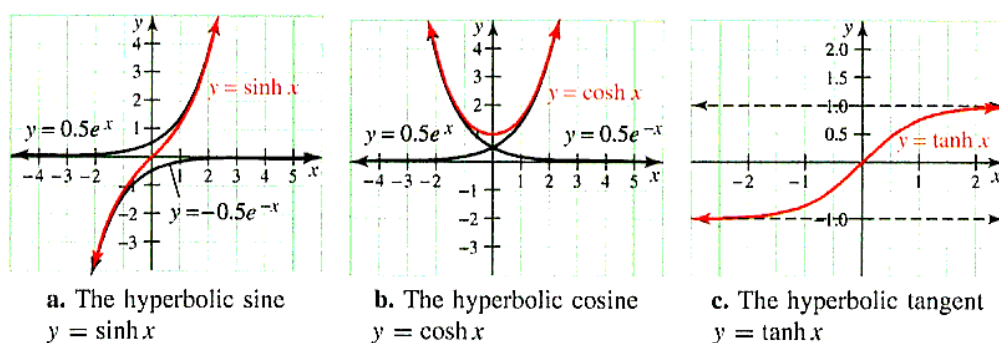


Figure 7.22 Graphs of the three basic hyperbolic functions

The list of properties in the following theorem suggests that the basic hyperbolic functions are analogous to the trigonometric functions.

Theorem 7.3 Properties of the hyperbolic functions

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sinh(-x) = -\sinh x$$

$\sinh x$ is odd

$$\cosh(-x) = \cosh x$$

$\cosh x$ is even

$$\tanh(-x) = -\tanh x$$

$\tanh x$ is odd

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

Proof: We will verify the first identity and leave the others for the reader.

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} \\ &= 1 \end{aligned}$$

A major difference between the trigonometric and hyperbolic functions is that the trigonometric functions are periodic, but the hyperbolic functions are not.

There are three additional hyperbolic functions: the **hyperbolic cotangent** ($\coth x$), the **hyperbolic secant** ($\operatorname{sech} x$), and the **hyperbolic cosecant** ($\operatorname{csch} x$). These functions are defined as follows:

$$\coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \quad \operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

TECHNOLOGY NOTE: Some of the software packages show these in simplified form as:

$$\coth x = \frac{e^{2x} + 1}{e^{2x} - 1} \quad \operatorname{sech} x = \frac{2e^x}{e^{2x} + 1} \quad \operatorname{csch} x = \frac{2e^x}{e^{2x} - 1}$$

Two identities involving these functions are

$$\operatorname{sech}^2 x = 1 - \tanh^2 x \quad \text{and} \quad \operatorname{csch}^2 x = \coth^2 x - 1$$

You will be asked to verify these identities in the problems.

Derivatives and Integrals Involving Hyperbolic Functions

Rules for differentiating the hyperbolic functions are listed in Theorem 7.4.

Theorem 7.4 Rules for differentiating the hyperbolic functions

Let u be a differentiable function of x . Then:

$$\begin{aligned} \frac{d}{dx}(\sinh u) &= \cosh u \frac{du}{dx} & \frac{d}{dx}(\cosh u) &= \sinh u \frac{du}{dx} \\ \frac{d}{dx}(\tanh u) &= \operatorname{sech}^2 u \frac{du}{dx} & \frac{d}{dx}(\coth u) &= -\operatorname{csch}^2 u \frac{du}{dx} \\ \frac{d}{dx}(\operatorname{sech} u) &= -\operatorname{sech} u \tanh u \frac{du}{dx} & \frac{d}{dx}(\operatorname{csch} u) &= -\operatorname{csch} u \coth u \frac{du}{dx} \end{aligned}$$

Proof: Each of these rules can be obtained by differentiating the exponential functions that make up the appropriate hyperbolic function. For example, to differentiate $\sinh x$, we use the definition of $\sinh x$:

$$\begin{aligned} \frac{d}{dx}(\sinh x) &= \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) \\ &= \frac{1}{2}e^x - \frac{1}{2}e^{-x}(-1) \\ &= \frac{1}{2}(e^x + e^{-x}) \\ &= \cosh x \end{aligned}$$

The proofs for the other derivatives can be handled similarly.

Example 1 Derivatives involving hyperbolic functions

Find $\frac{dy}{dx}$ for each of the following functions:

- a. $y = \cosh Ax$, A is a constant b. $y = \tanh(x^2 + 1)$ c. $y = \ln(\sinh x)$

Solution

- a. We have

$$\frac{d}{dx}(\cosh Ax) = \sinh(Ax) \frac{d}{dx}(Ax) = A \sinh Ax$$

b. We find that

$$\frac{d}{dx} [\tanh(x^2 + 1)] = \operatorname{sech}^2(x^2 + 1) \frac{d}{dx} (x^2 + 1) = 2x \operatorname{sech}^2(x^2 + 1)$$

c. Using the chain rule, with $u = \sinh x$, we obtain

$$\frac{d}{dx} [\ln(\sinh x)] = \frac{1}{\sinh x} \frac{d}{dx} (\sinh x) = \frac{1}{\sinh x} (\cosh x) = \coth x$$

Each differentiation formula for hyperbolic functions corresponds to an integration formula. These formulas are listed in the following theorem.

Theorem 7.5 Rules for integrating the hyperbolic functions

$$\begin{aligned} \int \sinh x \, dx &= \cosh x + C & \int \cosh x \, dx &= \sinh x + C \\ \int \operatorname{sech}^2 x \, dx &= \tanh x + C & \int \operatorname{csch}^2 x \, dx &= -\coth x + C \\ \int \operatorname{sech} x \tanh x \, dx &= -\operatorname{sech} x + C & \int \operatorname{csch} x \coth x \, dx &= -\operatorname{csch} x + C \end{aligned}$$

Proof: The proof of each of these formulas follows directly from the corresponding derivative formula. 

Example 2 Integration involving hyperbolic forms

Find each of the following integrals:

a. $\int \cosh^3 x \sinh x \, dx$ b. $\int x \operatorname{sech}^2(x^2) \, dx$ c. $\int \tanh x \, dx$

Solution

a. $\int \cosh^3 x \sinh x \, dx = \int u^3 \, du = \frac{u^4}{4} + C = \frac{1}{4} \cosh^4 x + C$

Let $u = \cosh x$, then $du = \sinh x \, dx$

b.
$$\begin{aligned} \int x \operatorname{sech}^2(x^2) \, dx &= \int \operatorname{sech}^2(x^2) (x \, dx) && \text{Let } u = x^2, \text{ then } du = 2x \, dx \\ &= \int \operatorname{sech}^2 u \left(\frac{1}{2} du \right) \\ &= \frac{1}{2} \tanh u + C \\ &= \frac{1}{2} \tanh x^2 + C \end{aligned}$$

c. $\int \tanh x \, dx = \int \frac{\sinh x \, dx}{\cosh x} = \int \frac{du}{u} = \ln |u| + C = \ln(\cosh x) + C$

Let $u = \cosh x$, then $du = \sinh x \, dx$

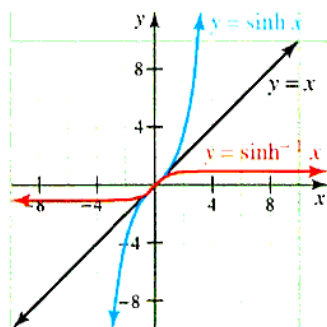


Figure 7.23 The graph of $y = \sinh^{-1} x$. The graph of $y = \sinh^{-1} x$ is obtained by reflecting $y = \sinh x$ in the line $y = x$

Inverse Hyperbolic Functions

Inverse hyperbolic functions are also of interest primarily because they enable us to express certain integrals in simple terms. Because $\sinh x$ is continuous and strictly monotonic (increasing), it is one-to-one and has an inverse, which is defined by

$$y = \sinh^{-1} x \quad \text{if and only if} \quad x = \sinh y$$

for all x and y . This is called the **inverse hyperbolic sine** function, and its graph is obtained by reflecting the graph of $y = \sinh x$ in the line $y = x$, as shown in Figure 7.23. Other inverse hyperbolic functions are defined similarly (see Problem 58).

Because the hyperbolic functions are defined in terms of exponential functions, we may expect to be able to express the inverse hyperbolic functions in terms of logarithmic functions. We summarize these relationships in the following theorem.

Theorem 7.6 Logarithmic formulas for inverse hyperbolic functions

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}), \text{ all } x \quad \operatorname{csch}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right), x \neq 0$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), x \geq 1 \quad \operatorname{sech}^{-1} x = \ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right), 0 < x \leq 1$$

$$\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}, |x| < 1 \quad \operatorname{coth}^{-1} x = \frac{1}{2} \ln \frac{x+1}{x-1}, |x| > 1$$

Proof: We will prove the first formula and leave the next two formulas for you to verify (see Problem 59). The other formulas are proved similarly.

Let $y = \sinh^{-1} x$; then its inverse is

$$x = \sinh y$$

$$x = \frac{1}{2}(e^y - e^{-y})$$

$$2x = e^y - \frac{1}{e^y}$$

$$e^{2y} - 2xe^y - 1 = 0$$

Quadratic formula with $a = 1$, $b = -2x$, $c = -1$

$$\begin{aligned} e^y &= \frac{2x \pm \sqrt{4x^2 + 4}}{2} \\ &= x \pm \sqrt{x^2 + 1} \end{aligned}$$

Because $e^y > 0$ for all y , the only solution is $e^y = x + \sqrt{x^2 + 1}$, and from the definition of logarithms,

$$y = \ln(x + \sqrt{x^2 + 1})$$

Differentiation and integration formulas involving inverse hyperbolic functions are listed in Theorem 7.7.

Theorem 7.7 Differentiation and integration of the inverse hyperbolic functions

$$\begin{aligned}
\frac{d}{dx}(\sinh^{-1} u) &= \frac{1}{\sqrt{1+u^2}} \frac{du}{dx} & \int \frac{du}{\sqrt{1+u^2}} &= \sinh^{-1} u + C \\
\frac{d}{dx}(\cosh^{-1} u) &= \frac{1}{\sqrt{u^2-1}} \frac{du}{dx} & \int \frac{du}{\sqrt{u^2-1}} &= \cosh^{-1} u + C \\
\frac{d}{dx}(\tanh^{-1} u) &= \frac{1}{1-u^2} \frac{du}{dx} & \int \frac{du}{1-u^2} &= \tanh^{-1} u + C \\
\frac{d}{dx}(\operatorname{csch}^{-1} u) &= \frac{-1}{|u| \sqrt{1+u^2}} \frac{du}{dx} & \int \frac{du}{u \sqrt{1+u^2}} &= -\operatorname{csch}^{-1} |u| + C \\
\frac{d}{dx}(\operatorname{sech}^{-1} u) &= \frac{-1}{u \sqrt{1-u^2}} \frac{du}{dx} & \int \frac{du}{u \sqrt{1-u^2}} &= -\operatorname{sech}^{-1} |u| + C \\
\frac{d}{dx}(\operatorname{coth}^{-1} u) &= \frac{1}{1-u^2} \frac{du}{dx} & \int \frac{du}{1-u^2} &= \operatorname{coth}^{-1} u + C
\end{aligned}$$

Proof: The derivative of each inverse hyperbolic function can be found either by differentiating the appropriate logarithmic function or by using the definition in terms of hyperbolic functions. We will show how this is done for $y = \sinh^{-1} x$, and then will leave it for you to apply the chain rule for u , a differentiable function of x . By definition of inverse, we see

$$x = \sinh y \quad \text{and} \quad \frac{dx}{dy} = \cosh y$$

Thus,

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}$$

since $\cosh^2 y - \sinh^2 y = 1$ and $\sinh y = x$. The integration formula that corresponds to this differentiation formula is

$$\int \frac{dx}{\sqrt{1+x^2}} = \sinh^{-1} x + C$$

The other differentiation and integration formulas follow similarly.

Example 3 Derivatives involving inverse hyperbolic functions

Find $\frac{dy}{dx}$ for **a.** $y = \sinh^{-1}(ax + b)$ **b.** $y = \cosh^{-1}(\sec x)$, $0 \leq x < \frac{\pi}{2}$

Solution

$$\begin{aligned}
\text{a.} \quad \frac{d}{dx} [\sinh^{-1}(ax + b)] &= \frac{1}{\sqrt{1 + (ax + b)^2}} \frac{d}{dx}(ax + b) \\
&= \frac{a}{\sqrt{1 + (ax + b)^2}}
\end{aligned}$$

$$\begin{aligned}
\text{b.} \quad \frac{d}{dx} [\cosh^{-1}(\sec x)] &= \frac{1}{\sqrt{\sec^2 x - 1}} \frac{d}{dx}(\sec x) \\
&= \frac{\sec x \tan x}{\sqrt{\tan^2 x}} && \tan x > 0 \text{ because } 0 \leq x < \pi/2 \\
&= \sec x
\end{aligned}$$

The result of Example 3b implies

$$\begin{aligned}\int \sec x \, dx &= \cosh^{-1}(\sec x) + C \\ &= \ln \left(\sec x + \sqrt{\sec^2 x - 1} \right) + C \\ &= \ln(\sec x + \tan x) + C\end{aligned}$$

Recall that the formula
 $\int \sec x \, dx = \ln |\sec x + \tan x| + C$
 was derived using algebraic
 procedures in Section 7.1.

Notice that we do not need absolute values because $0 \leq x < \frac{\pi}{2}$.

Example 4 Integral involving an inverse hyperbolic function

Evaluate $\int_0^1 \frac{dx}{\sqrt{1+x^2}}$.

Solution

$$\begin{aligned}\int_0^1 \frac{dx}{\sqrt{1+x^2}} &= [\sinh^{-1} x]_0^1 \\ &= \ln(x + \sqrt{x^2 + 1}) \Big|_0^1 \\ &= \ln(1 + \sqrt{2}) - \ln 1 \\ &= \ln(1 + \sqrt{2})\end{aligned}$$

It may be a worthwhile exercise to evaluate this integral using a trigonometric substitution to obtain the same result. 

PROBLEM SET 7.8

Level 1

Evaluate (correct to four decimal places) the indicated hyperbolic or inverse hyperbolic functions in Problems 1–12.

- | | | |
|--------------------|------------------------------------|----------------------|
| 1. $\sinh 2$ | 2. $\cosh 3$ | 3. $\tanh(-1)$ |
| 4. $\sinh^{-1} 0$ | 5. $\tanh^{-1} 0$ | 6. $\cosh^{-1} 1.5$ |
| 7. $\coth 1.2$ | 8. $\operatorname{sech} 1$ | 9. $\sinh(\ln 2)$ |
| 10. $\cosh(\ln 3)$ | 11. $\operatorname{sech}^{-1} 0.2$ | 12. $\coth^{-1}(-3)$ |

Find $\frac{dy}{dx}$ in Problems 13–28.

- | | |
|------------------------------|--|
| 13. $y = \sinh 3x$ | 14. $y = \sinh \sqrt{x}$ |
| 15. $y = \cosh(1 - 2x^2)$ | 16. $y = \cosh(2x^2 + 3x)$ |
| 17. $y = \sinh x^{-1}$ | 18. $y = \cosh^{-1} x^2$ |
| 19. $y = \sinh^{-1} x^3$ | 20. $y = x \tanh^{-1} 3x$ |
| 21. $y = \sinh^{-1}(\tan x)$ | 22. $y = \cosh^{-1}(\sec x)$ |
| 23. $y = \tanh^{-1}(\sin x)$ | 24. $y = \operatorname{sech} \left(\frac{1-x}{1+x} \right)$ |

25. $y = \frac{\sinh^{-1} x}{x}$ 26. $y = \sinh^{-1} x - \sqrt{1+x^2}$

27. $y = x \cosh^{-1} x - \sqrt{x^2 - 1}$

28. $x \cosh y = y \sinh x + 5$

Compute the integrals in Problems 29–44.

29. $\int x \cosh(1 - x^2) dx$ 30. $\int \frac{\operatorname{sech}^2(\ln x) dx}{x}$

31. $\int \frac{\sinh \frac{1}{x} dx}{x^2}$ 32. $\int \coth x \, dx$

33. $\int \frac{dt}{\sqrt{9t^2 - 16}}$ 34. $\int \frac{dx}{\sqrt{4x^2 + 16}}$

35. $\int \frac{\cos x \, dx}{\sqrt{1 + \sin^2 x}}$ 36. $\int \frac{dt}{36 - 16t^2}$

37. $\int \frac{x^2 \, dx}{1 - x^6}$ 38. $\int \frac{x \, dx}{\sqrt{1 + x^4}}$

39. $\int_2^3 \frac{dx}{1 - x^2}$ 40. $\int_0^{\frac{1}{2}} \frac{dx}{1 - x^2}$

41. $\int_1^2 \frac{e^x \, dx}{\sqrt{e^{2x} - 1}}$ 42. $\int_0^1 \frac{t^5 \, dt}{\sqrt{1 + t^{12}}}$

43. $\int_0^1 x \operatorname{sech}^2 x^2 \, dx$ 44. $\int_{-\ln 2}^{\ln 2} \sinh 3x \, dx$

Level 2

45. Show that
- $\tanh(x + y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$
 - $\sinh 2x = 2 \sinh x \cosh x$
 - $\cosh 2x = \cosh^2 x + \sinh^2 x$
46. Show that
- $-1 < \tanh x < 1$ for all x
 - $\lim_{x \rightarrow \infty} \tanh x = 1$
 - $\lim_{x \rightarrow -\infty} \tanh x = -1$
47. Determine where the graph of $y = \tanh x$ is rising and falling and where it is concave up and concave down. Sketch the graph and compare with Figure 7.22c.
48. Sketch the graph of $y = \coth x$. Be sure to show key features such as intercepts, maxima and minima, and points of inflection.
49. First show that $\cosh x + \sinh x = e^x$, and then use this result to prove that

$$(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$$

for positive integers n .

50. If $x = a \cosh t$ and $y = b \sinh t$ where a, b are positive constants and t is any number, show that

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

51. a. Verify that $y = a \cosh cx + b \sinh cx$ satisfies the differential equation

$$y'' - c^2 y = 0$$

- b. Use part a to find a solution of the differential equation

$$y'' - 4y = 0$$

subject to the initial conditions $y(0) = 1$, $y'(0) = 2$.

52. Find the volume V generated when the region bounded by the curves $y = \sinh x$, $y = \cosh x$, the y -axis, and the line $x = c$ ($c > 0$) is revolved about the x -axis. For what value of c does $V = 1$?
53. Find the length of the catenary

$$y = a \cosh \frac{x}{a}$$

between $x = -a$ and $x = a$.

54. Find the volume of the solid formed by revolving the region bounded by the curve $y = \tanh x$ on the interval $[0, 1]$ about the x -axis.
55. Find the surface area of the solid generated by revolving the curve $y = \cosh x$ on the interval $[-1, 1]$ about the x -axis.

Level 3

56. Prove the following formulas
- $\operatorname{sech}^2 x + \tanh^2 x = 1$
 - $\coth^2 x - \operatorname{csch}^2 x = 1$
57. Derive the differentiation formulas for $\cosh u$, $\tanh u$, and $\operatorname{sech} u$, where u is a differentiable function of x .
58. First give definitions for $\cosh^{-1} x$, $\tanh^{-1} x$, and $\operatorname{sech}^{-1} x$, and then derive the differentiation formulas for these functions by using these definitions.
59. Prove
- $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$, $x \geq 1$
 - $\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}$, $|x| < 1$
60. Show that

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1} \frac{x}{a} + C$$

for constant $a > 0$. State and prove a similar formula for

$$\int \frac{dx}{\sqrt{x^2 - a^2}}$$

with $x > a > 0$.

CHAPTER 7 REVIEW

Calculus is the most powerful weapon of thought yet devised by the wit of man.

William. B. Smith
Infinitesimal Analysis, (1898)

Proficiency Examination

Concept Problems

- Discuss the method of integration by substitution for both indefinite and definite integrals.
- What is the formula for integration by parts?
 - What is a reduction formula for integration?
- When should you consider a trigonometric substitution?
 - What are the substitutions to use when integrating rational trigonometric integrals?
- When should you consider the method of partial fractions?
- Outline a strategy for integration.
- What is a first-order linear differential equation? Outline a procedure for solving such an equation.
- What is an improper integral?
- Define the hyperbolic sine, hyperbolic cosine, and hyperbolic tangent functions.
- State the rules for differentiating and integrating the hyperbolic functions.
- What are the logarithmic formulas for the inverse hyperbolic functions?
- State the differentiation and integration formulas for the inverse hyperbolic functions.

Practice Problems

- Evaluate each of the given numbers.
 - $\tanh^{-1} 0.5$
 - $\sinh(\ln 3)$
 - $\coth^{-1} 2$

Find the integrals in Problems 13-18.

- $\int \frac{2x+3}{\sqrt{x^2+1}} dx$
- $\int x \sin 2x \, dx$
- $\int \sinh(1-2x) dx$
- $\int \frac{dx}{\sqrt{4-x^2}}$
- $\int \frac{x^2 dx}{(x^2+1)(x-1)}$
- $\int \frac{x^3 dx}{x^2-1}$

Evaluate the definite integrals in Problems 19-22.

- $\int_1^2 x \ln x^3 \, dx$
- $\int_2^3 \frac{dx}{(x-1)^2(x+2)}$
- $\int_3^4 \frac{dx}{2x-x^2}$
- $\int_0^{\frac{\pi}{4}} \sec^3 x \tan x \, dx$

Determine whether each improper integral in Problems 23-26 converges, and if it does, find its value.

- $\int_0^\infty x e^{-2x} dx$
- $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x \, dx}{\sqrt{\tan x}}$
- $\int_0^1 \frac{2x+3}{x^2(x-2)} dx$
- $\int_0^\infty e^{-x} \sin x \, dx$

27. Find $\frac{dy}{dx}$ for $y = \sqrt{\tanh^{-1} 2x}$.

28. Find the volume of the solid generated when the region under the curve

$$y = \frac{1}{\sqrt{9-x^2}}$$

on the interval $[0, 2]$ is revolved about the y-axis.

29. Solve the first-order linear differential equation

$$\frac{dy}{dx} + \frac{xy}{x+1} = e^{-x}$$

subject to the condition $y = 1$ when $x = 0$.

30. A tank contains 200 gal of saturated brine with 2 lb of salt per gallon. A salt solution containing 1.3 lb of salt per gallon flows in at the rate of 5 gal/min, and the uniform mixture flows out at 3 gal/min. Find the amount of salt in the tank (to the nearest tenth of a lb) after 1 hour.

Supplementary Problems*

Find the derivative $\frac{dy}{dx}$ in Problems 1-5.

1. $y = \tanh^{-1} \frac{1}{x}$

2. $y = x \cosh^{-1}(3x+1)$

3. $y = \frac{\sinh x}{e^x}$

4. $y = \sinh x \cosh x$

5. $y = x \sinh x + \sinh(e^x + e^{-x})$

Find the integrals in Problems 6-47.

6. $\int \cos^{-1} x \, dx$

7. $\int \frac{x^2 \, dx}{\sqrt{4-x^2}}$

8. $\int \frac{x \, dx}{x^2 - 2x + 5}$

9. $\int \frac{3x-2}{x^3-2x^2} \, dx$

10. $\int \frac{dx}{(x^2+x+1)^{3/2}}$

11. $\int \frac{dx}{\sqrt{x}(1+\sqrt[4]{x})}$

12. $\int \frac{\sqrt{9x^2-1} \, dx}{x}$

13. $\int x^2 \tan^{-1} x \, dx$

14. $\int \frac{dx}{\sin x + \tan x}$

15. $\int e^x \sqrt{4-e^{2x}} \, dx$

16. $\int \cos \frac{x}{2} \sin \frac{x}{3} \, dx$

17. $\int \frac{\sqrt{1+\frac{1}{x^2}}}{x^5} \, dx$

18. $\int \frac{\sin x \, dx}{\cos^5 x}$

19. $\int \sqrt{1+\sin x} \, dx$

20. $\int \cos x \ln(\sin x) \, dx$

21. $\int \sin(\ln x) \, dx$

22. $\int e^{2x} \operatorname{sech} e^{2x} \, dx$

23. $\int \frac{\sinh x \, dx}{2 + \cosh x}$

24. $\int \frac{\tanh^{-1} x \, dx}{1-x^2}$

25. $\int x^2 \cot^{-1} x \, dx$

26. $\int x(1+x)^{1/3} \, dx$

27. $\int \frac{x^2+2}{x^3+6x+1} \, dx$

28. $\int \frac{\sin x - \cos x}{(\sin x + \cos x)^{1/4}} \, dx$

29. $\int \cos \sqrt{x+2} \, dx$

30. $\int \sqrt{5+2\sin^2 x} \sin 2x \, dx$

31. $\int \frac{x^3+2x}{x^4+4x^2+3} \, dx$

32. $\int \frac{x \, dx}{\sqrt{5-x^2}}$

33. $\int \frac{\sqrt{5-x^2} \, dx}{x}$

34. $\int \frac{\sqrt{x^2+x} \, dx}{x}$

35. $\int x^3(x^2+4)^{-1/2} \, dx$

36. $\int \sin^2 \frac{x}{2} \cos^2 \frac{x}{2} \, dx$

37. $\int \sec^3 x \tan x \, dx$

38. $\int \sec^5 x \tan^2 x \, dx$

39. $\int \frac{x^{1/3} \, dx}{x^{1/2} + x^{2/3}}$

40. $\int \frac{\sin x \, dx}{\sin x + \cos x}$

41. $\int \frac{\cos x + \sin x}{1 + \cos x - \sin x} \, dx$

42. $\int \frac{dx}{x^{1/4} + 1}$

43. $\int \frac{e^x \, dx}{e^{x/3} - e^{x/2}}$

44. $\int \tan^4 x \, dx$

45. $\int \sec^4 x \, dx$

46. $\int \frac{5x^2 - 4x + 9}{x^3 - x^2 + 4x - 4} \, dx$

47. $\int \frac{x \, dx}{(x+1)(x+2)(x+3)}$

In Problems 48-53, solve the given differential equations.

48. $\frac{dy}{dx} = \sqrt{\frac{x^2-1}{y^2+1}}$

49. $\frac{dy}{dx} = \frac{1-y}{1+x}$

50. $\frac{dy}{dx} - \frac{y}{2x} = \frac{1}{x^{1/2}+1}$

51. $xy' - yx^{1/2} = 0$

52. $y' + (\tan x)y = \sec^3 x$

53. $(\cos^3 x)y' = \sin^3 x \cos y$

*The supplementary problems are presented in a somewhat random order, not necessarily in order of difficulty.

Determine whether the improper integral in Problems 54–59 converges, and if it does, find its value.

54. $\int_0^{\infty} 5e^{-2x} dx$

55. $\int_1^{\infty} e^{1-x} dx$

56. $\int_{-\infty}^{\infty} \frac{dx}{4+x^2}$

57. $\int_0^{\pi/2} \frac{\cos x}{\sqrt{\sin x}} dx$

58. $\int_1^{\infty} \frac{dx}{x^4+x^2}$

59. $\int_{-1}^1 \frac{dx}{(2x+1)^{1/3}}$

60. **Think Tank Problem** Each of the following equations may be either true or false. In each case, either show that the equation is generally true or find a number x for which it fails.

a. $\tanh\left(\frac{1}{2} \ln x\right) = \frac{x-1}{x+1}, x > 0$

b. $\sinh^{-1}(\tan x) = \tanh^{-1}(\sin x)$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

61. Compute $\int_0^1 x f''(3x) dx$, given that $f'(0)$ is defined, $f(0) = 1$, $f(3) = 4$, and $f'(3) = -2$.

62. What is $\int_0^{\pi/2} [f(x) + f''(x)] \cos x dx$ if $f(0)$ is defined, $f(\frac{\pi}{2}) = 5$, and $f'(0) = -1$?

63. **Newton's law of cooling** says that the temperature T of a body satisfies

$$\frac{dT}{dt} = k(T - T_0)$$

where T_0 is the temperature of the surrounding medium. Solve this equation for T .

64. **Modeling Problem** The psychologist L. L. Thurstone investigated the way people learn using the differential equation

$$\frac{dS}{dt} = \frac{2k}{\sqrt{m}} [S(1-S)]^{3/2}$$

where $S(t)$ is the state of the learner at time t , and k and m are positive constants depending on the learner and the nature of the task.* Solve this equation.

65. If $s > 0$, the integral function defined by

$$\Gamma(s) = \int_0^{\infty} e^{-t} t^{s-1} dt$$

is called the **gamma function**. It was introduced by Leonhard Euler in 1729 and has some useful properties.

- Show that $\Gamma(s)$ converges for all $s > 0$.
 - Show that $\Gamma(s+1) = s\Gamma(s)$.
 - Show that $\Gamma(n+1) = n!$ for any positive integer n .
- Find the centroid of a thin plate of constant density that occupies the region bounded by the graph of $y = \sin x + \cos x$, the coordinate axes, and the line $x = \frac{\pi}{4}$.
 - Repeat Problem 66 for a plate that occupies the region bounded by $y = \sec^2 x$, the coordinate axes, and the line $x = \frac{\pi}{3}$.
 - Find the volume obtained by revolving about the y -axis the region bounded by the curve $y = \sinh x$, the x -axis, and the line $x = 1$.
 - Find the volume (to the nearest hundredth) obtained by revolving about the y -axis the region bounded by the curve $y = \cosh x$, the y -axis, and the line $y = 2$.
 - The region bounded by the graph of $y = \sin x$ and the x -axis between $x = 0$ and $x = \pi$ is revolved about the x -axis to generate a solid. Find the volume of the solid and its surface area.
 - Find the length of the arc of the curve $y = \frac{4}{5}x^{5/4}$ that lies between $x = 0$ and $x = 1$.
 - Find the volume of the solid obtained by revolving about the x -axis the region bounded by the curve $y = (9-x^2)^{1/4}$ and the x -axis.
 - Find the volume of the solid formed by revolving about the y -axis the region bounded by the curve

$$y = \frac{1}{1+x^4}$$

and the x -axis between $x = 0$ and $x = 4$.

* L. L. Thurstone, "The Learning Function," J. of General Psychology 3(1930), pp. 469-493.

74. Find the volume of the solid formed by revolving about the x -axis the region bounded by the curve

$$y = \frac{2}{\sqrt{3x-2}}$$

the x -axis, and the lines $x = 1$ and $x = 2$.

75. Find the surface area of the solid generated by revolving the region bounded by the curve

$$y = e^x + \frac{1}{4}e^{-x}$$

on $[0, 1]$ about the x -axis.

76. Find the volume of the solid whose base is the region R bounded by the curve $y = e^x$ and the lines $y = 0$, $x = 0$, $x = 1$, if cross sections perpendicular to the x -axis are equilateral triangles.

77. Show that the area under $y = \frac{1}{x}$ on the interval $[1, a]$ equals the area under the same curve on $[k, ka]$ for any number $k > 0$.

78. Find the length of the curve $y = \ln(\csc x)$ on the interval $[\frac{\pi}{3}, \frac{\pi}{2}]$.

79. Find the length of the curve

$$x = \frac{ay^2}{b^2} - \frac{b^2}{8a} \ln \frac{y}{b}$$

between $y = b$ and $y = 2b$.

80. Derive the reduction formula

$$\int (a^2 - x^2)^n dx = \frac{x(a^2 - x^2)^n}{2n+1} + \frac{2a^2n}{2n+1} \int (a^2 - x^2)^{n-1} dx$$

Use this formula to find $\int (9 - x^2)^{5/2} dx$.

81. Derive the reduction formula

$$\int \frac{\sin^n x \, dx}{\cos^m x} = \frac{1}{m-1} \frac{\sin^{n-1} x}{\cos^{m-1} x} - \frac{n-1}{m-1} \int \frac{\sin^{n-2} x \, dx}{\cos^{m-2} x}$$

82. Derive a reduction formula for $\int \frac{\cos^n x \, dx}{\sin^m x}$.

83. Derive a reduction formula for $\int x^n (x^2 + a^2)^{-1/2} dx$.

84. Derive the reduction formula

$$\int \frac{dx}{x^n \sqrt{ax+b}} = \frac{-\sqrt{ax+b}}{(n-1)bx^{n-1}} - \frac{(2n-3)a}{(2n-2)b} \int \frac{dx}{x^{n-1} \sqrt{ax+b}}$$

85. The improper integral

$$\int_1^\infty \left[\frac{2Ax^3}{x^4+1} - \frac{1}{x+1} \right] dx$$

converges for exactly one value of the constant A . Find A and then compute the value of the integral.

86. Evaluate $\int_0^\infty \frac{\sqrt{x} \ln x \, dx}{(x+1)(x^2+x+1)}$ if it exists. *Hint:* Let $u = \frac{1}{x}$.

87. Spectroscopic measurements based on the Doppler effect yield an observed mass m_0 for a certain type of binary star. The true mass m is then estimated by $m = \frac{m_0}{I}$, where

$$I = \int_0^{\pi/2} \sin^4 x \, dx$$

Determine the number I .

88. Two substances, A and B , are being converted into a single compound C . In the laboratory, it is shown that the time rate of change of the amount x of compound C is proportional to the product of the amount of unconverted substances A and B . Thus,

$$\frac{dx}{dt} = k(a - x)(b - x)$$

for initial concentrations a and b , of A and B , respectively.

a. Solve this differential equation to express x in terms of time t . What happens to $x(t)$ as $t \rightarrow \infty$ if $b > a$? What if $a > b$?

b. Suppose $a = b$. What is $x(t)$? What happens to $x(t)$ as $t \rightarrow \infty$ in this case?

89. Let $f(x) = \begin{cases} \frac{1}{x^2} & \text{for } x \geq 1 \\ 1 & \text{for } -1 < x < 1 \\ e^{x+1} & \text{for } x \leq -1 \end{cases}$

Sketch the graph of f and evaluate

$$\int_{-\infty}^{\infty} f(x) dx$$

90. Evaluate the improper polar integral

$$\int_0^{\infty} \theta e^{-\theta} d\theta$$

91. Find the total area between the spirals

$$r = e^{-2\theta} \text{ and } r = e^{-5\theta}$$

for $\theta \geq 0$.

92. **Modeling Problem** The residents of a certain community have voted to discontinue the fluoridation of their water supply. The local reservoir currently holds 200 million gallons of fluoridated water containing 1,600 lb of fluoride. The fluoridated water flows out of the reservoir at the rate of 4 million gallons per day and is replaced by unfluoridated water at the same rate. At all times, the remaining fluoride in the reservoir is evenly distributed. Set up and solve a differential equation for the amount $F(t)$ of fluoride in the reservoir at time t . When will 90% of the fluoride be replaced?

93. **Modeling Problem** The Nuclear Regulatory Commission puts atomic waste into sealed containers and dumps them into the ocean. It is important to dump the containers in water shallow enough to ensure they do not break when they hit bottom. Suppose $s(t)$ is the depth of the container at time t and let W and B denote the container's weight and the constant buoyancy force, respectively. Assume there is a "drag force" proportional to velocity v .

a. Explain why the motion of the containers may be modeled by the differential equation

$$\frac{W}{g} \frac{dv}{dt} = W - B - kv$$

where g is the constant acceleration due to gravity. Solve this equation to express $v(t)$ in terms of W , k , B , and g .

b. Integrate the expression for $v(t)$ in part a to find $s(t)$.

c. Suppose $W = 1,125$ newtons (about 250 lb), $B = 1,100$ newtons, and $k = 0.64$ kg/s. If the container breaks when the impact speed exceeds 10 m/s, what is the maximum depth (to the nearest meter) for safe dumping?

94. **The Evans price-adjustment model** assumes that if there is an excess demand D over supply S in any time period, the price p changes at a rate proportional to the excess, $D - S$; that is,

$$\frac{dp}{dt} = k(D - S)$$

Suppose for a certain commodity, demand is linear,

$$D(p) = c - pd$$

and supply is cyclical

$$S(t) = a \sin(bt)$$

Solve the differential equation to express price $p(t)$ in terms of a, b, c , and d . What happens to $p(t)$ “in the long run” (as $t \rightarrow \infty$)? Use $p_0 = p(0)$.

- 95. Modeling Problem** A country has 5 billion dollars of paper currency. Each day about 10 million dollars comes into banks and 12 million is paid out. The government decides to issue new currency, and whenever an “old” bill comes into the bank, it is replaced by a “new” bill. Set up and solve a differential equation to model the currency replacement. How long will it take for 95% of the currency in circulation to be “new” bills?
- 96.** If n is a positive integer and a is a positive number, find

$$\int_0^{\infty} x^n e^{-ax} dx$$

- 97. Putnam Examination Problem** Show that

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4 dx}{1+x^2}$$

- 98. Putnam Examination Problem** Evaluate

$$\int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}$$

- 99. Putnam Examination Problem** Evaluate

$$\int_0^{\infty} t^{-1/2} \exp \left[-1985 \left(t + \frac{1}{t} \right) \right] dt$$

You may assume that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

CHAPTER 7 GROUP RESEARCH PROJECT

Working in small groups is typical of most work environments, and this book seeks to develop skills with group activities. We present a group project at the end of each chapter. These projects are to be done in groups of three or four students.

Buoy Design



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You have been hired as a special consultant by the U. S. Coast Guard to evaluate some proposed new designs for navigational aids (buoys).

The buoys are floating cans that need to be visible from some distance away, without rising too far out of the water. Each buoy has a circular cross section and will be fitted with a superstructure that carries equipment such as lights and batteries. To be acceptable, a fully equipped buoy must float with not less than 1.5 ft nor more than 3 ft of freeboard. (Freeboard is the distance from the water level to the top of the device.)

You should make the following assumptions:*

- A floating object will displace a volume of water whose weight equals the weight of the floating object. (This is Archimedes' principle.)
- Your devices will be floating in salt water, which weighs 64.0 lb/ft³.
- The buoys will be constructed of $\frac{1}{2}$ in.-thick sheet metal that weighs 490 lb/ft³.
- You can estimate an additional 20% in weight attributable to welds, bolts, and the like.
- Each buoy will be fitted with a superstructure and equipment weighing a total of 2,000 lb.
- Each design you are given to evaluate will be presented to you in the form of a curve $x = f(y)$, to be revolved about the y -axis.

Extended paper for further study. Write a report using the given assumptions to evaluate several designs for buoys, and formulate a conclusion about the design you recommend to the Coast Guard.

*This project is adapted from a computer project used at the U.S. Coast Guard Academy.